

A Double-Scaling Large- d Saddle of BFSS/BMN Matrix Quantum Mechanics

BFSS/BMN Matrix Quantum Mechanics I

Badis Ydri

Department of Physics, Badji Mokhtar Annaba University, Algeria

August 11, 2026

Abstract

We study the large- d dynamics of the mass-deformed bosonic BFSS $_{d+1}$ matrix quantum mechanics using a Hubbard-Stratonovich localization of the Yang-Mills interaction. After integrating out the matrix coordinates, the theory reduces to a holonomy-dependent effective action for an auxiliary adjoint kernel. We introduce a commuting-symmetric saddle and its maximally symmetric specialization, in which the interaction is encoded in a single dynamically generated mass shift k_0 . The resulting large- d description is a gauged matrix harmonic oscillator with self-consistent frequency $s^2 = m + k_0$, fixed by a gap equation.

We analyze the low-temperature X -space physics, the holonomy effective action, the Yang-Mills observable, and the associated phase structure. We then identify a correlated double-scaling limit in which $d \rightarrow \infty$, $m \rightarrow \infty$, and $\kappa = m^{3/2}/d$ is held fixed. In this limit the Yang-Mills interaction and the explicit mass deformation remain parametrically balanced: the theory interpolates between the commutator-dominated BFSS regime and the mass-dominated Gaussian regime.

The double-scaled theory exhibits two complementary large- d regimes. At low temperature, the enhanced gap pushes the deconfinement scale upward and opens a parametrically large uniform-holonomy region, where the bulk dynamics behaves as weakly coupled BFSS $_2$ -type gauged harmonic-oscillator sectors. At the same time, the high-temperature branch reveals an overlap window in which the Gaussian description remains self-consistent while the commutator contribution per matrix pair is parametrically suppressed. The resulting dynamics is therefore BFSS $_2$ -like in its enlarged uniform-holonomy sector and IKKT-like in its almost-commuting matrix behavior.

Keywords: BFSS matrix quantum mechanics; BMN matrix model; large- d expansion; double-scaling limit; Hubbard-Stratonovich localization; matrix harmonic oscillator; holonomy effective action; deconfinement transition; Yang-Mills observable; IKKT matrix model; Molien-Weyl integral; emergent geometry.

Contents

1	Introduction and Goal	3
1.1	Generalities:BFSS/BMN systems	3
1.2	Goal and overview	5
1.3	Organization of the paper	7
2	Hubbard–Stratonovich localization of the BFSS/BMN action	8
2.1	Action and observables	8
2.2	Action in terms of large- d effective variables	9
2.3	Localization: Hubbard–Stratonovich decoupling	12
2.4	Effective action	14
3	Saddle-point equation and the maximally symmetric ansatz	15
3.1	Saddle point equation	15
3.2	Maximally symmetric ansatz	17
3.3	Scope of the maximally symmetric ansatz	20
4	Low-temperature X-space physics of the large-d saddle	21
4.1	Gap equation at low temperature	21
4.2	Careful treatment of the holonomy	23
4.3	Bosonic effective action	25
4.4	Free energy	29
4.5	Extent of space	31
4.6	Consistency between the saddle equation and the extent of space	32
4.7	Low-temperature evaluation and mass regimes	32
5	Holonomy physics at low-T	34
5.1	Holonomy effective action	34
5.2	Confined phase and critical point	35
5.3	Effect of a mass deformation on the transition temperature	36
5.4	Deconfined phase and eigenvalue distributions	36
5.5	Physical interpretation of the holonomy phases	38
5.6	Black-string dictionary and the critical-dimension question	38
6	Yang–Mills observable	44
6.1	Large- d saddle value of the Yang–Mills observable	44
6.2	The small- m and large- m behaviors	46
7	Double-scaling saddle and high-temperature branch	48
7.1	The Yang–Mills-mass-deformation balance	48
7.2	Observables at fixed large κ : nearly commuting, localized matrices	49

7.3	The high-temperature limit	51
7.4	Size of the auxiliary-field fluctuations at high temperature	54
8	On supersymmetric extensions and Molien–Weyl approximations	55
8.1	Saddle point equation with split masses	55
8.2	Gaussian Molien–Weyl approximation	59
8.3	Supersymmetric completion and Monte Carlo simulations	61
9	Conclusion	66
10	Acknowledgments	67
A	Fourier derivation of the holonomic effective action	68
A.1	Coordinate contribution	68
A.2	Vandermonde contribution	69

1 Introduction and Goal

1.1 Generalities:BFSS/BMN systems

The BFSS_{*d*+1} models form a central class of matrix quantum mechanics: one-dimensional $U(N)$ gauge theories with d adjoint matrix coordinates X_a , obtained by dimensional reduction of ten-dimensional $\mathcal{N} = 1$ super Yang–Mills theory [1]. Their Euclidean action may be written schematically as

$$S_{\text{BFSS}}^{\text{E}} = \frac{1}{g^2} \int_0^\beta dt \, \text{Tr} \left[\frac{1}{2} (D_t X_a)^2 - \frac{1}{4} [X_a, X_b]^2 \right] + S_{\text{BFSS,F}}^{\text{E}}, \quad D_t = \partial_t - i[A_t, \cdot], \quad (1.1)$$

where $S_{\text{BFSS,F}}^{\text{E}}$ contains the fermionic kinetic and Yukawa terms required by supersymmetry.

Supersymmetry restricts the allowed matrix quantum mechanics to $D_{\text{YM}} = d+1 = 10, 6, 4, 3, 2$, by the Fierz identity analysis of Baake, Reinicke and Rittenberg [2]; the corresponding supermembrane or M-theory dimensions are $D_{\text{M}} = d+2 = 11, 7, 5, 4, 3$.

The holographic regime is obtained in the planar 't Hooft limit [3],

$$N \rightarrow \infty, \quad g^2 \rightarrow 0, \quad \lambda = g^2 N = \text{fixed}. \quad (1.2)$$

This is the natural large- N setting for holography [4, 5], and at strong coupling it is expected to admit a dual weakly curved classical supergravity description.

The central example is the $d = 9$ BFSS₁₀ model, or M-(atrix) theory, which gives a large- N nonperturbative formulation of M-theory in the infinite-momentum frame [6]. It describes the low-energy worldvolume dynamics of N coincident D0-branes [7, 9]. At strong coupling and large N , its thermodynamics is dual to the near-horizon black 0-brane geometry of type IIA

supergravity [8], which is the low-energy limit of type IIA superstring theory. Equivalently, type IIA supergravity is obtained from eleven-dimensional supergravity [10] by compactification on S^1 [11], and reducing the eleven-dimensional M-wave along the compact direction gives the ten-dimensional black 0-brane solution [12, 13]. This places BFSS₁₀ at the intersection of D0-brane dynamics, black 0-brane supergravity, light-cone supermembranes [14–16], and superparticles in maximally supersymmetric pp-wave backgrounds [17, 18].

In this interpretation, the matrices X_a are noncommutative transverse coordinates of the D0-branes: their diagonal components encode brane positions, while their off-diagonal components describe open strings stretched between different branes, becoming light when the branes coincide. See, for example, [19] and the pedagogical accounts [20, 21].

In the decoupling limit, the BFSS₁₀ matrix quantum mechanics is related by gauge/gravity duality to type IIA string theory on the near-horizon black 0-brane background [22–24]. In the large- N , strong-coupling regime, this becomes a weakly curved type IIA supergravity description, while the gauge theory side remains a nonperturbative matrix quantum mechanics which admits a nonperturbative lattice regularization [25]. Thus the black 0-brane correspondence provides a concrete nonperturbative setting for quantum gravity and black-hole thermodynamics.

This duality has been tested extensively by Monte Carlo simulations [26–30] and analytic methods [31–33]. In particular, BFSS₁₀/black 0-brane thermodynamics relates quantum-gravity corrections on the gravity side to finite- N and finite-coupling effects in the matrix model. For a pedagogical review, see [34].

More generally, one may consider the lower-dimensional BFSS _{$d+1$} matrix quantum mechanical models associated with the other allowed supersymmetric Yang–Mills dimensions, besides the maximally supersymmetric BFSS₁₀ case. These models describe lower-dimensional DLCQ dynamics of D0-branes and the associated gauge/gravity dualities. They also provide the corresponding matrix regularizations of light-cone supermembranes and are naturally related to the light-cone quantization of superparticles in maximally supersymmetric pp-wave backgrounds.

Although simpler than BFSS₁₀, these lower-dimensional models retain many of its essential dynamical features: confinement–deconfinement behavior, holonomy eigenvalue condensation, gauge/gravity-inspired matrix dynamics, and emergent geometric phases. They therefore provide analytically and numerically more tractable laboratories for studying the mechanisms underlying BFSS holography and matrix quantum geometry.

A second natural generalization is obtained by adding maximally supersymmetric mass deformations to the BFSS _{$d+1$} models. The prototype is the BMN plane-wave matrix model [35], with the general classification given in [36, 37]. These deformations add quadratic matrix masses, Myers-type cubic interactions [38], and the corresponding fermionic mass terms required by supersymmetry. Schematically,

$$S_{\text{BMN}}^{\text{E}} = S_{\text{BFSS}}^{\text{E}} + \frac{1}{g^2} \int_0^\beta dt \, \text{Tr} \left[\mu_1 X_a^2 + \mu_2 \epsilon_{ijk} X_i X_j X_k \right] + S_{\text{BMN,F}}^{\text{def}}, \quad (1.3)$$

where $S_{\text{BMN},\text{F}}^{\text{def}}$ denotes the fermionic part of the mass deformation.

The allowed cases are $d = 2, 3, 5, 9$, corresponding to $\text{BMN}_{3,4,6,10}$, while the special $d = 1$ case, BMN_2 , is discussed in [37, 65]. These deformations generally break the transverse $SO(d)$ symmetry into smaller rotational factors: the quadratic masses may split the matrix directions, while the Myers term selects an $SO(3)$ sector.

The mass term may be viewed as a cosmological, or curvature, scale: it lifts the flat directions of BFSS and replaces the flat target-space interpretation by a curved pp-wave background. On the gravity side, this is the distinction between the undeformed black 0-brane/type IIA setting and the deformed pp-wave setting. In the maximally supersymmetric case, the half-BPS sectors are related to LLM bubbling geometries [39]. Some Monte Carlo studies of the BMN model can be found in [40, 41] and references therein. For related recent developments, see also the preprints [77, 78].

In summary, each BFSS_{d+1} model admits a corresponding BMN deformation preserving maximal supersymmetry. These deformations describe supermembranes and superparticles in maximally supersymmetric pp-wave backgrounds. The corresponding classification is summarized in Table 1.

Model	D_{YM}	Splitting of $SO(D_{\text{YM}} - 1)$	Superalgebra	Deformation parameter
$\mathcal{N} = 16$	10	$SO(6) \times SO(3)$	$\mathfrak{su}(2 4)$	μ
$\mathcal{N} = 8$ type I	6	$SO(3) \times SO(2)$	$\mathfrak{su}(2 2)$	μ
$\mathcal{N} = 8$ type II	6	$SO(4)$	$\mathfrak{su}(2 1) \oplus \mathfrak{su}(2 1)$	μ
$\mathcal{N} = 4$ type I	4	$SO(3)$	$\mathfrak{su}(2 1)$	μ_1, μ_2
$\mathcal{N} = 4$ type II	4	$SO(2)$	$\text{Clifford}_4(\mathbb{R})$	μ
$\mathcal{N} = 2$	3	$SO(2)$	$\text{Clifford}_2(\mathbb{R})$	μ
$\mathcal{N} = 1 + 1$	2	$SO(1, 2)$	$\mathfrak{osp}(1 2, \mathbb{R})$	$\Lambda(t), \rho(t)$

Table 1: Classification of massive supersymmetric Yang–Mills quantum mechanics models and their deformation parameters.

1.2 Goal and overview

We study here the mass-deformed BFSS_{d+1} matrix model using a combined *large- d* and *saddle-point* analysis, with the aim of isolating the dominant low-energy degrees of freedom and clarifying their geometric and gravitational interpretation. Our approach is motivated by earlier large- d studies of matrix quantum mechanics, which show that the dynamics simplifies dramatically and becomes governed by a small set of collective variables, most notably the holonomy around the Euclidean time circle [42–45].

The first step of our analysis is to integrate out the matrix degrees of freedom X_a at large d using a Hubbard–Stratonovich localization and a saddle-point approximation. This leads to an

effective description in terms of a dynamically generated mass scale determined self-consistently by a gap equation. At leading order in $1/d$, we show that the interacting BFSS model reduces to a *gauged matrix harmonic oscillator*. This observation builds on earlier results demonstrating that the matrix harmonic oscillator already captures the essential thermodynamic and spectral features of the BFSS model in the low-temperature regime [30, 46].

After integrating out the X_a matrices, the remaining dynamics is encoded in an effective action for the holonomy eigenvalues. This holonomy effective action consists of a universal Van-dermonde repulsion term together with an attractive contribution generated by the massive adjoint fields. Its structure is closely related to unitary matrix models of the Gross–Witten–Wadia type [47, 48], and can be derived using standard methods for gauge theories on a circle, following Polyakov’s treatment of eigenvalue measures and holonomies [49].

Within this effective description we compute physical observables such as the free energy and the *extent of space*, defined by $\langle \text{Tr } X_a^2 \rangle$. In the Gaussian saddle, these observables are controlled by the dynamically generated mass and can be evaluated explicitly. We show that the saddle-point expectation values reproduce the classical constraints implied by the Hubbard–Stratonovich equations of motion, providing a nontrivial consistency check of the large- d approximation.

The resulting phase structure is governed by the competition between eigenvalue repulsion and attraction. At low temperatures the holonomy eigenvalues are uniformly distributed, corresponding to a confining phase in the gauge-theory sense. As the temperature increases, the uniform distribution becomes unstable and gives way to a non-uniform or gapped distribution, signaling a deconfinement transition. This transition is closely related to the Hagedorn phenomenon in string theory [50–53] and to the Gregory–Laflamme instability of black strings [54, 55].

A central result here is that a positive mass deformation pushes the deconfinement transition to parametrically higher temperatures. Combined with the large- d limit, this allows one to keep the system deep in the uniform, confining phase even for moderately large d . In this regime, BFSS_{d+1} behaves effectively as a collection of BFSS_2 -like sectors governed by matrix quantum mechanics with an AdS_2 dual description. This provides a controlled setting in which the Yang–Mills phase of the IKKT model can be identified with a confining, string-like phase encoding emergent geometry in a nontrivial way.

From the gravitational perspective, the uniform holonomy phase corresponds to a black-string-like configuration, while the gapped phase corresponds to a localized black hole in the dual description [56]. Thus our analysis shows how mass deformations and large- d dynamics can suppress the black-hole phase and stabilize the stringy, confining phase over a wide range of parameters. In this way, BFSS_2 -like physics dominates the low-energy sector of higher-dimensional BFSS models, and geometry emerges through AdS_2 quantum mechanics rather than higher-dimensional black-hole dynamics.

At the same time, the double-scaled saddle has a complementary IKKT-like aspect. The low-temperature analysis already shows that the matrices are localized near the origin and

that the commutator contribution per fixed matrix pair is parametrically suppressed. This interpretation becomes sharper after combining the low-temperature and high-temperature saddle descriptions: there is an overlap window in which both Gaussian descriptions are self-consistent, while the matrices remain localized and approximately commuting. Thus the system is not driven by strong noncommutativity, even though the original Yang–Mills interaction is retained self-consistently through the dynamically generated mass. Together with the enlarged BFSS₂-like uniform-holonomy regime, this shows that the double-scaling limit interpolates between BFSS₂-like AdS₂ quantum mechanics and IKKT-like commuting-matrix geometry.

The present work is part of a broader program on one-dimensional BFSS/BMN matrix gauge theories with an arbitrary number d of noncommuting matrix coordinates. The program combines large- d Gaussian reduction, Molien–Weyl singlet counting, endpoint and Wishart–Stiefel formulations, and Monte Carlo methods, with the broader aim of clarifying matrix quantum mechanics and its possible interpretation as matrix quantum gravity. Related aspects are developed in [71–76].

For completeness, the first arXiv version of the present work, which contains additional material not retained in this manuscript, is available in Ref. [70].

1.3 Organization of the paper

The paper is organized as follows. Section 2 develops the localized large- d formulation of the mass-deformed bosonic BFSS _{$d+1$} model, including the holonomy variables, the adjoint projector, and the Hubbard–Stratonovich kernel. Section 3 derives the saddle-point equation and introduces the commuting-symmetric and maximally symmetric ansätze, together with a discussion of the scope of the latter. Section 4 studies the low-temperature X -space physics of the saddle, including the self-consistent matrix harmonic oscillator, the auxiliary-field fluctuations, the free energy, and the extent of space.

Section 5 derives the low-temperature holonomy effective action, analyzes the associated phase structure, and discusses its black-string interpretation. Section 6 evaluates the Yang–Mills observable at the large- d saddle and studies its small- and large-mass limits. Section 7 introduces the double-scaling limit in which the Yang–Mills interaction and the explicit mass deformation remain parametrically balanced. It then analyzes the nearly commuting regime, the high-temperature Gaussian branch, and the size of the auxiliary-field fluctuations in that regime.

Section 8 discusses split-mass saddles, Gaussian Molien–Weyl approximations, and the supersymmetric completion of the large- d bosonic Gaussian theory, together with its purely holonomic and bosonized lattice formulations. The main results and their physical implications are summarized in Section 9. Finally, Appendix A gives the Fourier derivation of the holonomy effective action.

2 Hubbard–Stratonovich localization of the BFSS/BMN action

For earlier large- d treatments of bosonic matrix quantum mechanics, including the auxiliary-field reduction, the dynamically generated mass gap, and the associated thermal phase structure, see Refs. [30, 42, 43, 57] and references therein.

2.1 Action and observables

We consider the Euclidean mass-deformed bosonic BFSS $_{d+1}$ model

$$S = N \int_0^\beta dt \operatorname{Tr} \left[\frac{1}{2} (D_t X_a)^2 + \frac{m}{2} X_a^2 - \frac{\alpha}{4} [X_a, X_b]^2 \right], \quad (\alpha = 1 \text{ at the end}), \quad (2.1)$$

with partition function and free energy

$$Z(\alpha) = \int \mathcal{D}X e^{-S(\alpha)}, \quad F(\alpha) = -\frac{1}{\beta} \log Z(\alpha). \quad (2.2)$$

A primary observable is the quartic Yang–Mills term in this *original* BFSS theory,

$$\left\langle \frac{N}{4} \int_0^\beta dt \operatorname{Tr} [X_a, X_b]^2 \right\rangle = \left. \frac{\partial \ln Z(\alpha)}{\partial \alpha} \right|_{\alpha=1} = -\beta \left. \frac{\partial F(\alpha)}{\partial \alpha} \right|_{\alpha=1}. \quad (2.3)$$

A second key observable is the extent of space,

$$R^2 \equiv \frac{1}{N\beta} \left\langle \int_0^\beta dt \operatorname{Tr} X_a(t) X_a(t) \right\rangle. \quad (2.4)$$

In the large- d saddle, the mass-deformed BFSS $_{d+1}$ is approximated by a gauged matrix harmonic oscillator with a dynamically generated effective mass m_{eff} determined self-consistently by the gap equation. In this Gaussian saddle, R^2 can be obtained either from the holonomy-dependent propagator, or equivalently from the MHO free energy through

$$\frac{\partial F_{\text{MHO}}}{\partial m_{\text{eff}}} = \frac{N^2}{2} R^2, \quad (2.5)$$

(up to the same normalization conventions used in the quadratic action).

The internal energy of the system can be expressed as a linear combination of the Yang–Mills observable and the extent of space [57]. The internal energy is defined by

$$E = -\frac{\partial}{\partial \beta} \ln Z, \quad Z(\beta) = \int \mathcal{D}X e^{-S}. \quad (2.6)$$

To isolate the explicit β -dependence, rescale

$$t = \beta\tau, \quad X_a(t) = \sqrt{\beta} \tilde{X}_a(\tau), \quad \tau \in [0, 1]. \quad (2.7)$$

Then the action becomes

$$S = N \int_0^1 d\tau \operatorname{Tr} \left[\frac{1}{2} (D_\tau \tilde{X}_a)^2 + \frac{\beta^2 m}{2} \tilde{X}_a^2 - \frac{\beta^3 \alpha}{4} [\tilde{X}_a, \tilde{X}_b]^2 \right], \quad (2.8)$$

so that (up to β -independent measure factors)

$$-\frac{\partial}{\partial \beta} \ln Z = \left\langle \frac{\partial S}{\partial \beta} \right\rangle = N \int_0^1 d\tau \operatorname{Tr} \left[\beta m \tilde{X}_a^2 - \frac{3\beta^2 \alpha}{4} [\tilde{X}_a, \tilde{X}_b]^2 \right]. \quad (2.9)$$

Returning to the original variables gives the compact identity

$$E = -\frac{3\alpha}{\beta} \left\langle \frac{N}{4} \int_0^\beta dt \operatorname{Tr} [X_a, X_b]^2 \right\rangle + N^2 m R^2. \quad (2.10)$$

An alternative formula follows from the Ward identity. Indeed, on a lattice with spacing a and $\Lambda = \beta/a$ sites, the path integral is invariant under the infinitesimal rescaling $X_a \rightarrow (1 + \varepsilon)X_a$. Since the Jacobian contributes $\varepsilon d\Lambda (N^2 - 1)$, while the action varies as $\delta S = 2\varepsilon S - 2\varepsilon \alpha \hat{\mathcal{O}}_{\text{YM}}$ with

$$\hat{\mathcal{O}}_{\text{YM}} \equiv \left\langle \frac{N}{4} \int_0^\beta dt \operatorname{Tr} [X_a, X_b]^2 \right\rangle, \quad (2.11)$$

one obtains

$$2\langle S \rangle - 2\alpha \left\langle \frac{N}{4} \int_0^\beta dt \operatorname{Tr} [X_a, X_b]^2 \right\rangle = d(\Lambda(N^2 - 1)). \quad (2.12)$$

(In a continuum treatment, the right-hand side is replaced by the corresponding regulated counting of bosonic degrees of freedom.)

In the following, we will compute the free energy F and extent of space R^2 explicitly in the presence of holonomy, and then evaluate the Yang–Mills expectation value via (2.3) within the large- d saddle approximation. These observables determine the holonomy effective action and the phase structure, and quantify the regime in which BFSS $_{d+1}$ is accurately captured by the effective gauged matrix harmonic oscillator.

2.2 Action in terms of large- d effective variables

The computation proceeds, as we will discuss in detail, in several steps. First, we rewrite the action in a form suitable for the large- d analysis as follows:

- **Kinetic term:** Since we are on a Euclidean time circle of circumference β , all bosonic fields are periodic: $X_a(t+\beta) = X_a(t)$. This periodicity quantizes the temporal momentum, so the Fourier expansion is

$$X_a^{ij}(t) = \frac{1}{\sqrt{\beta}} \sum_{n \in \mathbb{Z}} X_a^{ij}(n) e^{i\omega_n t}, \quad \omega_n = \frac{2\pi n}{\beta}. \quad (2.13)$$

X_a are Hermitian matrices, $X_a^{ji}(-n) = [X_a^{ij}(n)]^*$.

The gauge field enters only through the temporal component $A_0(t)$. In one-dimensional gauge theory, A_0 has no dynamics of its own—it is an auxiliary field enforcing Gauss' law. However, the path-ordered exponential

$$U = \mathcal{P} \exp \left(-i \int_0^\beta dt A_0(t) \right) \quad (2.14)$$

around the circle (the Polyakov loop or holonomy) is gauge-invariant up to conjugation. Indeed, the holonomy matrix U transforms by conjugation under a periodic gauge transformation $g(t)$ with $g(0) = g(\beta)$:

$$U \longrightarrow g(0) U g^{-1}(0). \quad (2.15)$$

Thus the matrix U itself is not gauge-invariant, but its physical information—namely its spectrum, or equivalently the set of holonomy angles $\{\theta_i\}$ —remains unchanged.

However, by a gauge transformation that is periodic in Euclidean time, one can always bring $A_0(t)$ to a time-independent, diagonal form:

$$A_0 = -\frac{1}{\beta} \text{diag}(\theta_1, \dots, \theta_N), \quad U = \text{diag}(e^{i\theta_1}, \dots, e^{i\theta_N}), \quad \sum_{i=1}^N \theta_i = 0 \quad (\text{for } SU(N)). \quad (2.16)$$

This is the static diagonal (Polyakov) gauge. The only residual information in A_0 is the set of eigenvalues $\{\theta_i\}$, i.e. the holonomy angles; all fluctuations of $A_0(t)$ away from this constant diagonal form can be gauged away.

With this gauge choice, the covariant derivative $D_t X_a = \partial_t X_a - i[A_0, X_a]$ acting on Fourier modes shifts the Matsubara frequencies by color-dependent phases:

$$(D_t X_a)^{ij}(n) = \frac{i}{\sqrt{\beta}} \sum_{n, ij} \left(\omega_n + \frac{\theta_i - \theta_j}{\beta} \right) X_a^{ij}(n) e^{i\omega_n t}. \quad (2.17)$$

Thus, in the kinetic term one always encounters the combination $\omega_n + (\theta_i - \theta_j)/\beta$. The kinetic term then reads

$$\frac{N}{2} \int_0^\beta dt \text{Tr}(D_t X_a)^2 = \frac{N}{2} \sum_{a=1}^d \sum_{n \in \mathbb{Z}} \sum_{i,j=1}^N \left(\frac{2\pi n + \theta_i - \theta_j}{\beta} \right)^2 X_a^{ij}(n) X_a^{ji}(-n). \quad (2.18)$$

Each adjoint mode (i, j) thus experiences the shifted Matsubara frequency $\omega_n \mapsto \omega_n + (\theta_i - \theta_j)/\beta$. In particular, the trace component ($i = j$) corresponds to the overall $U(1)$ mode and can be removed by inserting the adjoint projector

$$P_{ij,kl} = \delta_{il} \delta_{jk} - \frac{1}{N} \delta_{ij} \delta_{kl}, \quad (2.19)$$

which projects onto traceless $SU(N)$ matrices. The projected kinetic term reads

$$S_{\text{kin}} = \frac{N}{2} \sum_{a=1}^d \sum_{n \in \mathbb{Z}} \sum_{i,j=1}^N X_a^{ij}(-n) \left(\left(\omega_n + \frac{\theta_i - \theta_j}{\beta} \right)^2 P_{ij,kl} \right) X_a^{kl}(n). \quad (2.20)$$

- **Mass term:** We expand the matrix coordinates as

$$X_a = \sum_{\mu=1}^{N^2-1} x_a^\mu \Lambda_\mu, \quad [\Lambda_\mu, \Lambda_\nu] = 2i f_{\mu\nu\lambda} \Lambda_\lambda, \quad \text{Tr}(\Lambda_\mu \Lambda_\nu) = 2 \delta_{\mu\nu}. \quad (2.21)$$

The quadratic term then takes the form

$$\begin{aligned} \frac{Nm}{2} \int_0^\beta dt \text{Tr} X_a^2 &= Nm \int_0^\beta dt (x_a^\mu x_a^\mu), \quad x_a^\mu = \frac{1}{2} \text{Tr}(X_a \Lambda_\mu) \\ &= \frac{Nm}{2} \int_0^\beta dt X_a^{ij} X_a^{kl} P_{ij,kl}, \quad P_{ij,kl} = \frac{1}{2} \sum_\mu \Lambda_\mu^{ji} \Lambda_\mu^{lk} = \delta_{il} \delta_{jk} - \frac{1}{N} \delta_{ij} \delta_{kl} \\ &= \frac{Nm}{2} \sum_{a=1}^d \sum_{n \in \mathbb{Z}} \sum_{i,j,k,l=1}^N X_a^{ij}(-n) P_{ij,kl} X_a^{kl}(n). \end{aligned} \quad (2.22)$$

Again $P_{ij,kl}$ is the adjoint projector, which removes the trace ($i = j$) and enforces the $SU(N)$ sector.

- **Quartic coupling:** Similarly, the quartic action takes the form

$$-\frac{N\alpha}{4} \int_0^\beta dt \text{Tr}[X_a, X_b]^2 = -\frac{N\alpha}{4} \lambda_{\mu\rho\nu\sigma} \int_0^\beta dt (x_a^\mu x_a^\rho)(x_b^\nu x_b^\sigma), \quad (2.23)$$

with the coefficient

$$\lambda_{\mu\rho\nu\sigma} = \frac{1}{2} \text{Tr} \left([\Lambda_\mu, \Lambda_\nu][\Lambda_\rho, \Lambda_\sigma] + [\Lambda_\mu, \Lambda_\sigma][\Lambda_\rho, \Lambda_\nu] \right). \quad (2.24)$$

Let $\mu_{\mu\rho\nu\sigma}$ denote the inverse of $\lambda_{\mu\rho\nu\sigma}$. Then we must have the pairwise identities

$$\mu_{\mu\rho\nu\sigma} \lambda_{\nu\sigma\alpha\beta} = \delta_{\mu\alpha} \delta_{\rho\beta}, \quad \lambda_{\mu\rho\nu\sigma} \mu_{\nu\sigma\alpha\beta} = \delta_{\mu\alpha} \delta_{\rho\beta}. \quad (2.25)$$

For $N = 2$ (so $f_{\mu\nu\kappa} = \varepsilon_{\mu\nu\kappa}$), we compute

$$\begin{aligned} \lambda_{\mu\rho\nu\sigma} &= -4(f_{\mu\nu\kappa} f_{\rho\sigma\kappa} + f_{\mu\sigma\kappa} f_{\rho\nu\kappa}) \\ &= -4(2\delta_{\mu\rho} \delta_{\nu\sigma} - \delta_{\mu\sigma} \delta_{\nu\rho} - \delta_{\mu\nu} \delta_{\rho\sigma}), \quad (SU(2)). \end{aligned} \quad (2.26)$$

An ansatz for the inverse kernel μ is given by

$$\mu_{\mu\rho\nu\sigma} = a \delta_{\mu\rho} \delta_{\nu\sigma} + b \delta_{\mu\sigma} \delta_{\nu\rho} + c \delta_{\mu\nu} \delta_{\rho\sigma}. \quad (2.27)$$

Solving (2.25) gives $a = -b = -c = -1/16$, so that

$$\mu_{\mu\rho\nu\sigma} = \frac{1}{16} \left(-\delta_{\mu\rho}\delta_{\nu\sigma} + \delta_{\mu\sigma}\delta_{\nu\rho} + \delta_{\mu\nu}\delta_{\rho\sigma} \right). \quad (SU(2)). \quad (2.28)$$

Moreover, we can check the two contractions

$$\lambda_{\mu\rho\nu\sigma} \delta_{\nu\sigma} = -16 \delta_{\mu\rho}, \quad \mu_{\mu\rho\nu\sigma} \delta_{\nu\sigma} = -\frac{1}{16} \delta_{\mu\rho}. \quad (2.29)$$

For generic N , we will have

$$\lambda_{\mu\rho\nu\sigma} \delta_{\nu\sigma} = -8N \delta_{\mu\rho}, \quad \mu_{\mu\rho\nu\sigma} \delta_{\nu\sigma} = -\frac{1}{8N} \delta_{\mu\rho}. \quad (2.30)$$

• **Total action:** The total action can then be written as

$$\begin{aligned} S[X] &= \frac{N}{2} \sum_{a=1}^d \sum_{n \in \mathbb{Z}} \sum_{i,j=1}^N X_a^{ij}(-n) \left(\left(\omega_n + \frac{\theta_i - \theta_j}{\beta} \right)^2 P_{ij,kl} + m P_{ij,kl} \right) X_a^{kl}(n) \\ &\quad - \frac{N\alpha}{4} \lambda_{\mu\rho\nu\sigma} \int_0^\beta dt Y_{\mu\rho} Y_{\nu\sigma}. \end{aligned} \quad (2.31)$$

Here we have introduced the bilinear

$$Y_{\mu\rho}(t) = \sum_{a=1}^d x_a^\mu(t) x_a^\rho(t). \quad (2.32)$$

2.3 Localization: Hubbard–Stratonovich decoupling

Next, we introduce an auxiliary symmetric kernel $k_{\mu\rho} = k_{\rho\mu}$ and consider the following Gaussian identity (a completion of the square)

$$\begin{aligned} 0 &= \frac{N}{4\alpha} \int_0^\beta dt \mu_{\mu\rho\nu\sigma} \left(k_{\mu\rho} + \alpha \lambda_{\mu\rho\alpha\beta} Y_{\alpha\beta} \right) \left(k_{\nu\sigma} + \alpha \lambda_{\nu\sigma\gamma\delta} Y_{\gamma\delta} \right) \\ &\quad - \frac{N\alpha}{4} \int_0^\beta dt \lambda_{\mu\rho\nu\sigma} Y_{\mu\rho} Y_{\nu\sigma} - \frac{N}{2} \int_0^\beta dt k_{\mu\rho} Y_{\mu\rho} - \frac{N}{4\alpha} \int_0^\beta dt \mu_{\mu\rho\nu\sigma} k_{\mu\rho} k_{\nu\sigma}. \end{aligned} \quad (2.33)$$

In other words, the path integral is extended to include the auxiliary field $k_{\mu\rho}$:

$$Z = \int \mathcal{D}X e^{-S[X]} = \int \mathcal{D}X \mathcal{D}k e^{-S_{\text{loc}}[X,k]}, \quad (2.34)$$

so that we are now integrating over both the original matrices X_a and the new auxiliary variable $k_{\mu\rho}$. The localization action is given explicitly by

$$S_{\text{loc}}[X, k] = S[X] + \frac{N}{4\alpha} \int_0^\beta dt \mu_{\mu\rho\nu\sigma} \left(k_{\mu\rho} + \alpha \lambda_{\mu\rho\alpha\beta} Y_{\alpha\beta} \right) \left(k_{\nu\sigma} + \alpha \lambda_{\nu\sigma\gamma\delta} Y_{\gamma\delta} \right). \quad (2.35)$$

Explicitly, we have

$$\begin{aligned} S_{\text{loc}} = & \frac{N}{2} \sum_{a=1}^d \sum_{n \in \mathbb{Z}} \sum_{i,j=1}^N X_a^{ij}(-n) \left[\left(\omega_n + \frac{\theta_i - \theta_j}{\beta} \right)^2 P_{ij,kl} + m P_{ij,kl} \right] X_a^{kl}(n) \\ & + \frac{N}{2} \int_0^\beta dt k_{\mu\rho} Y_{\mu\rho} + \frac{N}{4\alpha} \mu_{\mu\rho\nu\sigma} \int_0^\beta dt k_{\mu\rho} k_{\nu\sigma}. \end{aligned} \quad (2.36)$$

Strictly speaking, the auxiliary kernel introduced in the Hubbard–Stratonovich step is a function of Euclidean time, since it couples directly to the time-dependent bilinear $Y_{\mu\rho}(t)$. Thus the auxiliary field is not *a priori* constant in t . However, in the large- d limit (or at the saddle point of the effective action after integrating out X_a) the dominant configurations of $k_{\mu\rho}(t)$ are time-independent. This justifies the use of a constant $k_{\mu\rho}(t)$ in the action, which then reads

$$\begin{aligned} S_{\text{loc}} = & \frac{N}{2} \sum_{a=1}^d \sum_{n \in \mathbb{Z}} \sum_{i,j=1}^N X_a^{ij}(-n) \left[\left(\omega_n + \frac{\theta_i - \theta_j}{\beta} \right)^2 P_{ij,kl} + m P_{ij,kl} + k_{ij,kl} \right] X_a^{kl}(n) \\ & + \frac{N}{4\alpha} \beta \mu_{\mu\rho\nu\sigma} k_{\mu\rho} k_{\nu\sigma}, \end{aligned} \quad (2.37)$$

with the adjoint-to-matrix map

$$k_{ij,kl} = \frac{1}{4} k_{\mu\rho} \Lambda_\mu^{ji} \Lambda_\rho^{lk}. \quad (2.38)$$

Thus the interaction has been *localized* into a linear coupling of k to the bilinear $Y_{\mu\rho}$ plus a Gaussian weight for k . Integrating out X_a is now Gaussian and yields an effective action for k ; the saddle point of that effective action is the real starting point of the large- d analysis.

Moreover, note that $k_{ij,kl}$ is traceless in each index pair:

$$\begin{aligned} \delta_{ij} k_{ij,kl} &= \frac{1}{4} k_{\mu\rho} (\delta_{ij} \Lambda_\mu^{ji}) \Lambda_\rho^{lk} = \frac{1}{4} k_{\mu\rho} \text{Tr}(\Lambda_\mu) \Lambda_\rho^{lk} = 0, \\ \delta_{kl} k_{ij,kl} &= \frac{1}{4} k_{\mu\rho} \Lambda_\mu^{ji} (\delta_{kl} \Lambda_\rho^{lk}) = \frac{1}{4} k_{\mu\rho} \Lambda_\mu^{ji} \text{Tr}(\Lambda_\rho) = 0. \end{aligned} \quad (2.39)$$

Now apply the projector from the right:

$$\begin{aligned} (kP)_{ij,kl} &= k_{ij,rs} P_{rs,kl} \\ &= k_{ij,rs} \left(\delta_{rk} \delta_{sl} - \frac{1}{N} \delta_{rs} \delta_{kl} \right) = k_{ij,kl} - \frac{1}{N} (\delta_{rs} k_{ij,rs}) \delta_{kl} = k_{ij,kl}. \end{aligned} \quad (2.40)$$

Similarly, applying the projector from the left gives

$$(Pk)_{ij,kl} = P_{ij,rs} k_{rs,kl} = \left(\delta_{ir} \delta_{js} - \frac{1}{N} \delta_{ij} \delta_{rs} \right) k_{rs,kl} = k_{ij,kl} - \frac{1}{N} \delta_{ij} (\delta_{rs} k_{rs,kl}) = k_{ij,kl}. \quad (2.41)$$

Combining both results, we obtain

$$(PkP)_{ij,kl} = k_{ij,kl}. \quad (2.42)$$

2.4 Effective action

Integrating out the Gaussian matrices X_a in (2.37) yields the effective action for the auxiliary kernel k :

$$S_{\text{eff}}[k] = \frac{d}{2} \sum_n \text{Tr} \log(P \mathcal{W}(n) P) + \frac{N\beta}{4\alpha} \mu_{\mu\rho\nu\sigma} k_{\mu\rho} k_{\nu\sigma}, \quad (2.43)$$

with

$$\mathcal{W}_{ij,kl}(n) = \left(\omega_n + \frac{\theta_i - \theta_j}{\beta} \right)^2 P_{ij,kl} + m P_{ij,kl} + k_{ij,kl}. \quad (2.44)$$

In the free part, we have also used the fact that P commutes with the quadratic kernel $\mathcal{W}_{ij}^{(0)}(n)$ defined by

$$\mathcal{W}_{ij,kl}^{(0)}(n) = A_{ij} P_{ij,kl}, \quad A_{ij} \equiv \left(\omega_n + \frac{\theta_i - \theta_j}{\beta} \right)^2 + m, \quad P_{ij,kl} = \delta_{il} \delta_{jk} - \frac{1}{N} \delta_{ij} \delta_{kl}. \quad (2.45)$$

Indeed, we have

$$\mathcal{W}^{(0)}(n) P = P \mathcal{W}^{(0)}(n). \quad (2.46)$$

The proof goes as follows. First, we compute

$$\begin{aligned} (P \mathcal{W}^{(0)})_{ij,rs} &= P_{ij,kl} \mathcal{W}_{kl,rs}^{(0)} = P_{ij,kl} A_{kl} P_{kl,rs} = \left(\delta_{ik} \delta_{jl} - \frac{1}{N} \delta_{ij} \delta_{kl} \right) A_{kl} \left(\delta_{kr} \delta_{ls} - \frac{1}{N} \delta_{kl} \delta_{rs} \right) \\ &= T_1 + T_2 + T_3 + T_4, \end{aligned} \quad (2.47)$$

with

$$T_1 = \delta_{ik} \delta_{jl} A_{kl} \delta_{kr} \delta_{ls} = A_{ij} \delta_{ir} \delta_{js}, \quad (2.48)$$

$$T_2 = -\delta_{ik} \delta_{jl} A_{kl} \frac{1}{N} \delta_{kl} \delta_{rs} = -\frac{1}{N} \delta_{ij} \delta_{rs} A_{ii} = -\frac{1}{N} (\omega_n^2 + m) \delta_{ij} \delta_{rs}, \quad (2.49)$$

$$T_3 = -\frac{1}{N} \delta_{ij} \delta_{kl} A_{kl} \delta_{kr} \delta_{ls} = -\frac{1}{N} \delta_{ij} \delta_{rs} A_{rs}, \quad (2.50)$$

$$T_4 = +\frac{1}{N^2} \delta_{ij} \delta_{kl} A_{kl} \delta_{kl} \delta_{rs} = \frac{1}{N} \delta_{ij} \delta_{rs} (\omega_n^2 + m), \quad (2.51)$$

where we used $A_{ii} = \omega_n^2 + m$ and $\sum_k A_{kk} = N(\omega_n^2 + m)$. Hence $T_2 + T_4 = 0$ and

$$(P\mathcal{W}^{(0)})_{ij,rs} = A_{ij} \delta_{ir} \delta_{js} - \frac{1}{N} \delta_{ij} \delta_{rs} A_{rs}. \quad (2.52)$$

On the other hand, we compute

$$(\mathcal{W}^{(0)}P)_{ij,rs} = \mathcal{W}_{ij,kl}^{(0)} P_{kl,rs} = A_{ij} P_{ij,rs} = A_{ij} \left(\delta_{ir} \delta_{js} - \frac{1}{N} \delta_{ij} \delta_{rs} \right). \quad (2.53)$$

Therefore $(P\mathcal{W}^{(0)})_{ij,rs} = (\mathcal{W}^{(0)}P)_{ij,rs}$.

3 Saddle-point equation and the maximally symmetric ansatz

The saddle-point and self-consistent Gaussian methods used in this section are closely related to the large- d analyses of Refs. [30, 42, 43, 57] and references therein.

3.1 Saddle point equation

Motivated by the fact that the free quadratic kernel is compatible with the adjoint projector,

$$P\Delta^{(0)} = \Delta^{(0)}P, \quad (3.1)$$

we impose the analogous requirement on the Hubbard–Stratonovich interaction kernel. We therefore adopt the factorized ansatz

$$k_{ij,kl} \equiv k_{ij} P_{ij,kl} = P_{ij,kl} k_{kl}. \quad (3.2)$$

This choice makes the interaction kernel act as an effective mass term in the adjoint sector, rather than as a new kinetic operator. For consistency of the two factorizations in (3.2), one must have

$$k_{ij} = k_{ji}, \quad k_{ii} = k_{jj} \quad \forall i, j. \quad (3.3)$$

Indeed,

$$(kP)_{ij,rs} = k_{ij} P_{ij,mn} P_{mn,rs} = k_{ij} \left(\delta_{ir} \delta_{js} - \frac{1}{N} \delta_{ij} \delta_{rs} \right), \quad (3.4)$$

while

$$(Pk)_{ij,rs} = P_{ij,mn} k_{mn} P_{mn,rs} = k_{ji} \delta_{ir} \delta_{js} - \frac{1}{N} \delta_{ij} \delta_{rs} k_{ii} - \frac{1}{N} \delta_{ij} \delta_{rs} k_{rr} + \frac{1}{N^2} \delta_{ij} \delta_{rs} \sum_m k_{mm}. \quad (3.5)$$

Using $k_{ij} = k_{ji}$ and the equality of all diagonal entries, this becomes

$$(Pk)_{ij,rs} = k_{ij} \left(\delta_{ir} \delta_{js} - \frac{1}{N} \delta_{ij} \delta_{rs} \right) = (kP)_{ij,rs}. \quad (3.6)$$

Thus

$$Pk = kP. \quad (3.7)$$

Since the free quadratic kernel also commutes with P , the full quadratic operator satisfies

$$P\mathcal{W}(n) = \mathcal{W}(n)P. \quad (3.8)$$

Therefore, on the projected adjoint subspace, P acts as the identity and

$$P\mathcal{W}(n)P = P\mathcal{W}(n) = \mathcal{W}(n)P. \quad (3.9)$$

Hence the trace-log in (2.43) is understood as a trace over the projected $SU(N)$ sector:

$$\text{Tr} \log(P\mathcal{W}(n)P) = \text{Tr}_{\text{adj}} \log \mathcal{W}(n). \quad (3.10)$$

The ansatz (3.2) also gives the factorized form

$$\mathcal{W}_{ij,kl}(n) = \bar{A}_{ij}(n) P_{ij,kl} = P_{ij,kl} \bar{A}_{lk}(n), \quad \bar{A}_{ij}(n) \equiv A_{ij}(n) + k_{ij}. \quad (3.11)$$

The second equality contains $\bar{A}_{lk}$, not $\bar{A}_{kl}$, because the projector $P_{ij,kl}$ reverses the bi-index pair. The projected trace therefore reduces to the sum over adjoint bi-index modes,

$$\text{Tr}_{\text{adj}} \log \mathcal{W}(n) = \sum_{i,j=1}^N P_{ij,ji} \log \bar{A}_{ij}(n). \quad (3.12)$$

Thus the effective action becomes

$$S_{\text{eff}}[k] = \frac{d}{2} \sum_{n \in \mathbb{Z}} \sum_{i,j=1}^N P_{ij,ji} \log \bar{A}_{ij}(n) + \frac{N\beta}{4\alpha} \mu_{\mu\rho\nu\sigma} k_{\mu\rho} k_{\nu\sigma}. \quad (3.13)$$

Recall now the adjoint-to-matrix map

$$k_{ij,kl} = \frac{1}{4} k_{\mu\rho} \Lambda_{\mu}^{ji} \Lambda_{\rho}^{lk}, \quad \text{Tr}(\Lambda_{\mu} \Lambda_{\nu}) = 2 \delta_{\mu\nu}. \quad (3.14)$$

Varying S_{eff} with respect to the *matrix* variables k_{ij} gives

$$0 = \frac{\partial S_{\text{eff}}}{\partial k_{ij}} = \frac{d}{2} \sum_{n \in \mathbb{Z}} P_{ij,ji} \frac{1}{\bar{A}_{ij}(n)} + \frac{N\beta}{2\alpha} \mu_{\mu\rho\nu\sigma} k_{\nu\sigma} \frac{\partial k_{\mu\rho}}{\partial k_{ij}}. \quad (3.15)$$

We use the inverse of the above map (obtained by contracting with the generators and using the commuting-symmetric ansatz $k_{ij,kl} = P_{ij,kl} k_{ij}$),

$$k_{\mu\rho} = \Lambda_\mu^{ij} k_{ij,kl} \Lambda_\rho^{kl} = \Lambda_\mu^{ij} P_{ij,kl} k_{ij} \Lambda_\rho^{kl}, \quad (3.16)$$

to obtain the variation

$$\frac{\partial k_{\mu\rho}}{\partial k_{i_1 j_1}} = \Lambda_\mu^{ij} P_{ij,kl} \Lambda_\rho^{kl} \delta_{ii_1} \delta_{jj_1} = \Lambda_\mu^{i_1 j_1} P_{i_1 j_1,kl} \Lambda_\rho^{kl}. \quad (3.17)$$

Hence (3.15) becomes

$$0 = \frac{d}{2} \sum_{n \in \mathbb{Z}} P_{ij,ji} \frac{1}{\bar{A}_{ij}(n)} + \frac{N\beta}{2\alpha} \mu_{\mu\rho\nu\sigma} k_{\nu\sigma} \Lambda_\mu^{ij} P_{ij,kl} \Lambda_\rho^{kl}. \quad (3.18)$$

Since P projects onto traceless matrices and each Λ_ρ is traceless, we have $P_{ij,kl} \Lambda_\rho^{kl} = \Lambda_\rho^{ji}$, and therefore the saddle equation simplifies to

$$0 = \frac{d}{2} \sum_{n \in \mathbb{Z}} P_{ij,ji} \frac{1}{\bar{A}_{ij}(n)} + \frac{N\beta}{2\alpha} \mu_{\mu\rho\nu\sigma} k_{\nu\sigma} \Lambda_\mu^{ij} \Lambda_\rho^{ji}. \quad (3.19)$$

The Matsubara sum is standard [58, 59]:

$$\sum_{n \in \mathbb{Z}} \frac{1}{\left(\omega_n + \frac{\theta_i - \theta_j}{\beta}\right)^2 + \tilde{k}_{ij}} = \beta \frac{1}{2\sqrt{\tilde{k}_{ij}}} \frac{\sinh(\beta\sqrt{\tilde{k}_{ij}})}{\cosh(\beta\sqrt{\tilde{k}_{ij}}) - \cos(\theta_i - \theta_j)}, \quad \tilde{k}_{ij} = k_{ij} + m. \quad (3.20)$$

Substituting this back we obtain the saddle-point (gap) equation in the commuting-symmetric ansatz:

$$0 = \frac{d\beta}{4} P_{ij,ji} \frac{1}{\sqrt{\tilde{k}_{ij}}} \frac{\sinh(\beta\sqrt{\tilde{k}_{ij}})}{\cosh(\beta\sqrt{\tilde{k}_{ij}}) - \cos(\theta_i - \theta_j)} + \frac{N\beta}{2\alpha} \mu_{\mu\rho\nu\sigma} k_{\nu\sigma} \Lambda_\mu^{ij} \Lambda_\rho^{ji}. \quad (3.21)$$

Equations (3.2), (3.13), and (3.21) encapsulate the commuting-symmetric ansatz, the corresponding simplification of the effective action in the constant commuting sector, and the resulting saddle-point (gap) condition obtained after performing the Matsubara summation.

3.2 Maximally symmetric ansatz

We now pass from the commuting-symmetric saddle used above to the maximally symmetric ansatz. The constancy of $k_{\mu\rho}(t)$ in Euclidean time and the factorization of the interaction kernel through the adjoint projector have already been used in deriving (3.13) and (3.21). The new ingredient in the present subsection is the final strengthening of this ansatz: uniformity in color space, $k_{ij} = k_0$.

Therefore, we package together several consistency requirements:

(i) *Constancy in Euclidean time (commutativity).*

$$k_{\mu\rho}(t) = k_{\mu\rho}, \quad k_{\mu\rho} = \Lambda_\mu^{ij} k_{ij,kl} \Lambda_\rho^{kl} \quad (\text{time-independent}). \quad (3.22)$$

This condition has already been assumed in passing to the constant large- d saddle and in deriving the effective action. Also, since $k_{\mu\rho}$ is an adjoint kernel, its matrix representative lies in the $SU(N)$ sector; thus $PkP = k$ follows from adjoint support and is not an additional assumption.

(ii) *Factorization with the adjoint projector (symmetric properties).*

$$k_{ij,kl} = k_{ij} P_{ij,kl} = P_{ij,kl} k_{kl}, \quad k_{ij} = k_{ji}, \quad (3.23)$$

From this one obtains

$$k_{ij} = k_{ji} \quad (\text{symmetry}), \quad k_{ii} = k_{jj} \quad \forall i, j \quad (\text{all diagonals equal}). \quad (3.24)$$

This is precisely the symmetric ansatz already used above: it makes the interaction kernel compatible with the same adjoint projector structure as the free quadratic kernel. In turn, the four-index structure collapses to a two-index one, and k_{ij} behaves as a holonomy-dependent effective mass term.

(iii) *Uniformity in color space.*

$$k_{ij} = k_0 \quad \forall i, j. \quad (3.25)$$

This is the genuinely new step of the maximally symmetric ansatz. Consequently the remaining two-index structure collapses to a single scalar mass parameter,

$$k_{ij,kl} = k_0 P_{ij,kl} \quad \text{with } k_0 \in \mathbb{R}. \quad (3.26)$$

Thus, in the terminology used here, the commuting-symmetric saddle consists of (i) and (ii), while the maximally symmetric ansatz is obtained by adding the uniformity condition (iii).

In the literature this is often termed simply the “symmetric ansatz”: symmetric in Euclidean time (constant), symmetric in indices ($k_{ij} = k_{ji}$), and effectively uniform in matrix (color) space.

Starting now from

$$0 = \frac{d\beta}{4} P_{ij,ji} \frac{1}{\sqrt{\tilde{k}_{ij}}} \frac{\sinh(\beta\sqrt{\tilde{k}_{ij}})}{\cosh(\beta\sqrt{\tilde{k}_{ij}}) - \cos(\theta_i - \theta_j)} + \frac{N\beta}{2\alpha} \mu_{\mu\rho\nu\sigma} k_{\nu\sigma} \Lambda_\mu^{ij} \Lambda_\rho^{ji}, \quad (3.27)$$

we now impose the maximally symmetric ansatz $k_{ij} = k_0$. At this stage we keep the (i, j) dependence explicit; the final scalar gap equation is obtained only after projecting or summing over the color indices, or after specifying the holonomy saddle. Using

$$\text{Tr} \Lambda_\mu \Lambda_\rho = 2 \delta_{\mu\rho}, \quad k_{ij,kl} = k_0 P_{ij,kl} \iff k_{\nu\sigma} = 2k_0 \delta_{\nu\sigma}, \quad (3.28)$$

and the identity

$$\mu_{\mu\rho\nu\sigma} \delta_{\nu\sigma} = -\frac{1}{8N} \delta_{\mu\rho}, \quad (3.29)$$

the second term in (3.27) gives

$$\begin{aligned} \frac{N\beta}{2\alpha} \mu_{\mu\rho\nu\sigma} k_{\nu\sigma} \Lambda_{\mu}^{ij} \Lambda_{\rho}^{ji} &= \frac{N\beta}{2\alpha} \mu_{\mu\rho\nu\sigma} (2k_0 \delta_{\nu\sigma}) \Lambda_{\mu}^{ij} \Lambda_{\rho}^{ji} \\ &= \frac{N\beta}{2\alpha} \left(-\frac{1}{8N} \delta_{\mu\rho} \right) (2k_0) \Lambda_{\mu}^{ij} \Lambda_{\rho}^{ji} \\ &= \frac{N\beta}{2\alpha} \left(-\frac{1}{8N} \right) (2k_0) (2P_{ij,ji}) \\ &= -\frac{\beta k_0}{4\alpha} P_{ij,ji}. \end{aligned} \quad (3.30)$$

In the third line, we used

$$\sum_{\mu} \Lambda_{\mu}^{ij} \Lambda_{\mu}^{ji} = 2 P_{ij,ji}. \quad (3.31)$$

Hence the maximally symmetric ansatz gives the pairwise saddle condition

$$0 = \frac{\beta P_{ij,ji}}{4} \left[\frac{d}{\sqrt{\tilde{k}_0}} \frac{\sinh(\beta \sqrt{\tilde{k}_0})}{\cosh(\beta \sqrt{\tilde{k}_0}) - \cos(\theta_i - \theta_j)} - \frac{k_0}{\alpha} \right]. \quad (3.32)$$

For the components surviving the adjoint projection, this is equivalently written as

$$0 = \frac{d}{\sqrt{\tilde{k}_0}} \frac{\sinh(\beta \sqrt{\tilde{k}_0})}{\cosh(\beta \sqrt{\tilde{k}_0}) - \cos(\theta_i - \theta_j)} - \frac{k_0}{\alpha}. \quad (3.33)$$

The uniform ansatz removes thus the color dependence of the auxiliary mass k_{ij} , but it does not remove the holonomy dependence carried by $\theta_i - \theta_j$. Thus the final scalar gap equation for k_0 is obtained only after a further treatment of the holonomy sector, for example by summing over i, j , averaging over the holonomy distribution, or evaluating at a chosen holonomy saddle.

For example, in the center-symmetric, or confining, holonomy saddle the eigenvalues are uniformly distributed on the unit circle,

$$\theta_i = \frac{2\pi}{N} \left(i - \frac{N+1}{2} \right), \quad i = 1, \dots, N, \quad (3.34)$$

and therefore

$$\theta_i - \theta_j = \frac{2\pi}{N} (i - j). \quad (3.35)$$

In this case the residual holonomy dependence is converted into a discrete color average, giving

$$\frac{k_0}{\alpha} = \frac{d}{N^2 - 1} \sum_{i,j=1}^N P_{ij,ji} \frac{1}{\sqrt{\tilde{k}_0}} \frac{\sinh(\beta \sqrt{\tilde{k}_0})}{\cosh(\beta \sqrt{\tilde{k}_0}) - \cos\left(\frac{2\pi(i-j)}{N}\right)}. \quad (3.36)$$

By contrast, setting all $\theta_i = 0$ corresponds to the collapsed, or deconfined, holonomy saddle.

3.3 Scope of the maximally symmetric ansatz

It is useful to derive the scalar gap equation once more by imposing the maximally symmetric ansatz directly in the effective action. This provides a complementary derivation of the result above and clarifies the precise scope of the ansatz.

Substituting

$$k_{ij,kl} = k_0 P_{ij,kl}, \quad k_{\mu\rho} = 2k_0 \delta_{\mu\rho}, \quad s^2 = m + k_0, \quad (3.37)$$

into (3.13), and using again $\mu_{\mu\rho\nu\sigma}\delta_{\nu\sigma} = -\delta_{\mu\rho}/8N$, one obtains, up to terms independent of k_0 ,

$$S_{\text{eff}}^{\text{max}}(k_0; U) = \frac{d}{2} \sum_{n \in \mathbb{Z}} \sum_{i,j=1}^N P_{ij,ji} \log \left[\left(\omega_n + \frac{\theta_i - \theta_j}{\beta} \right)^2 + s^2 \right] - \frac{\beta(N^2 - 1)}{8\alpha} k_0^2. \quad (3.38)$$

Varying with respect to k_0 gives

$$0 = \frac{d}{2} \sum_{n \in \mathbb{Z}} \sum_{i,j=1}^N \frac{P_{ij,ji}}{\left(\omega_n + \frac{\theta_i - \theta_j}{\beta} \right)^2 + s^2} - \frac{\beta(N^2 - 1)}{4\alpha} k_0. \quad (3.39)$$

After performing the Matsubara sum, this becomes

$$\frac{d}{s} \mathcal{R}(s; U) = \frac{k_0}{\alpha} = \frac{s^2 - m}{\alpha}, \quad (3.40)$$

where

$$\mathcal{R}(s; U) = \frac{1}{N^2 - 1} \sum_{i,j=1}^N P_{ij,ji} \frac{\sinh(\beta s)}{\cosh(\beta s) - \cos(\theta_i - \theta_j)}. \quad (3.41)$$

This is precisely the projector-weighted projection of the pairwise saddle equation (3.33) onto the maximally symmetric variation $\delta k_{ij} = \delta k_0$, which sums over the adjoint color modes with weight $P_{ij,ji}$. The pairwise derivation and the direct scalar variation are therefore complementary: the former retains the residual (i, j) -dependent holonomy structure, whereas the latter yields the single equation obeyed within the restricted k_0 sector.

The important point is that the maximally symmetric ansatz concerns the Hubbard–Stratonovich kernel, not the Polyakov-loop holonomy. It sets the dynamically generated mass shift equal in all adjoint color directions, but the eigenvalues θ_i remain dynamical and enter the gap equation through $\mathcal{R}(s; U)$. The ansatz therefore does not impose a uniform holonomy distribution.

Indeed, within the same maximally symmetric auxiliary sector, the leading holonomy action derived below is

$$\frac{S_{\text{hol}}}{N^2} = \sum_{n \geq 1} \frac{1 - de^{-n\beta s}}{n} |u_n|^2 + \dots, \quad u_n = \frac{1}{N} \text{Tr } U^n. \quad (3.42)$$

The confined saddle is characterized by $u_n = 0$, whereas nonuniform and gapped deconfined saddles have nonzero Polyakov moments. The uniform distribution loses stability when

$$1 - de^{-\beta s} = 0. \quad (3.43)$$

Consequently, condition (iii) does not exclude deconfinement: the confined, nonuniform, and gapped holonomy phases are all compatible with the same scalar auxiliary saddle.

The motivation for the ansatz is particularly clear at low temperature. Using

$$\frac{\sinh(\beta s)}{\cosh(\beta s) - \cos \phi} = 1 + 2 \sum_{n \geq 1} e^{-n\beta s} \cos(n\phi), \quad (3.44)$$

one finds

$$\mathcal{R}(s; U) = 1 + 2 \sum_{n \geq 1} C_n(U) e^{-n\beta s}, \quad C_n(U) = \frac{|\text{Tr } U^n|^2 - 1}{N^2 - 1}. \quad (3.45)$$

The leading zero-temperature saddle is therefore color independent, while its holonomy dependence is exponentially suppressed. This supports the use of the symmetry-preserving scalar kernel as the leading large- d , low-temperature saddle, as in the large- d analysis of Ref. [30].

The ansatz is nevertheless a restriction of the full auxiliary-field saddle space. More general color-dependent solutions, or nonsinglet fluctuations of $k_{\mu\rho}$, may affect subleading corrections and the detailed nonlinear structure of the deconfined branch. The present work does not claim to classify such saddles; it shows instead that the maximally symmetric auxiliary saddle itself is fully compatible with both confined and deconfined holonomy configurations.

4 Low-temperature X -space physics of the large- d saddle

4.1 Gap equation at low temperature

Starting from the exact gap equation

$$\frac{d}{\sqrt{\tilde{k}_0}} \frac{\sinh(\beta \sqrt{\tilde{k}_0})}{\cosh(\beta \sqrt{\tilde{k}_0}) - \cos \phi} = \frac{k_0}{\alpha}, \quad \phi \equiv \theta_i - \theta_j, \quad s \equiv \sqrt{\tilde{k}_0} > 0, \quad \tilde{k}_0 = k_0 + m. \quad (4.1)$$

we expand the thermal factor for $\beta \rightarrow \infty$:

$$\frac{\sinh(\beta s)}{\cosh(\beta s) - \cos \phi} = \frac{\frac{1}{2}e^{\beta s}(1 - e^{-2\beta s})}{\frac{1}{2}e^{\beta s}(1 - 2\cos \phi e^{-\beta s} + e^{-2\beta s})} = 1 + 2\cos \phi e^{-\beta s} + \mathcal{O}(e^{-2\beta s}). \quad (4.2)$$

Inserting (4.2) into (4.1) gives

$$\frac{d}{s} \left[1 + 2\cos \phi e^{-\beta s} + \mathcal{O}(e^{-2\beta s}) \right] = \frac{s^2 - m}{\alpha}. \quad (4.3)$$

Let $s = s_0 + \delta s$, with s_0 the $T = 0$ solution and $\delta s = \mathcal{O}(e^{-\beta s_0})$ the first thermal correction. At $T = 0$ we have $d/s_0 = (s_0^2 - m)/\alpha$, i.e.

$$s_0^3 - m s_0 = \alpha d. \quad (4.4)$$

Keeping only terms linear in δs and $e^{-\beta s_0}$, and neglecting $\delta s e^{-\beta s_0} = \mathcal{O}(e^{-2\beta s_0})$, Eq. (4.3) gives

$$\underbrace{\left(\frac{d}{s_0} - \frac{s_0^2 - m}{\alpha}\right)}_{=0} + \left(-\frac{d}{s_0^2} - \frac{2s_0}{\alpha}\right) \delta s + \frac{2d}{s_0} \cos \phi e^{-\beta s_0} = 0. \quad (4.5)$$

Using

$$-\frac{d}{s_0^2} - \frac{2s_0}{\alpha} = -\frac{3s_0^2 - m}{\alpha s_0}, \quad (4.6)$$

we obtain

$$\delta s = \frac{2s_0(s_0^2 - m)}{3s_0^2 - m} \cos \phi e^{-\beta s_0}. \quad (4.7)$$

Hence the mass gap and the auxiliary kernel are

$$s = \sqrt{\tilde{k}_0} = s_0 \left[1 + 2 \frac{s_0^2 - m}{3s_0^2 - m} \cos \phi e^{-\beta s_0} + \mathcal{O}(e^{-2\beta s_0}) \right], \quad (4.8)$$

$$\tilde{k}_0 = s_0^2 \left[1 + 4 \frac{s_0^2 - m}{3s_0^2 - m} \cos \phi e^{-\beta s_0} + \mathcal{O}(e^{-2\beta s_0}) \right]. \quad (4.9)$$

In a center-symmetric (confining) holonomy one often has the projector-weighted average $\langle \cos \phi \rangle = 0$, in which case the first thermal correction vanishes and the leading thermal effect is $\mathcal{O}(e^{-2\beta s_0})$. However, if the holonomy is not center-symmetric, the $\cos \phi$ factor should be understood as the appropriate (projector-weighted) average over color pairs.

The large- d saddle point s_0 is determined by the cubic equation (4.4) In Cardano form $s_0^3 + p s_0 + q = 0$ we have $p = -m$ and $q = -a$. The discriminant is

$$\Delta = \left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3 = \frac{a^2}{4} - \frac{m^3}{27}, \quad (4.10)$$

and the (real) root can be written as

$$s_0 = \left(\frac{a}{2} + \sqrt{\Delta}\right)^{1/3} + \left(\frac{a}{2} - \sqrt{\Delta}\right)^{1/3}. \quad (4.11)$$

We now expand (4.11) at large d , i.e. at large $a = \alpha d$ with fixed m . Introduce the small parameter

$$\varepsilon \equiv \frac{m}{a^{2/3}}. \quad (4.12)$$

First expand $\sqrt{\Delta}$:

$$\begin{aligned}\sqrt{\Delta} &= \frac{a}{2} \sqrt{1 - \frac{4m^3}{27a^2}} = \frac{a}{2} \left[1 - \frac{1}{2} \frac{4m^3}{27a^2} - \frac{1}{8} \left(\frac{4m^3}{27a^2} \right)^2 + \dots \right] \\ &= \frac{a}{2} - \frac{m^3}{27a} - \frac{m^6}{729a^3} + \dots\end{aligned}\quad (4.13)$$

Therefore

$$\frac{a}{2} + \sqrt{\Delta} = a - \frac{m^3}{27a} - \frac{m^6}{729a^3} + \dots = a \left[1 - \frac{m^3}{27a^2} - \frac{m^6}{729a^4} + \dots \right], \quad (4.14)$$

$$\frac{a}{2} - \sqrt{\Delta} = \frac{m^3}{27a} + \frac{m^6}{729a^3} + \dots = \frac{m^3}{27a} \left[1 + \frac{m^3}{27a^2} + \dots \right]. \quad (4.15)$$

Taking cube roots and using $(1+u)^{1/3} = 1 + \frac{1}{3}u - \frac{1}{9}u^2 + \dots$ gives

$$\left(\frac{a}{2} + \sqrt{\Delta} \right)^{1/3} = a^{1/3} \left[1 - \frac{1}{3} \frac{m^3}{27a^2} + \mathcal{O}\left(\frac{m^6}{a^4}\right) \right] = a^{1/3} - \frac{m^3}{81a^{5/3}} + \dots, \quad (4.16)$$

$$\left(\frac{a}{2} - \sqrt{\Delta} \right)^{1/3} = \left(\frac{m^3}{27a} \right)^{1/3} \left[1 + \frac{1}{3} \frac{m^3}{27a^2} + \dots \right] = \frac{m}{3a^{1/3}} + \frac{m^4}{243a^{7/3}} + \dots. \quad (4.17)$$

Adding (4.16) and (4.17) yields the large- d expansion of the positive root:

$$s_0 = a^{1/3} + \frac{m}{3a^{1/3}} - \frac{m^3}{81a^{5/3}} + \frac{m^4}{243a^{7/3}} + \mathcal{O}\left(\frac{m^6}{a^{11/3}}\right), \quad a = \alpha d. \quad (4.18)$$

Equivalently, in terms of $\varepsilon = m/a^{2/3}$,

$$s_0 = a^{1/3} \left(1 + \frac{\varepsilon}{3} - \frac{\varepsilon^3}{81} + \frac{\varepsilon^4}{243} + \dots \right), \quad \varepsilon = \frac{m}{(\alpha d)^{2/3}}. \quad (4.19)$$

The domain of validity is $\varepsilon \ll 1$, i.e.

$$\frac{m}{(\alpha d)^{2/3}} \ll 1. \quad (4.20)$$

4.2 Careful treatment of the holonomy

Starting from the maximally symmetric pairwise gap condition, we can restore the projector and average over the color indices as follows. With

$$s \equiv \sqrt{\tilde{k}_0} > 0, \quad \tilde{k}_0 = k_0 + m, \quad (4.21)$$

the scalar gap equation is written as

$$\frac{d}{s} \mathcal{R}(s; U) = \frac{k_0}{\alpha} = \frac{s^2 - m}{\alpha}, \quad (4.22)$$

where the projector-weighted holonomy kernel is

$$\mathcal{R}(s; U) = \frac{1}{N^2 - 1} \sum_{i,j=1}^N P_{ij,ji} \frac{\sinh(\beta s)}{\cosh(\beta s) - \cos(\theta_i - \theta_j)}. \quad (4.23)$$

Here

$$P_{ij,ji} = 1 - \frac{1}{N} \delta_{ij}, \quad \sum_{i,j} P_{ij,ji} = N^2 - 1. \quad (4.24)$$

The thermal kernel admits the standard low-temperature expansion

$$\frac{\sinh(\beta s)}{\cosh(\beta s) - \cos(\theta_i - \theta_j)} = 1 + 2 \sum_{p \geq 1} e^{-p\beta s} \cos(p(\theta_i - \theta_j)). \quad (4.25)$$

Therefore

$$\mathcal{R}(s; U) = 1 + 2 \sum_{p \geq 1} \mathcal{C}_p(U) e^{-p\beta s}, \quad (4.26)$$

with the projector-weighted holonomy moments

$$\begin{aligned} \mathcal{C}_p(U) &= \frac{1}{N^2 - 1} \sum_{i,j=1}^N P_{ij,ji} \cos(p(\theta_i - \theta_j)) \\ &= \frac{|\text{Tr } U^p|^2 - 1}{N^2 - 1}. \end{aligned} \quad (4.27)$$

Thus the first correction is controlled not by a single $\cos \phi$, but by the projector-weighted moment $\mathcal{C}_1(U)$. To first thermal order,

$$\mathcal{R}(s; U) = 1 + 2\mathcal{C}_1(U) e^{-\beta s} + \mathcal{O}(e^{-2\beta s}). \quad (4.28)$$

The zero-temperature gap s_0 is determined by (4.4). Writing $s = s_0 + \delta s$, with $\delta s = \mathcal{O}(e^{-\beta s_0})$, we obtain

$$\delta s = \frac{2s_0(s_0^2 - m)}{3s_0^2 - m} \mathcal{C}_1(U) e^{-\beta s_0} + \mathcal{O}(e^{-2\beta s_0}). \quad (4.29)$$

Hence

$$s = s_0 \left[1 + 2 \frac{s_0^2 - m}{3s_0^2 - m} \mathcal{C}_1(U) e^{-\beta s_0} + \mathcal{O}(e^{-2\beta s_0}) \right], \quad (4.30)$$

$$\tilde{k}_0 = s_0^2 \left[1 + 4 \frac{s_0^2 - m}{3s_0^2 - m} \mathcal{C}_1(U) e^{-\beta s_0} + \mathcal{O}(e^{-2\beta s_0}) \right]. \quad (4.31)$$

This makes the role of the holonomy explicit. For the collapsed, deconfined saddle $U = \mathbf{1}_N$, one has $\mathcal{C}_1 = 1$. For a center-symmetric confining holonomy, $\text{Tr } U = 0$, and therefore

$$\mathcal{C}_1 = -\frac{1}{N^2 - 1}, \quad (4.32)$$

which vanishes only in the strict large- N limit. Thus the first thermal correction is exactly absent at leading planar order in the center-symmetric phase, but at finite N there remains a projector-induced correction of order $1/N^2$.

4.3 Bosonic effective action

Let us start from the localized action:

$$S_{\text{loc}}[X, k] = \frac{N}{2} \sum_{a=1}^d \sum_{n \in \mathbb{Z}} \sum_{i,j=1}^N X_a^{ij}(-n) \left[\left(\omega_n + \frac{\theta_i - \theta_j}{\beta} \right)^2 P_{ij,kl} + m P_{ij,kl} \right] X_a^{kl}(n) \\ + \frac{N}{2} \int_0^\beta dt k_{\mu\rho}(t) Y_{\mu\rho}(t) + \frac{N}{4\alpha} \int_0^\beta dt \mu_{\mu\rho\nu\sigma} k_{\mu\rho}(t) k_{\nu\sigma}(t), \quad (4.33)$$

with

$$Y_{\mu\rho}(t) = \sum_{a=1}^d x_a^\mu(t) x_a^\rho(t), \quad (4.34)$$

and

$$k_{\mu\rho} = \Lambda_\mu^{ij} k_{ij,kl} \Lambda_\rho^{kl}, \quad k_{ij,kl}(t) = \frac{1}{4} k_{\mu\rho}(t) \Lambda_\mu^{ji} \Lambda_\rho^{lk}, \quad \mu_{\mu\rho\nu\sigma} \lambda_{\nu\sigma\alpha\beta} = \delta_{\mu\alpha} \delta_{\rho\beta}. \quad (4.35)$$

We expand around the time-independent saddle

$$k_{\mu\rho}(t) = k_{0\mu\rho} + \delta k_{\mu\rho}(t), \quad k_{0\mu\rho} = \text{const.} \quad (4.36)$$

It is important to remember that k_0 is a stationary point of the auxiliary-field effective action obtained after integrating out the coordinate matrices:

$$e^{-S_{\text{eff}}[k]} = \int \mathcal{D}X e^{-S_{\text{loc}}[X, k]}, \quad \left. \frac{\delta S_{\text{eff}}[k]}{\delta k_{\mu\rho}(t)} \right|_{k=k_0} = 0. \quad (4.37)$$

It is not, in general, a stationary point of $S_{\text{loc}}[X, k]$ for every fixed configuration X .

In the maximally symmetric sector one has $k_{0ij,kl} = k_0 P_{ij,kl}$, or equivalently $k_{0\mu\rho} = 2k_0 \delta_{\mu\rho}$. Hence the k_0 -piece in $\frac{N}{2} \int kY$ produces precisely a shift of the quadratic kernel by $k_0 P$. Therefore we rewrite

$$S_{\text{loc}}[X, k_0 + \delta k] = S_{\text{MHO}}[X; \theta, m_{\text{eff}}] + S_k[k_0] + S_{\text{int}}[X, \delta k] + S_{\delta k}^{(2)}[\delta k], \quad (4.38)$$

with the *effective mass*

$$m_{\text{eff}} = s^2 = \tilde{k}_0 = m + k_0, \quad (4.39)$$

and

$$S_{\text{MHO}}[X; \theta, m_{\text{eff}}] = \frac{N}{2} \sum_{a=1}^d \sum_{n \in \mathbb{Z}} \sum_{i,j=1}^N X_a^{ij}(-n) \left[\left(\omega_n + \frac{\theta_i - \theta_j}{\beta} \right)^2 P_{ij,kl} + m_{\text{eff}} P_{ij,kl} \right] X_a^{kl}(n). \quad (4.40)$$

The three remaining contributions to (4.38) are:

- **Zeroth-order term.** Evaluating the purely k_0 -dependent piece gives

$$\begin{aligned} S_k[k_0] &\equiv \frac{N}{4\alpha} \int_0^\beta dt \mu_{\mu\rho\nu\sigma} k_{0\mu\rho} k_{0\nu\sigma} \\ &= -\frac{\beta}{8\alpha} (N^2 - 1) k_0^2, \end{aligned} \quad (4.41)$$

where we used $\mu_{\mu\rho\nu\sigma} \delta_{\nu\sigma} = -(8N)^{-1} \delta_{\mu\rho}$ and the fact that $k_{0\mu\rho} \propto \delta_{\mu\rho}$ lives in the adjoint sector.

- **Linear order term.** The terms linear in the fluctuation δk combine into

$$\begin{aligned} S_{\text{int}}[X, \delta k] &\equiv \frac{N}{2} \int_0^\beta dt \delta k_{\mu\rho}(t) Y_{\mu\rho}(t) \\ &\quad + \frac{N}{2\alpha} \int_0^\beta dt \mu_{\mu\rho\nu\sigma} k_{0\nu\sigma} \delta k_{\mu\rho}(t) \\ &= \frac{N}{2} \int_0^\beta dt \delta k_{\mu\rho}(t) \tilde{Y}_{\mu\rho}(t), \end{aligned} \quad (4.42)$$

with

$$\tilde{Y}_{\mu\rho} \equiv Y_{\mu\rho} + \frac{1}{\alpha} \mu_{\mu\rho\nu\sigma} k_{0\nu\sigma} = Y_{\mu\rho} - \frac{k_0}{4\alpha N} \delta_{\mu\rho}. \quad (4.43)$$

Since k_0 is a saddle point of $S_{\text{eff}}[k]$, stationarity requires the expectation value of the coefficient of δk to vanish. The resulting gap equation can therefore be written as the self-consistency condition

$$\langle Y_{\mu\rho} \rangle_{k_0} = -\frac{1}{\alpha} \mu_{\mu\rho\nu\sigma} k_{0\nu\sigma} = \frac{k_0}{4\alpha N} \delta_{\mu\rho}. \quad (4.44)$$

Accordingly,

$$S_{\text{int}}[X, \delta k] = \frac{N}{2} \int_0^\beta dt \delta k_{\mu\rho}(t) [Y_{\mu\rho}(t) - \langle Y_{\mu\rho} \rangle_{k_0}]. \quad (4.45)$$

The linear coupling does not vanish identically for each fluctuating configuration X . Rather,

$$\langle \tilde{Y}_{\mu\rho} \rangle_{k_0} = 0, \quad (4.46)$$

so that the linear term vanishes after the coordinate matrices have been integrated out, as required by the stationarity of $S_{\text{eff}}[k]$.

- **Second-order term.** The bare quadratic fluctuation term in the localized action is

$$S_{\delta k}^{(2)}[\delta k] \equiv \frac{N}{4\alpha} \int_0^\beta dt \mu_{\mu\rho\nu\sigma} \delta k_{\mu\rho}(t) \delta k_{\nu\sigma}(t). \quad (4.47)$$

After the X_a matrices have been integrated out, their connected correlators contribute to the quadratic kernel of the auxiliary effective action. Expanding around k_0 , one obtains

$$S_{\text{eff}}[k_0 + \delta k] = S_{\text{eff}}[k_0] + \frac{1}{2} \int_0^\beta dt \int_0^\beta dt' \delta k_{\mu\rho}(t) \Gamma_{\mu\rho,\nu\sigma}^{(2)}(t, t') \delta k_{\nu\sigma}(t') + \mathcal{O}(\delta k^3), \quad (4.48)$$

where

$$\Gamma_{\mu\rho,\nu\sigma}^{(2)}(t, t') = \frac{N}{2\alpha} \mu_{\mu\rho\nu\sigma} \delta(t - t') - \frac{N^2}{4} \left\langle \tilde{Y}_{\mu\rho}(t) \tilde{Y}_{\nu\sigma}(t') \right\rangle_{k_0, c}. \quad (4.49)$$

The one-loop integration over the auxiliary fluctuation is therefore controlled by the full Hessian $\Gamma^{(2)}$:

$$Z \simeq e^{-S_{\text{eff}}[k_0]} [\det \Gamma^{(2)}]^{-1/2}. \quad (4.50)$$

Thus, integrating out δk at fixed X would not yield only an irrelevant constant: it would generate interactions involving $Y - \langle Y \rangle_{k_0}$. These interactions describe the subleading corrections to the Gaussian saddle.

We now estimate the two contributions to the auxiliary-field Hessian (4.49). The first is the bare kernel inherited from the quadratic k^2 term in the localized action, while the second is the polarization kernel generated by the connected fluctuations of the coordinate matrices.

Consider a time-independent fluctuation in the maximally symmetric channel,

$$k = k_0 + \delta k, \quad s^2 = m + k_0. \quad (4.51)$$

In this channel, and suppressing the holonomy dependence and common adjoint multiplicity, the effective action takes the schematic form

$$S_{\text{eff}}(k) = (N^2 - 1) \left[-\frac{\beta}{8\alpha} k^2 + \frac{d}{2} \sum_{n \in \mathbb{Z}} \log(\omega_n^2 + m + k) \right]. \quad (4.52)$$

Expanding around k_0 gives

$$\begin{aligned} S_{\text{eff}}(k_0 + \delta k) &= S_{\text{eff}}(k_0) + (N^2 - 1) \delta k \left[-\frac{\beta k_0}{4\alpha} + \frac{d}{2} \sum_{n \in \mathbb{Z}} \frac{1}{\omega_n^2 + s^2} \right] \\ &\quad - (N^2 - 1) (\delta k)^2 \left[\frac{\beta}{8\alpha} + \frac{d}{4} \sum_{n \in \mathbb{Z}} \frac{1}{(\omega_n^2 + s^2)^2} \right] + \mathcal{O}(\delta k^3). \end{aligned} \quad (4.53)$$

The coefficient of the linear fluctuation vanishes by the static auxiliary-field saddle equation

$$-\frac{\beta k_0}{4\alpha} + \frac{d}{2} \sum_{n \in \mathbb{Z}} \frac{1}{\omega_n^2 + s^2} = 0 \quad \Longleftrightarrow \quad \frac{k_0}{\alpha} = \frac{d}{s} \coth\left(\frac{\beta s}{2}\right). \quad (4.54)$$

In the zero-temperature limit, and upon using $k_0 = s^2 - m$, this saddle equation is precisely equivalent to the cubic gap equation (4.4).

The two terms in the second line of (4.53) are precisely the two contributions to the static Hessian (4.49). The coordinate polarization is governed by

$$\begin{aligned} \sum_{n \in \mathbb{Z}} \frac{1}{(\omega_n^2 + s^2)^2} &= \frac{\beta}{4s^3} \coth\left(\frac{\beta s}{2}\right) + \frac{\beta^2}{8s^2} \operatorname{csch}^2\left(\frac{\beta s}{2}\right) \\ &= \frac{\beta}{4s^3} [1 + \mathcal{O}(\beta s e^{-\beta s})], \quad \beta s \gg 1. \end{aligned} \quad (4.55)$$

It therefore scales parametrically as

$$\Gamma_{\text{stat}, X}^{(2)} \sim \frac{d\beta}{s^3}, \quad (4.56)$$

up to its numerical coefficient, sign, and adjoint tensor eigenvalue.

The bare contribution scales as β/α . To leading low-temperature order, the cubic gap equation (4.4) gives

$$\Gamma_{\text{stat}, \text{bare}}^{(2)} \sim \frac{d\beta}{s^3} \frac{s^2}{s^2 - m}, \quad (4.57)$$

again up to the corresponding numerical coefficient and tensor eigenvalue. Thus, provided $s^2 - m$ is of order s^2 , both parts of the static Hessian are parametrically of order $d\beta/s^3$.

For a stable eigenmode of the complete tensor kernel, understood along the appropriate steepest-descent contour of the Hubbard-Stratonovich integral, one may therefore write

$$\Gamma_{\text{stat}}^{(2)} \sim \frac{d\beta}{s^3} \mathcal{C}\left(\frac{m}{s^2}\right) [1 + \mathcal{O}(\beta s e^{-\beta s})], \quad (4.58)$$

where the dimensionless coefficient $\mathcal{C}$ combines the bare and polarization contributions. This is a parametric estimate of an eigenvalue of the full Hessian, rather than an equality for the complete tensor-valued kernel.

Provided that $\mathcal{C} \neq 0$ and that no parametrically soft mode is present, the variance of the corresponding static fluctuation along the stable steepest-descent direction is controlled parametrically by the inverse Hessian:

$$\langle (\delta k)^2 \rangle \sim \frac{s^3}{d\beta} \frac{1}{\mathcal{C}(m/s^2)}. \quad (4.59)$$

Consequently,

$$\frac{\sqrt{\langle(\delta k)^2\rangle}}{s^2} \sim \frac{1}{\sqrt{d\beta s}} \frac{1}{\sqrt{\mathcal{C}(m/s^2)}}. \quad (4.60)$$

The static auxiliary fluctuations are therefore suppressed at large d , with an additional suppression in the low-temperature regime $\beta s \gg 1$.

Consequently, at leading order in the combined large- d and low-temperature saddle expansion, one freezes the auxiliary field at $k = k_0$. The effective action governing the coordinate matrices then reduces to

$$\begin{aligned} S_{\text{eff}}[X; \theta] \big|_{\text{leading saddle}} &= S_{\text{MHO}}[X; \theta, m_{\text{eff}}] \\ &= N \int_0^\beta dt \text{Tr} \left[\frac{1}{2} (D_t X_a)^2 + \frac{m_{\text{eff}}}{2} X_a^2 \right]. \end{aligned} \quad (4.61)$$

In particular, the *dominant large- d contribution* is given by a gauged matrix harmonic oscillator with the dynamically generated effective mass

$$m_{\text{eff}} = s^2 = \tilde{k}_0 = m + k_0. \quad (4.62)$$

Thus, at leading order in the large- d expansion, the original commutator-squared interaction is encoded in a self-consistent mass term, while fluctuations of δk generate the subleading non-Gaussian corrections.

4.4 Free energy

At the large- d saddle, the localized theory reduces to a gauged matrix harmonic oscillator with effective mass

$$s^2 \equiv m_{\text{eff}} = m + k_0, \quad (4.63)$$

where k_0 is determined by the gap equation. Integrating out the Gaussian matrices X_a yields an effective action for the Polyakov loop eigenvalues $\{\theta_i\}$.

We start from the action

$$S_{\text{MHO}}[X; \theta, m_{\text{eff}}] = \frac{N}{2} \sum_{a=1}^d \sum_{n \in \mathbb{Z}} \sum_{i,j=1}^N X_a^{ij}(-n) \left[\left(\omega_n + \frac{\phi_{ij}}{\beta} \right)^2 P_{ij,kl} + s^2 P_{ij,kl} \right] X_a^{kl}(n), \quad (4.64)$$

with

$$\phi_{ij} = \theta_i - \theta_j. \quad (4.65)$$

Each adjoint mode (i, j) experiences shifted Matsubara frequencies $\omega_n \rightarrow \omega_n + \phi_{ij}/\beta$, and thus for each matrix element this produces the Gaussian determinant

$$\prod_{n \in \mathbb{Z}} \left[\left(\omega_n + \frac{\phi_{ij}}{\beta} \right)^2 + s^2 \right] \propto \cosh(\beta s) - \cos \phi_{ij}, \quad \omega_n = \frac{2\pi n}{\beta}. \quad (4.66)$$

The result is derived by expressing the determinant in logarithmic form and performing the Matsubara sum (3.20). Explicitly, set

$$x = \beta s, \quad \phi = \phi_{ij}. \quad (4.67)$$

Up to an overall β -dependent normalization independent of s and ϕ , we have

$$\prod_{n \in \mathbb{Z}} \left[\left(\omega_n + \frac{\phi}{\beta} \right)^2 + s^2 \right] \propto \prod_{n \in \mathbb{Z}} [(2\pi n + \phi)^2 + x^2]. \quad (4.68)$$

Let

$$\mathcal{D}(x, \phi) = \prod_{n \in \mathbb{Z}} [(2\pi n + \phi)^2 + x^2]. \quad (4.69)$$

Then

$$\frac{\partial}{\partial x} \log \mathcal{D}(x, \phi) = 2x \sum_{n \in \mathbb{Z}} \frac{1}{(2\pi n + \phi)^2 + x^2}. \quad (4.70)$$

Using the standard Matsubara sum (3.20) in the form

$$\sum_{n \in \mathbb{Z}} \frac{1}{(2\pi n + \phi)^2 + x^2} = \frac{1}{2x} \frac{\sinh x}{\cosh x - \cos \phi}, \quad (4.71)$$

we obtain

$$\frac{\partial}{\partial x} \log \mathcal{D}(x, \phi) = \frac{\sinh x}{\cosh x - \cos \phi} = \frac{\partial}{\partial x} \log(\cosh x - \cos \phi). \quad (4.72)$$

Hence

$$\mathcal{D}(x, \phi) \propto \cosh x - \cos \phi. \quad (4.73)$$

Restoring $x = \beta s$, this gives

$$\prod_{n \in \mathbb{Z}} \left[\left(\omega_n + \frac{\phi_{ij}}{\beta} \right)^2 + s^2 \right] \propto \cosh(\beta s) - \cos \phi_{ij}. \quad (4.74)$$

The proportionality constant is independent of s and ϕ_{ij} , and therefore contributes only an irrelevant additive constant to the free energy.

Therefore, after integrating out the coordinate modes $X_a^{ij}(n)$, the bosonic matrix harmonic-oscillator partition function is, up to an overall normalization,

$$\log Z_{\text{MHO}}(\theta; s) = -\frac{d}{2} \sum_{i,j=1}^N P_{ij,ji} \log \left(\cosh(\beta s) - \cos \phi_{ij} \right) + \text{const.} \quad (4.75)$$

The corresponding free energy is

$$F_{\text{MHO}}(\theta; s) = -\frac{1}{\beta} \log Z_{\text{MHO}} = \frac{d}{2\beta} \sum_{i,j=1}^N P_{ij,ji} \log \left(\cosh(\beta s) - \cos(\theta_i - \theta_j) \right) + \text{const.} \quad (4.76)$$

4.5 Extent of space

The extent of space is defined as

$$R^2 = \frac{1}{N\beta} \left\langle \int_0^\beta dt \text{Tr} X_a(t) X_a(t) \right\rangle. \quad (4.77)$$

For the Gaussian theory obtained at the large- d saddle point, the momentum-space propagator in the presence of holonomy is

$$\left\langle X_a^{ij}(n) X_a^{kl}(-n) \right\rangle = \frac{1}{N} \frac{P_{ij,kl}}{\left(\omega_n + \frac{\phi_{ij}}{\beta} \right)^2 + s^2}. \quad (4.78)$$

The equal-time (coincident) correlator follows again from the standard Matsubara sum

$$\sum_{n \in \mathbb{Z}} \frac{1}{\left(\omega_n + \frac{\phi_{ij}}{\beta} \right)^2 + s^2} = \frac{\beta}{2s} \frac{\sinh(\beta s)}{\cosh(\beta s) - \cos \phi_{ij}}. \quad (4.79)$$

Therefore, the extent of space is given explicitly by

$$R^2(\theta; s) = \frac{d}{N^2} \sum_{i,j} P_{ij,ji} \frac{1}{2s} \frac{\sinh(\beta s)}{\cosh(\beta s) - \cos \phi_{ij}}. \quad (4.80)$$

As a consistency check, the same quantity can be extracted from the free energy of the matrix harmonic oscillator via

$$\frac{\partial F_{\text{MHO}}}{\partial s^2} = \frac{N^2}{2} R^2, \quad (4.81)$$

up to the same normalization conventions used in the action.

4.6 Consistency between the saddle equation and the extent of space

At the large- d saddle point, the auxiliary effective action satisfies (4.44), namely

$$\langle Y_{\mu\rho} \rangle_{k_0} = -\frac{1}{\alpha} \mu_{\mu\rho\nu\sigma} k_{0\nu\sigma} = \frac{k_0}{4\alpha N} \delta_{\mu\rho}. \quad (4.82)$$

This relation follows from the stationarity of the auxiliary-field effective action, or equivalently from the vanishing expectation value of the coefficient of the term linear in δk .

Evaluating the same expectation value in the Gaussian saddle and invoking rotational symmetry in adjoint space, we obtain

$$\langle Y_{\mu\rho} \rangle_{k_0} = \delta_{\mu\rho} \frac{1}{N^2 - 1} \sum_{\lambda} \langle Y_{\lambda\lambda} \rangle_{k_0} = \delta_{\mu\rho} \frac{N}{2} \frac{R^2}{N^2 - 1}. \quad (4.83)$$

Equating (4.82) and (4.83) yields

$$R^2 = \left(1 - \frac{1}{N^2}\right) \frac{k_0}{2\alpha}. \quad (4.84)$$

Using $\sum_{i,j} P_{ij,ji} = N^2 - 1$, the zero-temperature Gaussian result (see below)

$$R^2 = \left(1 - \frac{1}{N^2}\right) \frac{d}{2s} \quad (4.85)$$

immediately implies

$$\frac{k_0}{\alpha} = \frac{d}{s}. \quad (4.86)$$

Upon using $k_0 = s^2 - m$, this is precisely equivalent to the saddle-point gap equation (4.4).

Thus (4.82) is not an additional assumption. Rather, it expresses the consistency between the expectation-value saddle equation of the auxiliary effective action and the extent of space computed in the emergent Gaussian theory. At zero temperature, the two evaluations agree at leading order in the large- d saddle expansion, while in the low-temperature regime the thermal corrections are exponentially suppressed. This is consistent with the gauged matrix harmonic oscillator providing the leading large- d , low-temperature description of the original BFSS model.

4.7 Low-temperature evaluation and mass regimes

In the low-temperature confined phase ($\beta s \gg 1$), the dominant holonomy saddle is uniform, $\rho(\theta) = 1/2\pi$. For a general function $f(\phi_{ij})$, the adjoint projector $P_{ij,ji} = 1 - \delta_{ij}/N$ gives

$$\begin{aligned} \frac{1}{N^2} \sum_{i,j} P_{ij,ji} f(\phi_{ij}) &= \frac{1}{N^2} \sum_{i,j} f(\phi_{ij}) - \frac{1}{N^2} f(0) \\ &\longrightarrow \int_{-\pi}^{\pi} \frac{d\phi}{2\pi} f(\phi) - \frac{1}{N^2} f(0). \end{aligned} \quad (4.87)$$

For the matrix harmonic oscillator determinant, we take

$$f(\phi) = \log(\cosh x - \cos \phi), \quad x = \beta s. \quad (4.88)$$

The required quantities are

$$\int_{-\pi}^{\pi} \frac{d\phi}{2\pi} \log(\cosh x - \cos \phi) = x - \log 2, \quad x > 0, \quad (4.89)$$

$$\begin{aligned} f(0) &= \log(\cosh x - 1) \\ &= x - \log 2 + 2 \log(1 - e^{-x}). \end{aligned} \quad (4.90)$$

Hence, Eq. (4.87) becomes

$$\begin{aligned} \frac{1}{N^2} \sum_{i,j} P_{ij,ji} \log(\cosh x - \cos \phi_{ij}) &\longrightarrow \left(1 - \frac{1}{N^2}\right)(x - \log 2) - \frac{2}{N^2} \log(1 - e^{-x}) \\ &= \left(1 - \frac{1}{N^2}\right)(x - \log 2) + \mathcal{O}\left(\frac{e^{-x}}{N^2}\right). \end{aligned} \quad (4.91)$$

Since the total free energy contains an overall factor N^2 , its leading low-temperature form is therefore

$$F_{\text{MHO}} = (N^2 - 1) \frac{d}{2} s + \text{const} + \mathcal{O}(\beta^{-1} e^{-\beta s}), \quad (\beta s \gg 1). \quad (4.92)$$

The additive constant is independent of s .

Similarly, in the same uniform low-temperature saddle, the extent of space is

$$R^2 = \left(1 - \frac{1}{N^2}\right) \frac{d}{2s} + \mathcal{O}(e^{-\beta s}), \quad (\beta s \gg 1). \quad (4.93)$$

To the displayed order, Eqs. (4.92) and (4.93) satisfy

$$\frac{\partial F_{\text{MHO}}}{\partial s^2} = \frac{N^2}{2} R^2, \quad (4.94)$$

as expected.

The dependence on the bare mass m enters entirely through the zero-temperature gap equation (4.4), which yields two distinct regimes:

(i) *Small mass*, $m \ll (\alpha d)^{2/3}$. Writing $\varepsilon = m/(\alpha d)^{2/3} \ll 1$, one finds

$$s_0 = (\alpha d)^{1/3} \left(1 + \frac{\varepsilon}{3} + \mathcal{O}(\varepsilon^2)\right). \quad (4.95)$$

Hence,

$$F_{\text{MHO}} = (N^2 - 1) \frac{d}{2} (\alpha d)^{1/3} \left(1 + \frac{\varepsilon}{3} + \cdots\right), \quad (4.96)$$

$$R^2 = \left(1 - \frac{1}{N^2}\right) \frac{d^{2/3}}{2\alpha^{1/3}} \left(1 - \frac{\varepsilon}{3} + \cdots\right). \quad (4.97)$$

In this regime the extent of space scales as $R^2 \sim d^{2/3}$ and the free energy as $F \sim d^{4/3}$, reflecting a genuinely interacting large- d saddle.

(ii) *Large mass*, $m \gg (\alpha d)^{2/3}$. The positive root admits

$$s_0 = \sqrt{m} + \frac{\alpha d}{2m} + \mathcal{O}(m^{-5/2}), \quad (4.98)$$

leading to

$$F_{\text{MHO}} = (N^2 - 1) \frac{d}{2} \sqrt{m} \left(1 + \frac{\alpha d}{2m^{3/2}} + \cdots \right), \quad (4.99)$$

$$R^2 = \left(1 - \frac{1}{N^2} \right) \frac{d}{2\sqrt{m}} \left(1 - \frac{\alpha d}{2m^{3/2}} + \cdots \right). \quad (4.100)$$

Here the matrices are tightly localized near the origin, $R^2 \sim d/\sqrt{m}$, and the theory approaches a truly Gaussian regime dominated by the quadratic term.

Thus, at low temperature the free energy and the extent of space are entirely controlled by the gap $s_0(m, d)$: the small- m regime retains strong BFSS interactions with $d^{4/3}$ scaling, while large m smoothly suppresses both R^2 and interaction effects.

5 Holonomy physics at low- T

5.1 Holonomy effective action

After the reduction to the gauged matrix harmonic oscillator, the remaining gauge dynamics is encoded in the holonomy eigenvalues θ_i , or equivalently in the Polyakov moments

$$u_n = \frac{1}{N} \sum_{j=1}^N e^{in\theta_j}, \quad u_{-n} = u_n^*, \quad q = e^{-\beta s}. \quad (5.1)$$

Integrating out the d Gaussian coordinate matrices gives an attractive contribution to the holonomy potential, while the Vandermonde determinant from the Haar measure gives a universal repulsive contribution. Dropping θ -independent constants, these are

$$S_X[\theta] = -d N^2 \sum_{n \geq 1} \frac{q^n}{n} |u_n|^2, \quad (5.2)$$

$$S_{\text{Vdm}}[\theta] = N^2 \sum_{n \geq 1} \frac{1}{n} |u_n|^2. \quad (5.3)$$

Hence the holonomy effective action is

$$S_{\text{hol}}[\theta] = N^2 \sum_{n \geq 1} \frac{1 - dq^n}{n} |u_n|^2 + \text{const}, \quad q = e^{-\beta s}. \quad (5.4)$$

The Fourier derivation of (5.2), (5.3), and (5.4) is given in Appendix A.

5.2 Confined phase and critical point

The coefficient $1 - dq^n$ displays the competition between the repulsive Vandermonde measure and the attractive coordinate determinant. The first Polyakov mode becomes marginal when $dq = 1$, giving the Gaussian large- d deconfinement, or Hagedorn, criterion. Indeed, the holonomy effective action (5.4) is positive definite provided

$$1 - dq^n > 0 \quad \forall n \geq 1. \quad (5.5)$$

In this regime the unique minimum is

$$u_n = 0 \quad (n \geq 1), \quad (5.6)$$

corresponding to the uniform (center-symmetric) eigenvalue density

$$\rho(\theta) = \frac{1}{2\pi}, \quad \theta \in [-\pi, \pi]. \quad (5.7)$$

Since the most stringent condition comes from the lowest mode, confinement reduces to

$$dq < 1. \quad (5.8)$$

The confined saddle becomes marginal when the coefficient of the $n = 1$ mode vanishes, namely

$$1 - dq = 0. \quad (5.9)$$

Recall that

$$\begin{aligned} q &= e^{-\beta s}, \quad s^2 = m_{\text{eff}}^2 = m^2 + k_0^2, \\ s &= s_0 \left[1 + \frac{2}{3} \frac{s_0^2 - m^2}{3s_0^2 - m^2} \langle \cos \phi \rangle e^{-\beta s_0} + \dots \right], \\ s_0 &= (\alpha d)^{1/3} \left[1 + \frac{\epsilon}{3} - \frac{\epsilon^3}{81} + \frac{\epsilon^4}{243} + \dots \right], \quad \epsilon = \frac{m}{(\alpha d)^{2/3}}. \end{aligned} \quad (5.10)$$

Here s_0 is determined by the zero-temperature gap equation (4.4).

In the center-symmetric phase one has $\langle \cos \phi \rangle = 0$, so the leading thermal correction to s vanishes and $s \simeq s_0$. Setting $\alpha = 1$ at the end of the calculation, the critical temperature is therefore fixed by

$$\beta_c s_0 = \log d \quad \Longleftrightarrow \quad T_c = \frac{s_0}{\log d}. \quad (5.11)$$

Explicitly, we have

$$T_c = \frac{d^{1/3}}{\log d} \left[1 + \frac{\epsilon}{3} - \frac{\epsilon^3}{81} + \frac{\epsilon^4}{243} + \dots \right]. \quad (5.12)$$

Thus, at large d with fixed m , one finds

$$T_c \rightarrow \infty \quad (d = 1, \mathbf{BFSS}_2), \quad (5.13)$$

whereas for $d > 1$,

$$T_c \sim \frac{d^{1/3}}{\log d} \gg 1 \quad (d > 1, \mathbf{BFSS}_{d+1}). \quad (5.14)$$

This explains why the low-temperature dynamics of the $\mathbf{BFSS}_d$ model is accurately captured by the gauged matrix harmonic oscillator: for a wide range $T < T_c$ the system remains in the confined phase.

5.3 Effect of a mass deformation on the transition temperature

The previous results apply in the regime of small mass,

$$m \ll (\alpha d)^{2/3}, \quad (5.15)$$

where the dynamically generated scale is controlled by $s_0 \sim (\alpha d)^{1/3}$. In the opposite, large-mass regime,

$$m \gg (\alpha d)^{2/3}, \quad (5.16)$$

the positive root of the zero-temperature gap equation (4.4) admits the asymptotic expansion

$$s_0 = \sqrt{m} + \frac{\alpha d}{2m} + \mathcal{O}(m^{-5/2}), \quad m \gg (\alpha d)^{2/3}. \quad (5.17)$$

Substituting this result into the general expression (5.11) for the critical temperature, the latter becomes

$$T_c = \frac{\sqrt{m}}{\log d} \left[1 + \frac{\alpha d}{2m^{3/2}} + \mathcal{O}(m^{-3}) \right], \quad d > 1. \quad (5.18)$$

Thus, a positive mass deformation increases the gap s and pushes the deconfinement instability to parametrically higher temperatures. In particular, for moderately large values of d (e.g. $d = 2$ in $\mathbf{BFSS}_3$), one may keep the system in the uniform, center-symmetric (stringy / Gaussian) phase over all temperatures of interest by choosing m sufficiently large, while still retaining quantitative control within the large- d saddle-point framework.

5.4 Deconfined phase and eigenvalue distributions

For $d > 1$ (equivalently $T > T_c$), the uniform distribution becomes unstable. Introducing a continuous eigenvalue density $\rho(\theta)$ and its moments

$$u_n = \int_{-\pi}^{\pi} d\theta \rho(\theta) e^{in\theta}, \quad \rho(\theta) \geq 0, \quad \int_{-\pi}^{\pi} \rho(\theta) d\theta = 1, \quad (5.19)$$

At leading order, the holonomy effective action (5.4) can be written as a functional of ρ :

$$S_{\text{hol}}[\rho] = -\frac{N^2}{2} \int_{-\pi}^{\pi} d\theta \int_{-\pi}^{\pi} d\phi \rho(\theta) \rho(\phi) \log\left(2 \sin \frac{\theta - \phi}{2}\right)^2 - N^2 d \sum_{n \geq 1} \frac{q^n}{n} u_n u_{-n} + \text{const.} \quad (5.20)$$

We extremize S_{hol} subject to $\int \rho = 1$ by introducing a Lagrange multiplier λ :

$$0 = \frac{\delta}{\delta \rho(\theta)} \left(S_{\text{hol}}[\rho] + \lambda \int_{-\pi}^{\pi} d\theta \rho(\theta) \right). \quad (5.21)$$

Varying (5.20) gives

$$0 = - \int_{-\pi}^{\pi} d\phi \rho(\phi) \log\left(2 \sin \frac{\theta - \phi}{2}\right)^2 - d \sum_{n \geq 1} \frac{q^n}{n} (u_{-n} e^{in\theta} + u_n e^{-in\theta}) + \lambda, \quad (5.22)$$

where we used

$$\frac{\delta u_n}{\delta \rho(\theta)} = e^{in\theta}, \quad \frac{\delta u_{-n}}{\delta \rho(\theta)} = e^{-in\theta}. \quad (5.23)$$

Taking one derivative with respect to θ removes the constant λ and yields

$$-\text{P.V.} \int_{-\pi}^{\pi} d\phi \rho(\phi) \frac{\partial}{\partial \theta} \log\left(2 \sin \frac{\theta - \phi}{2}\right)^2 = d \sum_{n \geq 1} q^n (u_{-n} i e^{in\theta} - u_n i e^{-in\theta}). \quad (5.24)$$

Using

$$\frac{\partial}{\partial \theta} \log\left(2 \sin \frac{\theta - \phi}{2}\right)^2 = \cot \frac{\theta - \phi}{2}, \quad (5.25)$$

and writing the right-hand side as $2d \sum_{n \geq 1} q^n \text{Im}(u_n e^{-in\theta})$, we obtain

$$-\text{P.V.} \int_{-\pi}^{\pi} d\phi \rho(\phi) \cot \frac{\theta - \phi}{2} = 2d \sum_{n \geq 1} q^n \text{Im}(u_n e^{-in\theta}). \quad (5.26)$$

For the relevant saddles one may take $\rho(\theta) = \rho(-\theta)$, hence $u_n \in \mathbb{R}$ and $\text{Im}(u_n e^{-in\theta}) = -u_n \sin(n\theta)$, giving the large- N saddle-point equation

$$\text{P.V.} \int_{-\pi}^{\pi} d\phi \rho(\phi) \cot \frac{\theta - \phi}{2} = 2d \sum_{n \geq 1} q^n u_n \sin(n\theta). \quad (5.27)$$

Just above T_c , only the $n = 1$ mode condenses and the density is well approximated by

$$\rho(\theta) = \frac{1}{2\pi} \left(1 + 2u_1 \cos \theta \right), \quad |u_1| \ll 1, \quad (5.28)$$

which continuously deforms the uniform distribution. At higher temperature, positivity of ρ enforces a gapped solution of (5.27), corresponding to clumping of eigenvalues around $\theta = 0$.

5.5 Physical interpretation of the holonomy phases

It is important to stress the correct physical meaning of the holonomy saddle. When the eigenvalues $\{\theta_i\}$ are *uniformly distributed* on the thermal circle,

$$\rho(\theta) = \frac{1}{2\pi}, \quad u_n = 0 \quad (n \geq 1), \quad (5.29)$$

the Polyakov loop vanishes and the system is in a *center-symmetric, confined phase*. In the D0-brane interpretation, this corresponds to a *stringy phase* in which the D0-branes are delocalized along the Euclidean time circle and no single temporal position is preferred.

By contrast, a *black-hole phase* requires *localization* of the D0-branes at a point on the thermal circle. This corresponds to a *non-uniform* eigenvalue distribution,

$$\rho(\theta) \neq \frac{1}{2\pi}, \quad u_1 \neq 0, \quad (5.30)$$

with a spectral gap in $\rho(\theta)$. Such a saddle breaks center symmetry and signals deconfinement in the matrix model, which is the correct gauge-theory dual of a black zero-brane geometry.

Therefore:

- **Uniform holonomy** $\Rightarrow$ string / Hagedorn-like phase,
- **Non-uniform (gapped) holonomy** $\Rightarrow$ black-hole phase.

In BFSS₂, the uniform phase persists at all temperatures, consistent with the absence of a finite-temperature black hole and the emergence of an AdS₂ stringy regime. For $d > 1$, the deconfinement transition corresponds to the onset of D0-brane localization and black-hole formation.

5.6 Black-string dictionary and the critical-dimension question

The holonomy phases admit a sharper gravitational interpretation when the one-dimensional model is regarded as the high-temperature reduction of thermal 1 + 1-dimensional Yang-Mills theory on a spatial circle [51, 53]. Denoting the thermal and spatial circumferences of the parent theory by $\tilde{\beta}$ and L , respectively, the limit $\tilde{\beta} \rightarrow 0$ removes the nonzero temporal Kaluza-Klein modes and, with antiperiodic thermal boundary conditions, the fermions. The surviving one-dimensional coordinate is the original spatial coordinate. Consequently,

$$\beta_{1d} = L, \quad U_{1d} = \mathcal{P} \exp \left(i \int_0^L dx A_x \right) = W_x, \quad (5.31)$$

so the Polyakov loop of the reduced model is the spatial Wilson line of the parent 1 + 1-dimensional theory.

After T-duality, the D1-branes are mapped to D0-branes, and the phases θ_i of the spatial Wilson line W_x are interpreted as the angular positions of the individual D0-branes on the dual circle [34, 66]. The resulting gravitational dictionary is therefore

$$\begin{aligned} \rho(\theta) = 1/2\pi & \iff \text{uniform black string,} \\ \rho(\theta) \text{ non-uniform with full support} & \iff \text{non-uniform black string,} \\ \rho(\theta) \text{ gapped and localized} & \iff \text{localized black hole.} \end{aligned} \tag{5.32}$$

In particular, the weak deformation

$$\rho(\theta) = \frac{1}{2\pi} (1 + 2u_1 \cos \theta + \dots) \tag{5.33}$$

is the matrix counterpart of the Gregory–Laflamme deformation of a uniform black string. The condition

$$1 - de^{-\beta s} = 0 \tag{5.34}$$

therefore identifies the onset of the corresponding linear instability. In the full two-dimensional phase diagram, the two high-temperature transitions between uniform, non-uniform, and gapped distributions may join the direct low-temperature black-string/black-hole transition at a multicritical region [51]. See also [34, 66].

The multicritical organization displayed in Fig. 1 concerns the global thermal and coupling phase diagram. A related but more refined phenomenon occurs for neutral Kaluza–Klein black strings in ordinary vacuum Einstein gravity. Consider a uniform black string in D spacetime dimensions, extended along a compact coordinate $z \sim z + L$, and define

$$n = D - 4. \tag{5.35}$$

At the Gregory–Laflamme threshold, the uniform string develops a static linear zero mode with wave number k_{GL} . Introducing its amplitude ε as a perturbative parameter, one seeks a family of nearby static solutions of the full nonlinear Einstein equations of the form

$$g_{\mu\nu}(\varepsilon) = g_{\mu\nu}^{\text{UBS}} + \varepsilon h_{\mu\nu}^{(1)} + \varepsilon^2 h_{\mu\nu}^{(2)} + \dots \tag{5.36}$$

The wave number and thermodynamic quantities are simultaneously allowed to shift along this family,

$$k(\varepsilon) = k_{\text{GL}} + k_2 \varepsilon^2 + \dots, \quad M(\varepsilon) = M_{\text{GL}} + M_2 \varepsilon^2 + \dots \tag{5.37}$$

This perturbative continuation in the space of static solutions generates the weakly non-uniform black-string branch that bifurcates from the uniform branch at the Gregory–Laflamme point; and it should not be confused with the real-time evolution of an unstable string.

In the large- D effective description, the corresponding mass-density profile may be written as

$$\frac{m(z)}{\bar{m}} = 1 + \varepsilon \cos(k(\varepsilon)z) + \frac{\varepsilon^2}{6} \cos(2k(\varepsilon)z) + \cdots, \quad (5.38)$$

where $z \sim z + L$ is the compact string coordinate and $\varepsilon = 0$ gives the uniform string. Since $k(\varepsilon) - k_{\text{GL}} = O(\varepsilon^2)$, one may equivalently replace $k(\varepsilon)$ by k_{GL} in (5.38) at the order displayed.

The effective density $m(z)$ is related to the local horizon radius by

$$r_{\text{h}}(z) = r_0 \left(\frac{m(z)}{\bar{m}} \right)^{1/n}, \quad n = D - 4, \quad (5.39)$$

and therefore, to linear order in the deformation,

$$\frac{r_{\text{h}}(z)}{r_0} = 1 + \frac{\varepsilon}{n} \cos(k_{\text{GL}}z) + O(\varepsilon^2). \quad (5.40)$$

Thus ε measures an order-one modulation of the effective mass density, whereas the corresponding geometrical modulation of the horizon radius is suppressed by $1/n$ at large D [67–69].

This construction makes the matrix/gravity correspondence more precise. Identifying the angular coordinate on the T-dual circle with the normalized string coordinate,

$$\theta = \frac{2\pi z}{L}, \quad (5.41)$$

the normalized eigenvalue density and the normalized gravitational mass density play analogous roles:

$$2\pi\rho(\theta) \iff \frac{m(z)}{\bar{m}}. \quad (5.42)$$

Hence the weak matrix deformation $2\pi\rho(\theta) = 1 + 2u_1 \cos\theta + \cdots$ is the counterpart of (5.38), with

$$\varepsilon \iff 2u_1 \quad (5.43)$$

at linear order, up to the normalization convention for the deformation amplitude. The first Polyakov moment u_1 is therefore the matrix analogue of the Gregory–Laflamme order parameter.

Sorkin showed that the branch emerging from the Gregory–Laflamme point changes character at a critical spacetime dimension $D_* \simeq 13$ [67]. Later large- D calculations locate the change more precisely near $D_* \simeq 13.5$, and show that both its thermodynamic and dynamical stability change there [68, 69].

For $D > D_*$, a stable weakly non-uniform string emerges continuously from the unstable uniform string, consistently with a second-order transition at M_{GL} . For $D < D_*$, the weakly

non-uniform branch is not the stable endpoint near M_{GL} , and a first-order transition to a finite-amplitude configuration is required. The latter may be a strongly non-uniform string or a localized black hole; the critical dimension controls the local branch near the Gregory–Laflamme point and does not by itself imply immediate horizon breakup.

The matrix analogue of this question requires the nonlinear effective action for the first Polyakov moment,

$$\frac{S_{\text{hol}}}{N^2} = S_0 + a_2(d, T)|u_1|^2 + a_4(d, T)|u_1|^4 + a_6(d, T)|u_1|^6 + \cdots, \quad a_2(d, T) = 1 - de^{-\beta s}. \quad (5.44)$$

For each d , the linear instability temperature $T_c(d)$ is determined by

$$a_2(d, T_c(d)) = 0. \quad (5.45)$$

A matrix critical dimension d_* , if it exists, would then be defined by

$$a_4(d_*, T_c(d_*)) = 0, \quad a_6(d_*, T_c(d_*)) > 0. \quad (5.46)$$

The sign of $a_4(d, T_c(d))$ on the two sides of d_* determines the character of the branch emerging from the uniform saddle. For $a_4 > 0$, a locally stable weakly non-uniform branch develops continuously from $u_1 = 0$, whereas for $a_4 < 0$ the weak branch is locally unstable and the transition requires a finite-amplitude configuration stabilized by the higher-order terms. The point $a_2 = a_4 = 0$, with $a_6 > 0$, is therefore the matrix analogue of a tricritical point.

The leading maximally symmetric saddle derived here fixes a_2 and hence the linear instability (5.34), but it does not yet determine the complete nonlinear coefficients. Establishing an analogue of the Sorkin critical dimension would require subleading $1/d$ corrections, the holonomy backreaction on the auxiliary saddle, and possibly more general color-dependent components of the auxiliary kernel.

Finally, two distinct notions of dimension enter the comparison. In the Sorkin problem, D is the total spacetime dimension of a neutral Kaluza–Klein black string in vacuum Einstein gravity and is treated as a variable parameter. In the physical matrix-model realization considered here, the dual black objects instead live in the fixed ten-dimensional bulk spacetime of type-II string theory.

The parameter d counts the adjoint matrix coordinates of the D0-brane constituents. For the maximally supersymmetric system,

$$D_{\text{bulk}} = 10, \quad d = 9, \quad (5.47)$$

and the nine matrices describe the nine spatial directions transverse to the D0-brane worldline. Equivalently, the parent D1-brane theory has eight transverse scalar matrices; after T-duality along its spatial circle, that circle becomes the ninth transverse D0-brane coordinate [34, 66].

The uniform, non-uniform, and localized phases are therefore phases of ten-dimensional charged and dilatonic black D0-brane geometries, while varying d defines a generalized family of matrix models. Except for the physical value $d = 9$, this variation should not be interpreted as varying the spacetime dimension of a fixed type-II gravitational theory. The comparison with the Sorkin critical dimension is consequently structural, rather than a numerical identification of D with d .

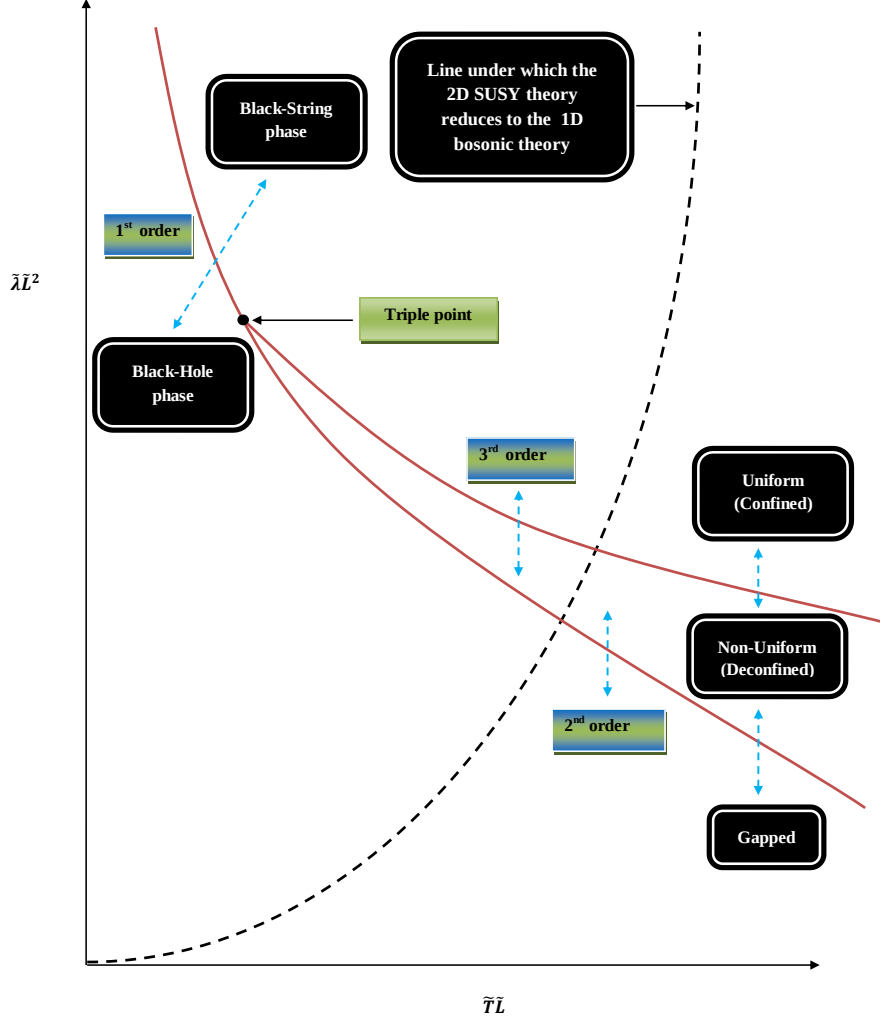


Figure 1: Schematic phase diagram of thermal 1 + 1-dimensional maximally supersymmetric Yang–Mills theory on a spatial circle and its high–temperature reduction to the one–dimensional bosonic matrix model. At low temperature, the uniform black–string and localized black–hole phases are separated by a first–order transition. At the putative multicritical point, this transition splits into a third–order uniform/non–uniform transition and a second–order non–uniform/gapped transition. The dashed curve indicates the boundary below which dimensional reduction to the one–dimensional bosonic theory is valid. The figure is schematic and is adapted from Refs. [34, 51, 66].

6 Yang–Mills observable

6.1 Large- d saddle value of the Yang–Mills observable

Recall that the bosonic Yang–Mills observable is defined by

$$\left\langle \frac{N}{4} \int_0^\beta dt \operatorname{Tr}[X_a, X_b]^2 \right\rangle = \left. \frac{\partial \ln Z(\alpha)}{\partial \alpha} \right|_{\alpha=1} = -\beta \left. \frac{\partial F(\alpha)}{\partial \alpha} \right|_{\alpha=1}. \quad (6.1)$$

At leading order in $1/d$ we evaluate the partition function by the maximally symmetric saddle $k_{\mu\rho}(t) = k_{0\mu\rho} = 2k_0 \delta_{\mu\rho}$, so that the localized theory reduces to the gauged matrix harmonic oscillator with

$$s^2 \equiv m_{\text{eff}} = m + k_0. \quad (6.2)$$

The corresponding large- d effective action (before integrating over θ) is

$$S_{\text{eff}}(\theta; \alpha) = S_{\text{Vdm}}[\theta] + S_{\text{MHO}}(\theta; s) + S_k[k_0] + \dots, \quad (6.3)$$

where S_{Vdm} is the Vandermonde contribution, $S_{\text{MHO}}(\theta; s)$ is the Gaussian determinant of X_a , and the dots denote subleading corrections (including the δk determinant, discussed below).

Using $\mu_{\mu\rho\nu\sigma} \delta_{\nu\sigma} = -(8N)^{-1} \delta_{\mu\rho}$ one finds

$$S_k[k_0] = \frac{N}{4\alpha} \int_0^\beta dt \mu_{\mu\rho\nu\sigma} k_{0\mu\rho} k_{0\nu\sigma} = -\frac{\beta}{8\alpha} (N^2 - 1) k_0^2. \quad (6.4)$$

Thus

$$S_{\text{eff}}(\theta; \alpha) = S_{\text{Vdm}}[\theta] + S_{\text{MHO}}(\theta; s) - \frac{\beta}{8\alpha} (N^2 - 1) k_0^2 + \dots. \quad (6.5)$$

At leading order (large- d saddle), we approximate the full free energy by

$$F(\theta; \alpha) \equiv \frac{1}{\beta} S_{\text{eff}}(\theta; \alpha) \simeq F_{\text{Vdm}}(\theta) + F_{\text{MHO}}(\theta; s) - \frac{N^2 - 1}{8\alpha} k_0^2, \quad (6.6)$$

where $k_0 \equiv k_0(\theta; \alpha)$ is fixed by the gap equation, i.e. by the stationarity of F with respect to k_0 :

$$0 = \frac{\partial F}{\partial k_0} = \frac{\partial F_{\text{MHO}}}{\partial s^2} \frac{\partial s^2}{\partial k_0} - \frac{N^2 - 1}{4\alpha} k_0 = \frac{\partial F_{\text{MHO}}}{\partial s^2} - \frac{N^2 - 1}{4\alpha} k_0. \quad (6.7)$$

(Here $\partial s^2 / \partial k_0 = 1$.)

Now differentiate the saddle free energy $F(\alpha) = F(\theta; \alpha, k_0(\alpha))$ with respect to α :

$$\frac{dF}{d\alpha} = \left. \frac{\partial F}{\partial \alpha} \right|_{k_0} + \left. \frac{\partial F}{\partial k_0} \right|_{\alpha} \frac{dk_0}{d\alpha}. \quad (6.8)$$

The key point is that the second term vanishes *at the saddle* by (6.7):

$$\left. \frac{\partial F}{\partial k_0} \right|_{\alpha} = 0 \quad \implies \quad \frac{dF}{d\alpha} = \left. \frac{\partial F}{\partial \alpha} \right|_{k_0}. \quad (6.9)$$

This is the envelope theorem in this context: *implicit* α -dependence through the optimized parameter $k_0(\alpha)$ does not contribute at leading saddle order.

Differentiating then (6.6) with respect to α at fixed saddle k_0 (envelope theorem at the stationary point of the k -effective potential) yields

$$-\beta \frac{\partial F}{\partial \alpha} \simeq -\beta \frac{\partial}{\partial \alpha} \left(-\frac{N^2-1}{8\alpha} k_0^2 \right) = -\frac{\beta}{8\alpha^2} (N^2-1) k_0^2. \quad (6.10)$$

Therefore, at leading order in the large- d saddle, the Yang-Mills observable becomes

$$\left\langle \frac{N}{4} \int_0^\beta dt \operatorname{Tr}[X_a, X_b]^2 \right\rangle = -\beta \left. \frac{dF}{d\alpha} \right|_{\alpha=1} \simeq -\frac{\beta}{8} (N^2-1) k_0^2, \quad (6.11)$$

with k_0 evaluated at the $\alpha = 1$ saddle. This is manifestly negative, consistent with $\operatorname{Tr}[X_a, X_b]^2 \leq 0$ for Hermitian X_a .

For completeness, let us also include the contribution of fluctuations around the maximally symmetric saddle. The quadratic fluctuation of the auxiliary field is

$$S_{\delta k}^{(2)}[\delta k] = \frac{N}{4\alpha} \int_0^\beta dt \mu_{\mu\rho\nu\sigma} \delta k_{\mu\rho}(t) \delta k_{\nu\sigma}(t). \quad (6.12)$$

Restricting, for simplicity, to the zero mode

$$\delta k_{\mu\rho}(t) = \delta k_{\mu\rho}, \quad (6.13)$$

one obtains

$$S_{\delta k,0}^{(2)} = \frac{N\beta}{4\alpha} \mu_{\mu\rho\nu\sigma} \delta k_{\mu\rho} \delta k_{\nu\sigma}. \quad (6.14)$$

Thus the zero-mode fluctuation produces only a Gaussian determinant. Indeed, the Gaussian integral gives

$$Z_{\delta k,0} \propto \left[\det_{\text{sym}} \left(\frac{N\beta}{2\alpha} \mu \right) \right]^{-1/2}, \quad (6.15)$$

where $\det_{\text{sym}}$ is taken over symmetric pairs $(\mu\rho)$ of dimension $D_{\text{sym}} = \frac{(N^2-1)N^2}{2}$. Hence

$$\log Z_{\delta k,0} = -\frac{1}{2} \log \det_{\text{sym}} \mu - \frac{D_{\text{sym}}}{2} \log \left(\frac{N\beta}{2\alpha} \right) + \text{const}, \quad (6.16)$$

so the induced shift in the free energy is an α -dependent *constant*,

$$\Delta F_{\delta k,0} = -\frac{1}{\beta} \log Z_{\delta k,0} = \frac{1}{2\beta} \log \det_{\text{sym}} \mu + \frac{D_{\text{sym}}}{2\beta} \log \left(\frac{N\beta}{2\alpha} \right) + \text{const}, \quad \frac{\partial}{\partial \alpha} \log Z_{\delta k,0} = \frac{D_{\text{sym}}}{2\alpha}. \quad (6.17)$$

In a properly normalized Hubbard-Stratonovich representation this Gaussian factor is absorbed into the overall HS prefactor: it is independent of θ and s , and thus does not affect the holonomy saddle, the phase transition, or any θ -dependent observable at leading order.

6.2 The small- m and large- m behaviors

By expressing k_0 in terms of the gap s , the Yang–Mills observable takes the form

$$\left\langle \frac{N}{4} \int_0^\beta dt \operatorname{Tr}[X_a, X_b]^2 \right\rangle = -\frac{\beta}{8} (N^2 - 1) (s^2 - m)^2. \quad (6.18)$$

The small- m and large- m regimes are obtained from the $T = 0$ gap equation (zero-temperature saddle) which is given by (4.4).

(i) *Small mass:* $m \ll a^{2/3}$. With $\varepsilon \equiv m/a^{2/3} \ll 1$ one has

$$s_0 = a^{1/3} \left(1 + \frac{\varepsilon}{3} - \frac{\varepsilon^3}{81} + \frac{\varepsilon^4}{243} + \cdots \right), \quad a = \alpha d, \quad (6.19)$$

so

$$k_0 = s_0^2 - m = a^{2/3} \left(1 - \frac{\varepsilon}{3} + \frac{\varepsilon^2}{9} - \frac{2\varepsilon^3}{81} + \cdots \right), \quad (\varepsilon \ll 1), \quad (6.20)$$

and inserting into (6.18) gives

$$\left\langle \frac{N}{4} \int_0^\beta dt \operatorname{Tr}[X_a, X_b]^2 \right\rangle \simeq \frac{\beta}{8} (N^2 - 1) (\alpha d)^{4/3} \left(1 - \frac{2\varepsilon}{3} + \frac{\varepsilon^2}{3} + \cdots \right) \Big|_{\alpha=1}. \quad (6.21)$$

(ii) *Large mass:* $m \gg a^{2/3}$. The positive root admits

$$s_0 = \sqrt{m} + \frac{a}{2m} - \frac{3a^2}{8m^{5/2}} + \cdots, \quad (m \gg a^{2/3}), \quad (6.22)$$

hence

$$k_0 = s_0^2 - m = \frac{a}{\sqrt{m}} - \frac{a^2}{2m^2} + \cdots, \quad (m \gg a^{2/3}), \quad (6.23)$$

and therefore

$$\left\langle \frac{N}{4} \int_0^\beta dt \operatorname{Tr}[X_a, X_b]^2 \right\rangle \simeq \frac{\beta}{8} (N^2 - 1) \left(\frac{(\alpha d)^2}{m} + \mathcal{O}(m^{-5/2}) \right) \Big|_{\alpha=1}. \quad (6.24)$$

Large positive m confines the matrices close to the origin, strongly suppressing commutators and hence $\operatorname{Tr}[X_a, X_b]^2$. In this regime the quadratic term dominates, and the theory is truly Gaussian. By contrast, at small m the quartic interaction dynamically generates a mass gap $k_0 \sim d^{2/3}$, making the Yang–Mills observable parametrically large at large d . The theory is then intrinsically non-perturbative, yet its dynamics are accurately captured by the Gaussian structure of the maximally symmetric large- d saddle, which encodes the essential physics of the BFSS model.

Let us also recall that at finite T the holonomy dependence enters through the MHO determinant,

$$F_{\text{MHO}}(\theta; s) = \frac{d}{2\beta} \sum_{i,j} P_{ij,ji} \log \left(\cosh(\beta s) - \cos(\theta_i - \theta_j) \right) + \text{const}, \quad (6.25)$$

and the holonomy effective action is obtained by combining F_{MHO} with the Vandermonde term. One finds, at leading order, the total action (see Appendix A)

$$S_{\text{hol}}[\theta] = N^2 \sum_{n \geq 1} \frac{1 - dq^n}{n} |u_n|^2 + \text{const}, \quad q = e^{-\beta s}, \quad u_n = \frac{1}{N} \text{Tr } U^n. \quad (6.26)$$

Thus the phase structure is controlled by dq :

$$dq < 1 : \quad u_n = 0 \quad \Rightarrow \quad \rho(\theta) = \frac{1}{2\pi} \quad (\text{uniform / confined}), \quad (6.27)$$

whereas for $dq > 1$ the uniform saddle is unstable and u_1 condenses, eventually yielding a gapped (clumped) distribution.

Across the transition, the holonomy changes from the uniform saddle to a non-uniform, eventually clumped, saddle. Equivalently, the Polyakov moments change from

$$u_n = 0 \quad (\text{uniform / confined}) \quad (6.28)$$

to

$$u_n \neq 0 \quad (\text{non-uniform / deconfined}). \quad (6.29)$$

In the gap equation this dependence enters through the projector-weighted holonomy moments

$$\mathcal{C}_p(U) = \frac{|\text{Tr } U^p|^2 - 1}{N^2 - 1} = \frac{N^2 |u_p|^2 - 1}{N^2 - 1}, \quad (6.30)$$

or equivalently through the kernel $\mathcal{R}(s; U)$ in (4.22)–(4.23). Therefore (6.18) should be understood as

$$\left\langle \frac{N}{4} \int_0^\beta dt \text{Tr} [X_a, X_b]^2 \right\rangle \simeq -\frac{\beta}{8} (N^2 - 1) \left(s^2(T; U) - m \right)^2. \quad (6.31)$$

Here $s(T; U)$ is evaluated in the dominant holonomy saddle: uniform in the confined phase and non-uniform in the deconfined phase. In practice, at low temperature the uniform saddle gives $s \simeq s_0$, up to exponentially small thermal corrections, so the small- m and large- m formulas (6.21) and (6.24) apply directly. In the deconfined phase, the condensed moments u_n , or equivalently the nonzero $\mathcal{C}_p(U)$, correct the gap and hence the Yang–Mills observable.

7 Double-scaling saddle and high-temperature branch

7.1 The Yang–Mills–mass-deformation balance

At the large- d saddle, the quartic interaction is encoded entirely in the dynamical mass shift k_0 , or equivalently in

$$s^2 = m + k_0. \quad (7.1)$$

The Yang–Mills and quadratic contributions to the energy density scale parametrically as

$$\mathcal{O}_{\text{YM}} \equiv \frac{1}{N\beta} \left\langle \int_0^\beta dt \frac{N}{4} \text{Tr}[X_a, X_b]^2 \right\rangle \sim -N k_0^2, \quad (7.2)$$

$$\mathcal{O}_{\text{quad}} \equiv \frac{1}{N\beta} \left\langle \int_0^\beta dt \frac{Nm}{2} \text{Tr} X_a^2 \right\rangle \sim Nm R^2 \sim Nm \frac{d}{2s}. \quad (7.3)$$

Our goal is to identify a regime in which the quartic interaction is neither negligible nor parametrically dominant over the quadratic term. Schematically, this “sweet spot” is defined by

$$\mathcal{O}_{\text{YM}} \sim \mathcal{O}_{\text{quad}}. \quad (7.4)$$

This condition precisely marks the crossover between the interaction-dominated regime, $m \ll (\alpha d)^{2/3}$, and the mass-dominated regime, $m \gg (\alpha d)^{2/3}$.

It therefore suggests choosing the mass deformation to scale as

$$m \sim d^{2/3}. \quad (7.5)$$

In this regime:

1. The gap s is enhanced, and hence the deconfinement instability is pushed to higher temperatures, even for moderate fixed d , such as $d = 2$ in BFSS₃. Equivalently, the uniform holonomy regime is enlarged. In this sense the theory behaves, over a parametrically wider range, as d effectively decoupled BFSS₂-type gauged matrix harmonic oscillators, since the holonomy-mediated interaction between the different matrix directions becomes weak at large d .
2. The Yang–Mills observable remains sizable, but no longer dominates parametrically over the quadratic term. Thus the double-scaling limit interpolates smoothly between the interaction-dominated regime and the mass-dominated Gaussian regime. In this sense the original BFSS _{$d+1$} interaction is retained self-consistently in the Gaussian saddle through the dynamically generated mass shift k_0 .

The BFSS₂-type factorization described above should therefore be understood as an effective bulk approximation. The endpoint dynamics and the full holonomy structure still retain information about the original BFSS _{$d+1$} theory.

3. In this double-scaling limit, the theory is therefore neither reduced to a purely free massive model nor driven by uncontrolled strong commutator dynamics. The matrices are in fact localized near the origin and weakly non-commuting: the commutator contribution per matrix pair is parametrically suppressed. This is the IKKT-like aspect of the double-scaled saddle, reproducing the characteristic Yang-Mills, or almost-commuting, phase familiar from the IKKT reduction.

We now impose the double-scaling condition

$$\kappa \equiv \frac{m^{3/2}}{\alpha d} \iff m = (\kappa \alpha d)^{2/3}, \quad (7.6)$$

and solve the $T = 0$ gap equation (4.4). Since $m \sim (\alpha d)^{2/3}$ in this limit, it is natural to scale

$$s_0 = (\alpha d)^{1/3} y, \quad m = (\alpha d)^{2/3} \kappa^{2/3}. \quad (7.7)$$

Substituting into (4.4) gives the *exact* reduced cubic equation for y ,

$$y^3 - \kappa^{2/3} y - 1 = 0, \quad (7.8)$$

which is independent of d . Hence the double-scaled saddle is

$$s(\kappa) = (\alpha d)^{1/3} y(\kappa), \quad k_0(\kappa) = s^2(\kappa) - m = (\alpha d)^{2/3} (y(\kappa)^2 - \kappa^{2/3}). \quad (7.9)$$

Writing (7.8) as $y^3 + py + q = 0$ with $p = -\kappa^{2/3}$ and $q = -1$, the discriminant is

$$\Delta_\kappa = \left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3 = \frac{1}{4} - \frac{\kappa^2}{27}. \quad (7.10)$$

A convenient real root is

$$y(\kappa) = \left(\frac{1}{2} + \sqrt{\Delta_\kappa}\right)^{1/3} + \left(\frac{1}{2} - \sqrt{\Delta_\kappa}\right)^{1/3}. \quad (7.11)$$

7.2 Observables at fixed large κ : nearly commuting, localized matrices

The critical temperature, extent of space and Yang-Mills observable are given respectively by

$$T_c(\kappa) = \frac{s(\kappa)}{\log d} = \frac{(\alpha d)^{1/3}}{\log d} y(\kappa). \quad (7.12)$$

$$R^2(\kappa) = \left(1 - \frac{1}{N^2}\right) \frac{d}{2s(\kappa)} + \mathcal{O}(e^{-\beta s}) = \left(1 - \frac{1}{N^2}\right) \frac{(\alpha d)^{2/3}}{2\alpha y(\kappa)} + \mathcal{O}(e^{-\beta d^{1/3} y}). \quad (7.13)$$

$$\left\langle -\frac{N}{4} \int_0^\beta dt \operatorname{Tr}[X_a, X_b]^2 \right\rangle_\kappa = \frac{N^2 - 1}{8\alpha^2} k_0(\kappa)^2 = \frac{N^2 - 1}{8\alpha^2} (\alpha d)^{4/3} \left(y(\kappa)^2 - \kappa^{2/3} \right)^2. \quad (7.14)$$

The entire dependence on the double-scaling parameter is encoded in the single dimensionless function $y(\kappa)$ solving (7.8), and the leading large- d scalings are

$$s \sim d^{1/3}, \quad (7.15)$$

and

$$T_c \sim \frac{d^{1/3}}{\log d}, \quad R^2 \sim d^{2/3}, \quad \langle -\operatorname{Tr}[X, X]^2 \rangle \sim d^{4/3}, \quad (7.16)$$

with κ -dependent prefactors given explicitly by Eqs. (7.12), (7.13), and (7.14), where the function $y(\kappa)$ is determined in closed form by the Cardano solution (7.11). In particular, for sufficiently large fixed κ one has the asymptotic behavior

$$y(\kappa) = \kappa^{1/3} + \frac{1}{2\kappa^{2/3}} + \mathcal{O}(\kappa^{-5/3}), \quad (\kappa \rightarrow \infty), \quad (7.17)$$

so that $y(\kappa)$ is large and close to $\kappa^{1/3}$. Consequently, the gap $s \propto d^{1/3}y(\kappa)$ increases with κ , which drives the critical temperature $T_c \propto s/\log d$ upward even for moderately small d .

In the double-scaling limit (we set $\alpha = 1$)

$$\kappa \equiv \frac{m^{3/2}}{d} = \text{fixed}, \quad m \sim d^{2/3}, \quad s \sim d^{1/3}, \quad k_0 = s^2 - m \sim d^{2/3}. \quad (7.18)$$

Thus, the quadratic and quartic terms in the BFSS action are indeed parametrically of the *same* order at large d , which means that this regime interpolates between the interaction-dominated and mass-dominated saddles and realizes the desired “sweet spot” discussed above.

The structure emerging in this limit is that of *localized, almost commuting* matrices. To make this explicit, it is convenient to pass from observables summed over a, b to quantities defined per matrix and per matrix pair, using rotational invariance in the $SO(d)$ index a .

By $SO(d)$ symmetry, each matrix contributes equally to the extent of space. Thus

$$\frac{1}{N\beta} \left\langle \int_0^\beta dt \operatorname{Tr} X_1^2 \right\rangle = \frac{R^2}{d} = \left(1 - \frac{1}{N^2} \right) \frac{1}{2s} + \mathcal{O}(e^{-\beta s}) \sim d^{-1/3}. \quad (7.19)$$

Each individual matrix X_a is therefore localized near the origin, with a typical width that shrinks as $d^{-1/3}$, even though the total extent grows as $R^2 \sim d^{2/3}$ due to the sum over d matrices.

Similarly, the Yang–Mills observable sums over $\sim d^2$ ordered pairs (a, b) . By $SO(d)$ invariance, the typical contribution per pair scales as

$$-\frac{1}{N\beta} \left\langle \int_0^\beta dt \operatorname{Tr}[X_1, X_2]^2 \right\rangle \sim -\frac{1}{d^2} \frac{1}{N\beta} \left\langle \int_0^\beta dt \operatorname{Tr}[X_a, X_b]^2 \right\rangle \sim d^{-2/3}. \quad (7.20)$$

Hence

$$-\left\langle \text{Tr}[X_1, X_2]^2 \right\rangle \propto d^{-2/3} \longrightarrow 0 \quad (d \rightarrow \infty), \quad (7.21)$$

so the matrices become asymptotically commuting in the large- d limit, despite the presence of a non-trivial self-consistent mass gap.

Taken together, Eqs. (7.19) and (7.21) describe a regime of *nearly commuting, localized matrices*: each X_a is clumped near the origin, while commutators are parametrically suppressed. This is precisely the qualitative behavior expected in the Yang–Mills (“commuting”) phase of the IKKT reduction, where geometry is encoded in an almost diagonal matrix configuration with small, controlled noncommutative fluctuations generated here by the double-scaled saddle.

7.3 The high-temperature limit

At sufficiently high temperature the theory is in the deconfined phase, and the holonomy eigenvalues cluster so that $\phi = \theta_i - \theta_j \simeq 0$. The exact gap equation

$$\frac{d}{s} \frac{\sinh(\beta s)}{\cosh(\beta s) - \cos \phi} = \frac{s^2 - m}{\alpha}, \quad s = \sqrt{\tilde{k}_0} > 0, \quad (7.22)$$

reduces to

$$\frac{d}{s} \frac{\sinh(\beta s)}{\cosh(\beta s) - 1} = \frac{d}{s} \coth\left(\frac{\beta s}{2}\right) = \frac{s^2 - m}{\alpha}. \quad (7.23)$$

For $\beta \rightarrow 0$ one has $\coth(\beta s/2) = 2/(\beta s) + \mathcal{O}(\beta s)$, and keeping the leading term gives

$$\frac{2d}{\beta s^2} = \frac{s^2 - m}{\alpha}. \quad (7.24)$$

Writing $y = s^2$, this yields a quadratic equation

$$y(y - m) = \frac{2\alpha d}{\beta}, \quad (7.25)$$

whose physical (positive) solution is

$$s^2 = \frac{1}{2} \left(m + \sqrt{m^2 + \frac{8\alpha d}{\beta}} \right), \quad k_0 = s^2 - m = \frac{1}{2} \left(-m + \sqrt{m^2 + \frac{8\alpha d}{\beta}} \right). \quad (7.26)$$

In the strict high-temperature limit $\beta \rightarrow 0$ this behaves as

$$s^2 \sim \sqrt{\frac{2\alpha d}{\beta}}, \quad s \sim \left(\frac{2\alpha d}{\beta} \right)^{1/4}, \quad k_0 \sim \sqrt{\frac{2\alpha d}{\beta}}. \quad (7.27)$$

In this regime the saddle point remains Gaussian, and the adjoint matrices are again governed by a matrix harmonic oscillator (M.H.O.) with a self-consistent frequency $s(\beta)$ determined

by (7.26). This contrasts with the low-temperature (uniform-holonomy) branch, where the M.H.O. frequency saturates to a temperature-independent value s_0 determined by the large- d gap equation $s_0^3 - ms_0 = \alpha d$, implying

$$m_{\text{eff}} \equiv s_0 \sim (\alpha d)^{1/3} \quad (T \rightarrow 0). \quad (7.28)$$

By contrast, the present high-temperature (deconfined) branch exhibits a genuinely temperature-dependent mass scale. In particular,

$$m_{\text{eff}}^2 \equiv s^2 \sim \sqrt{\frac{2\alpha d}{\beta}} \propto \sqrt{\alpha d T}, \quad m_{\text{eff}} \equiv s \sim (2\alpha d T)^{1/4} \quad (T \rightarrow \infty). \quad (7.29)$$

Thus, while both phases are described by the same M.H.O. structure, the low- T dynamics is controlled by a T -independent mass scale set by d (and m), whereas the high- T phase exhibits a qualitatively new scaling in which the oscillator frequency grows also with temperature.

In the M.H.O. approximation, integrating out d adjoint bosons of frequency s generates for the first holonomy Fourier mode u_1 a quadratic coefficient of the schematic form

$$a_1(\beta) = 1 - d e^{-\beta s(\beta)} + \dots. \quad (7.30)$$

Thus the confinement/deconfinement transition is still controlled by the condition

$$a_1(\beta_c) = 0 \quad \implies \quad d e^{-\beta_c s(\beta_c)} = 1 \quad \implies \quad \beta_c s(\beta_c) = \ln d. \quad (7.31)$$

This is the same criticality condition as in the low-temperature (uniform-holonomy) phase; however, while at low T the frequency s approaches a constant $s_0 \sim (\alpha d)^{1/3}$, here s itself depends on temperature through the high- T saddle. Using the high- T scaling $s(\beta) \sim (2\alpha d/\beta)^{1/4}$ from (7.27), the criticality condition becomes

$$\beta_c \left(\frac{2\alpha d}{\beta_c} \right)^{1/4} = \ln d \quad \implies \quad (2\alpha d)^{1/4} \beta_c^{3/4} = \ln d, \quad (7.32)$$

hence

$$\beta_c \sim \frac{(\ln d)^{4/3}}{(2\alpha d)^{1/3}}, \quad \implies \quad T_c^{\text{high}} = \frac{1}{\beta_c} \sim \frac{(2\alpha d)^{1/3}}{(\ln d)^{4/3}}. \quad (7.33)$$

Therefore, although the M.H.O. structure and the condition $\beta_c s(\beta_c) = \ln d$ are unchanged, the temperature dependence of the effective mass scale on the high- T branch leads to a parametrically different scaling of T_c compared to the low- T result

$$T_c^{\text{low}} \sim \frac{s_0}{\ln d} \sim \frac{(\alpha d)^{1/3}}{\ln d}. \quad (7.34)$$

In summary, the low-temperature analysis dictates a regime of validity for $T < T_c^{\text{low}}$, where the M.H.O. frequency saturates to the temperature-independent value $s_0 \sim (\alpha d)^{1/3}$. Conversely,

the high-temperature analysis applies for $T > T_c^{\text{high}}$, where the effective frequency grows with temperature as $s(\beta) \sim (2\alpha d/\beta)^{1/4}$. Since parametrically $T_c^{\text{high}} < T_c^{\text{low}}$, there exists an intermediate window

$$T_c^{\text{high}} < T < T_c^{\text{low}}, \quad (7.35)$$

in which both approximations overlap. In this regime the dynamics is consistently governed by a Gaussian saddle with a self-consistent mass, and is therefore dominated by approximately commuting matrices.

This can be seen more clearly by analyzing basic observables at the large- d Gaussian (M.H.O.) saddle, using their exact saddle-point expressions and comparing the low- and high-temperature branches.

At the large- d saddle point, the extent of space and the Yang-Mills term are given respectively by

$$R^2 \equiv \frac{1}{N\beta} \left\langle \int_0^\beta dt \text{Tr} X_a^2 \right\rangle = \frac{d}{2s}, \quad (7.36)$$

$$\left\langle \frac{N}{4} \int_0^\beta dt \text{Tr} [X_a, X_b]^2 \right\rangle = -\frac{N^2 - 1}{8\alpha^2} k_0^2, \quad k_0 = s^2 - m. \quad (7.37)$$

On the high-temperature (deconfined) branch one has $s \sim (2\alpha d T)^{1/4}$ and $k_0 \sim \sqrt{2\alpha d T}$, which implies

$$\frac{R^2}{d} \sim (2\alpha d T)^{-1/4}, \quad \frac{1}{d^2} \left\langle \frac{N}{4} \int_0^\beta dt \text{Tr} [X_a, X_b]^2 \right\rangle \sim -T(\alpha d)^{-1}. \quad (7.38)$$

At low temperature, the M.H.O. frequency saturates to $s_0 \sim (\alpha d)^{1/3}$ with $k_0 \sim (\alpha d)^{2/3}$, yielding

$$\frac{R^2}{d} \sim (\alpha d)^{-1/3}, \quad \frac{1}{d^2} \left\langle \frac{N}{4} \int_0^\beta dt \text{Tr} [X_a, X_b]^2 \right\rangle \sim -(\alpha d)^{-2/3}. \quad (7.39)$$

Thus, in both the low- T and high- T limits, the basic objects per matrix and per matrix pair are parametrically suppressed at large d (and, on the high-temperature branch, also with increasing T).

These results imply that, in both limits, the dynamics is dominated by approximately commuting matrices. Combined with the existence of the overlap window $T_c^{\text{high}} < T < T_c^{\text{low}}$, this establishes an extended temperature range in which the Gaussian/M.H.O. description is self-consistent and commuting-matrix configurations control the dynamics. Importantly, this window widens with increasing d , providing a clear signal for the emergence of IKKT-type physics in the large- d limit.

7.4 Size of the auxiliary-field fluctuations at high temperature

We now verify directly that the matrix harmonic oscillator description remains valid on the high-temperature branch. As in Sec. 4.3, the quadratic kernel governing the auxiliary-field fluctuations contains two contributions: the bare Hubbard–Stratonovich kernel and the polarization kernel induced by the connected fluctuations of the coordinate matrices.

Consider a time-independent fluctuation in the maximally symmetric channel,

$$k = k_0 + \delta k, \quad s^2 = m + k_0. \quad (7.40)$$

Suppressing the holonomy shifts and the common adjoint multiplicity, the effective action in this channel takes the schematic form

$$S_{\text{eff}}(k) = -\frac{\beta}{8\alpha}k^2 + \frac{d}{2} \sum_{n \in \mathbb{Z}} \log(\omega_n^2 + m + k). \quad (7.41)$$

The term linear in δk vanishes by the finite-temperature auxiliary saddle equation

$$-\frac{\beta k_0}{4\alpha} + \frac{d}{2} \sum_{n \in \mathbb{Z}} \frac{1}{\omega_n^2 + s^2} = 0 \quad \Longleftrightarrow \quad \frac{k_0}{\alpha} = \frac{d}{s} \coth\left(\frac{\beta s}{2}\right). \quad (7.42)$$

The quadratic term generated by the coordinate determinant is governed by the static polarization sum which at high temperature is given by

$$\begin{aligned} \sum_{n \in \mathbb{Z}} \frac{1}{(\omega_n^2 + s^2)^2} &= \frac{\beta}{4s^3} \coth\left(\frac{\beta s}{2}\right) + \frac{\beta^2}{8s^2} \text{csch}^2\left(\frac{\beta s}{2}\right) \\ &= \frac{1}{s^4} [1 + \mathcal{O}((\beta s)^4)], \quad \beta s \ll 1. \end{aligned} \quad (7.43)$$

The coordinate contribution to a static eigenvalue of the auxiliary Hessian therefore scales as

$$\Gamma_{\text{stat}, X}^{(2)} \sim \frac{d}{s^4}, \quad (7.44)$$

up to its numerical coefficient, sign, and adjoint tensor eigenvalue.

The bare contribution is proportional to β/α . Expanding the finite-temperature saddle equation for high temperature $\beta s \ll 1$, and using $k_0 = s^2 - m$, gives

$$\begin{aligned} \frac{\beta}{\alpha} &= \frac{2d}{s^2(s^2 - m)} [1 + \mathcal{O}((\beta s)^2)] \\ &= \frac{2d}{s^4} \frac{s^2}{s^2 - m} [1 + \mathcal{O}((\beta s)^2)]. \end{aligned} \quad (7.45)$$

Thus, provided $s^2 - m$ is of order s^2 , the bare and polarization contributions have the same parametric scaling. For a stable eigenmode of the complete tensor kernel, understood along the appropriate steepest-descent contour, one may therefore write

$$\Gamma_{\text{stat}}^{(2)} \sim \frac{d}{s^4} \mathcal{C}_{\text{HT}}\left(\frac{m}{s^2}\right) [1 + \mathcal{O}((\beta s)^2)], \quad (7.46)$$

where $\mathcal{C}_{\text{HT}}$ combines the bare and coordinate polarization contributions.

Provided that $\mathcal{C}_{\text{HT}} \neq 0$ and that no parametrically soft auxiliary mode is present, the variance of the corresponding static fluctuation is controlled parametrically by the inverse Hessian:

$$\langle (\delta k)^2 \rangle \sim \frac{s^4}{d} \frac{1}{\mathcal{C}_{\text{HT}}(m/s^2)}. \quad (7.47)$$

Consequently,

$$\frac{\sqrt{\langle (\delta k)^2 \rangle}}{s^2} \sim \frac{1}{\sqrt{d}} \frac{1}{\sqrt{\mathcal{C}_{\text{HT}}(m/s^2)}}. \quad (7.48)$$

The auxiliary-field fluctuations on the high-temperature branch are therefore suppressed as $d^{-1/2}$. Unlike the low-temperature regime, there is no additional suppression by βs , since the static Matsubara mode dominates the high-temperature polarization. The large- d limit nevertheless provides the required control of the Gaussian saddle, while auxiliary-field fluctuations generate the subleading non-Gaussian corrections to the matrix harmonic oscillator description.

Thus the matrix harmonic oscillator description remains valid on the high-temperature branch at leading order in the large- d expansion. In contrast with the low-temperature regime, high temperature does not provide an additional suppression of the auxiliary-field fluctuations; rather, their relative size remains of order $d^{-1/2}$. The control of the Gaussian saddle at high temperature is therefore supplied by the large number of coordinate matrices, while fluctuations of the auxiliary field generate the subleading non-Gaussian corrections.

8 On supersymmetric extensions and Molien–Weyl approximations

8.1 Saddle point equation with split masses

In many supersymmetric BFSS $_{d+1}$ models with mass deformations (BMN $_{d+1}$), the bosonic mass term breaks the $SO(d)$ rotational symmetry to $SO(n_1) \times SO(n_2)$ with $n_1 + n_2 = d$. The quadratic potential then takes the form

$$m_1 \sum_{a_1=1}^{n_1} \text{Tr} X_{a_1}^2 + m_2 \sum_{a_2=1}^{n_2} \text{Tr} X_{a_2}^2. \quad (8.1)$$

Let us then split the transverse index as $a = (a_1, a_2)$ with

$$a_1 = 1, \dots, n_1, \quad a_2 = 1, \dots, n_2, \quad n_1 + n_2 = d, \quad (8.2)$$

and assign masses m_1 and m_2 to the two flavors.

The Hubbard–Stratonovich localization proceeds exactly as in the $SO(d)$ –symmetric case; the only modification is that, after integrating out the bosonic matrices, the one–loop determinants appear with multiplicities n_1 and n_2 .

Indeed, in the constant- k sector, after integrating out all bosonic matrices, we obtain the effective action

$$S_{\text{eff}}[k] = \frac{n_1}{2} \sum_{n \in \mathbb{Z}} \text{Trlog}(P\mathcal{W}^{(1)}(n)P) + \frac{n_2}{2} \sum_{n \in \mathbb{Z}} \text{Trlog}(P\mathcal{W}^{(2)}(n)P) + \frac{N\beta}{4\alpha} \mu_{\mu\rho\nu\sigma} k_{\mu\rho} k_{\nu\sigma}, \quad (8.3)$$

with

$$\mathcal{W}_{ij,kl}^{(s)}(n) = \left(\omega_n + \frac{\theta_i - \theta_j}{\beta} \right)^2 P_{ij,kl} + m_s P_{ij,kl} + k_{ij,kl}, \quad s = 1, 2. \quad (8.4)$$

Imposing the commuting–symmetric ansatz

$$k_{ij,kl} \equiv k_{ij} P_{ij,kl} = P_{ij,kl} k_{kl}, \quad (8.5)$$

the quadratic kernels reduce to

$$\mathcal{W}^{(s)}(n)_{ij,kl} = \Delta_{ij}^{(s)}(n) P_{ij,kl} = P_{ij,kl} \Delta_{kl}^{(s)}(n), \quad (8.6)$$

$$\Delta_{ij}^{(s)}(n) \equiv \left(\omega_n + \frac{\theta_i - \theta_j}{\beta} \right)^2 + m_s + k_{ij}, \quad s = 1, 2. \quad (8.7)$$

Therefore the effective action simplifies exactly as before to

$$S_{\text{eff}}[k] = \frac{1}{2} \sum_{s=1}^2 n_s \sum_{n \in \mathbb{Z}} \sum_{i,j=1}^N P_{ij,ji} \log(\Delta_{ij}^{(s)}(n)) + \frac{N\beta}{4\alpha} \mu_{\mu\rho\nu\sigma} k_{\mu\rho} k_{\nu\sigma}. \quad (8.8)$$

Varying now (8.8) with respect to the variables k_{ij} gives

$$0 = \frac{\partial S_{\text{eff}}}{\partial k_{ij}} = \frac{1}{2} \sum_{s=1}^2 n_s \sum_{n \in \mathbb{Z}} P_{ij,ji} \frac{1}{\Delta_{ij}^{(s)}(n)} + \frac{N\beta}{2\alpha} \mu_{\mu\rho\nu\sigma} k_{\nu\sigma} \frac{\partial k_{\mu\rho}}{\partial k_{ij}}, \quad (8.9)$$

which simplifies, following the same steps as before, to

$$0 = \frac{1}{2} \sum_{s=1}^2 n_s \sum_{n \in \mathbb{Z}} P_{ij,ji} \frac{1}{\Delta_{ij}^{(s)}(n)} + \frac{N\beta}{2\alpha} \mu_{\mu\rho\nu\sigma} k_{\nu\sigma} \Lambda_{\mu}^{ij} \Lambda_{\rho}^{ji}. \quad (8.10)$$

Performing the Matsubara sums with

$$\tilde{k}_{ij}^{(s)} \equiv k_{ij} + m_s, \quad s = 1, 2, \quad (m_1 \text{ multiplicity } n_1, \quad m_2 \text{ multiplicity } n_2), \quad (8.11)$$

yields

$$\sum_{n \in \mathbb{Z}} \frac{1}{\left(\omega_n + \frac{\theta_i - \theta_j}{\beta}\right)^2 + \tilde{k}_{ij}^{(s)}} = \beta \frac{1}{2\sqrt{\tilde{k}_{ij}^{(s)}}} \frac{\sinh\left(\beta\sqrt{\tilde{k}_{ij}^{(s)}}\right)}{\cosh\left(\beta\sqrt{\tilde{k}_{ij}^{(s)}}\right) - \cos(\theta_i - \theta_j)}. \quad (8.12)$$

Substituting (8.12) into (8.10) gives the split-mass saddle equation:

$$0 = \frac{\beta}{4} P_{ij,ji} \sum_{s=1}^2 n_s \frac{1}{\sqrt{\tilde{k}_{ij}^{(s)}}} \frac{\sinh\left(\beta\sqrt{\tilde{k}_{ij}^{(s)}}\right)}{\cosh\left(\beta\sqrt{\tilde{k}_{ij}^{(s)}}\right) - \cos(\theta_i - \theta_j)} + \frac{N\beta}{2\alpha} \mu_{\mu\rho\nu\sigma} k_{\nu\sigma} \Lambda_{\mu}^{ij} \Lambda_{\rho}^{ji}. \quad (8.13)$$

Next, we impose the maximally symmetric ansatz:

$$k_{ij} = k_0 \quad \forall i, j \quad \implies \quad k_{ij,kl} = k_0 P_{ij,kl}. \quad (8.14)$$

With this maximally symmetric ansatz, the second term of (8.13) simplifies as before while the first term becomes a function of

$$\tilde{k}_{ij}^{(s)} \rightarrow \tilde{k}_0^{(s)} \equiv k_0 + m_s, \quad s_s \equiv \sqrt{\tilde{k}_0^{(s)}} = \sqrt{k_0 + m_s}. \quad (8.15)$$

Thus (8.13) reduces to

$$0 = \frac{\beta}{4} P_{ij,ji} \sum_{s=1}^2 n_s \frac{1}{s_s} \frac{\sinh(\beta s_s)}{\cosh(\beta s_s) - \cos \phi} - \frac{\beta k_0}{4\alpha} P_{ij,ji}, \quad \phi \equiv \theta_i - \theta_j. \quad (8.16)$$

Now sum over i, j and divide by $\sum_{i,j} P_{ij,ji} = N^2 - 1$. This gives the scalar maximally symmetric gap equation

$$\sum_{s=1}^2 n_s \frac{1}{s_s} \mathcal{R}(s_s; U) = \frac{k_0}{\alpha}, \quad s_s = \sqrt{k_0 + m_s}, \quad (8.17)$$

where the projector-weighted holonomy kernel is

$$\mathcal{R}(s_s; U) = \frac{1}{N^2 - 1} \sum_{i,j=1}^N P_{ij,ji} \frac{\sinh(\beta s_s)}{\cosh(\beta s_s) - \cos(\theta_i - \theta_j)}. \quad (8.18)$$

Equations (8.17) and (8.18) are the split-mass analogues of the original projector-averaged gap equation (4.22) and holonomy kernel (4.23). The crucial point is that it is a *sum of two thermal factors* with two different gaps s_1, s_2 .

The gap equation at low temperature is obtained following the same steps as before. We expand the T -dependent factor of (8.17) in the limit $\beta \rightarrow \infty$ as

$$\frac{\sinh(\beta s_s)}{\cosh(\beta s_s) - \cos \phi} = 1 + 2 \cos \phi e^{-\beta s_s} + \mathcal{O}(e^{-2\beta s_s}). \quad (8.19)$$

Insert (8.19) into (8.17):

$$\sum_{s=1}^2 n_s \frac{1}{s_s} \left[1 + 2 \cos \phi e^{-\beta s_s} + \mathcal{O}(e^{-2\beta s_s}) \right] = \frac{k_0}{\alpha}. \quad (8.20)$$

At strictly $T = 0$ (drop all $e^{-\beta s_s}$ terms) we get the *split-mass* $T = 0$ gap equation:

$$\frac{n_1}{s_1} + \frac{n_2}{s_2} = \frac{n_1}{\sqrt{k_0 + m_1}} + \frac{n_2}{\sqrt{k_0 + m_2}} = \frac{k_0}{\alpha}. \quad (8.21)$$

This replaces our cubic $s_0^3 - m s_0 = \alpha d$ and, in general, does *not* collapse to a single effective mass parameter.

The $T = 0$ maximally symmetric gap equation (8.21) can also be put in the following equivalent form

$$\frac{n_2}{\sqrt{s_1^2 + \Delta m}} = \frac{s_1^2 - m_1}{\alpha} - \frac{n_1}{s_1}, \quad \Delta m := m_2 - m_1. \quad (8.22)$$

Squaring once yields an eighth-degree polynomial equation for s_1 ,

$$(s_1^3 - m_1 s_1 - \alpha n_1)^2 (s_1^2 + \Delta m) - \alpha^2 n_2^2 s_1^2 = 0, \quad (8.23)$$

supplemented by the sign condition

$$\frac{s_1^2 - m_1}{\alpha} - \frac{n_1}{s_1} > 0, \quad (8.24)$$

which discards spurious solutions introduced by squaring.

Equivalently, the two-flavor saddle point equation, or more precisely the gap equation (8.21) or (8.23), is a *single* self-consistency condition for the Hubbard–Stratonovich mass parameter k_0 :

$$s_1^2 = m_1 + k_0, \quad s_2^2 = m_2 + k_0 \quad \implies \quad s_2^2 - s_1^2 = m_2 - m_1, \quad (8.25)$$

or, solving in terms of s_1 ,

$$k_0 = s_1^2 - m_1, \quad s_2 = \sqrt{s_1^2 + \Delta m}. \quad (8.26)$$

Thus the two gaps are not independent; both are determined by the same saddle k_0 . The cubic equation of the single-flavor case is simply replaced by an eighth-degree equation in the split-flavor case.

Now introduce the fractions $r_s = n_s/d$ (with $r_1 + r_2 = 1$) and the large- d scaling variables

$$s_* = (\alpha d)^{1/3}, \quad k_* = (\alpha d)^{2/3}. \quad (8.27)$$

Solving (8.21) at large d yields

$$k_0 = k_* - \frac{1}{3}\bar{m} + \mathcal{O}(d^{-2/3}), \quad \bar{m} = r_1 m_1 + r_2 m_2, \quad (8.28)$$

$$s_{1,0} = s_* + \frac{m_1}{2s_*} - \frac{\bar{m}}{6s_*} + \mathcal{O}(d^{-1}), \quad (8.29)$$

$$s_{2,0} = s_* + \frac{m_2}{2s_*} - \frac{\bar{m}}{6s_*} + \mathcal{O}(d^{-1}). \quad (8.30)$$

In particular,

$$s_{1,0} \sim s_{2,0} \sim (\alpha d)^{1/3}, \quad s_{2,0}^2 - s_{1,0}^2 = m_2 - m_1, \quad \frac{s_{2,0}^2 - s_{1,0}^2}{s_*^2} = \mathcal{O}(d^{-2/3}), \quad (8.31)$$

so the mass splitting is parametrically subleading in the large- d expansion. At leading order, both flavors share the same dynamically generated gap which swamps their bare masses.

Finally, by expanding the localized action about k_0 , the dominant bosonic theory is therefore a sum of two gauged matrix harmonic oscillators,

$$S_{\text{MHO}}^{\text{split}}[X; U] = N \int_0^\beta dt \text{Tr} \left[\frac{1}{2} \sum_{a=1}^d (D_t X_a)^2 + \frac{s_1^2}{2} \sum_{a_1=1}^{n_1} X_{a_1}^2 + \frac{s_2^2}{2} \sum_{a_2=1}^{n_2} X_{a_2}^2 \right]. \quad (8.32)$$

8.2 Gaussian Molien–Weyl approximation

The basic model of interest is the bosonic BFSS $_{d+1}$ theory with a mass deformation, namely the BMN $_{d+1}$ model, defined by

$$S = N \int_0^\beta dt \text{Tr} \left[\frac{1}{2} (D_t X_a)^2 + \frac{m}{2} X_a^2 - \frac{1}{4} [X_a, X_b]^2 \right]. \quad (8.33)$$

In the large- d limit, this interacting theory is well approximated by a gauged matrix harmonic oscillator,

$$S = N \int dt \text{Tr} \left[\frac{1}{2} (D_t X_a)^2 + \frac{1}{2} s^2 X_a^2 \right], \quad (8.34)$$

where the dynamically generated mass $s^2 = k_0 + m$ is determined by the cubic gap equation $s^3 - ms - \alpha d = 0$. Here, k_0 is the large- d saddle point.

In the presence of a split mass deformation, the quadratic and interaction terms are effectively replaced as

$$\frac{m_1}{2} \sum_{a_1=1}^{n_1} X_a^2 + \frac{m_2}{2} \sum_{a_2=1}^{n_2} X_a^2 - \frac{1}{4} [X_a, X_b]^2 \longrightarrow \frac{1}{2} s_1^2 \sum_{a_1=1}^{n_1} X_{a_1}^2 + \frac{1}{2} s_2^2 \sum_{a_2=1}^d X_{a_2}^2, \quad (8.35)$$

leading to the effective action

$$S = N \int dt \text{Tr} \left[\frac{1}{2} (D_t X_a)^2 + \frac{1}{2} s_1^2 X_{a_1}^2 + \frac{1}{2} s_2^2 X_{a_2}^2 \right]. \quad (8.36)$$

The two gaps are related by $s_2 = \sqrt{s_1^2 + \Delta m} = k_0 + m_2$, while $s_1 = k_0 + m_1$ is fixed by the eighth-degree equation

$$(s_1^3 - m_1 s_1 - n_1)^2 (s_1^2 + \Delta m) - n_2^2 s_1^2 = 0. \quad (8.37)$$

In the large- d limit the single-flavor gap behaves as $s \sim d^{1/3}$, and the two-flavor gaps become degenerate, $s_1 \sim s_2 \sim d^{1/3}$. Consequently, even in the split-mass case, the dynamics is well approximated at leading order by the single-mass action (8.34).

Moreover, in the absence of mass deformation, corresponding to the genuine BFSS $_{d+1}$ model, the large- d gap equation yields the exact relation $s^2 = d^{1/3}$.

Since the resulting theory is Gaussian, the integration over the matrices X_a can be performed exactly. Fixing the gauge on the thermal circle reduces the gauge field to a constant holonomy $g = \mathcal{P} \exp(i \int_0^\beta dt A_0)$, and the path integral reduces to a group integral over g . Diagonalizing $g = \text{diag}(z_1, \dots, z_N)$, the normal-ordered $SU(N)$ partition function can be written as a Molien–Weyl integral [60, 61]

$$Z_{N,d}(x) = \frac{1}{N!} \frac{1}{(1-x_1)^{n_1(N-1)} (1-x_2)^{n_2(N-1)}} \oint \prod_{i=1}^N \frac{dz_i}{2\pi i z_i} \Delta_A(-1, z) \frac{1}{\Delta_b^{n_2}(-x_2, z) \Delta_b^{n_1}(-x_1, z)}, \quad (8.38)$$

where the Vandermonde–Faddeev–Popov determinant Δ_A and the bosonic determinant Δ_B are built from

$$\Delta(x, z) = \prod_{i < j} \left(1 + x \frac{z_i}{z_j} \right) \prod_{i < j} \left(1 + x \frac{z_j}{z_i} \right), \quad (8.39)$$

and with the fugacities x_1 and x_2 defined in terms of the masses m_1 and m_2 by the equation

$$x_i = e^{-\beta m_i}. \quad (8.40)$$

These holonomy models are closely related to multitrace matrix models, in which the dynamics is entirely encoded in the eigenvalue distribution [62, 63].

In this language, the bosonic determinants (and, in supersymmetric extensions, the combined bosonic and fermionic determinants) play, for the holonomy angles, the role that the classical scalar potential $\text{Tr}(bM^2 + cM^4 + \dots)$ plays for the eigenvalues of M in multitrace matrix models. More generally, one may introduce an explicit eigenvalue potential $V(\theta)$ through a Boltzmann weight $\exp[-V(\theta)]$, leading to a broad class of scalar–gauge matrix models governed by hybrid scalar–holonomy dynamics.

8.3 Supersymmetric completion and Monte Carlo simulations

To construct the supersymmetric completion of the Gaussian models discussed above, it is instructive to work out explicitly the case of BMN_3 , and then extend the same construction to the allowed supersymmetric BFSS_{d+1} and BMN_{d+1} models listed in Sec. 1.1.

However, the meaning of the large- d construction in the supersymmetric case should be first stated carefully. Supersymmetry restricts the admissible values of d through the corresponding Fierz identities, and hence there is no continuous family of fully supersymmetric BFSS_{d+1} or BMN_{d+1} theories for arbitrary d . Nevertheless, the large- d limit used here is literal at the level of the bosonic sector: one first determines the self-consistent large- d bosonic gap mass s^2 , or the corresponding set of gap masses in the split-mass case. The resulting Gaussian bosonic theory is then completed supersymmetrically, by analogy with the large-mass Gaussian limit of the corresponding supersymmetric model, through the addition of the fermionic sector and the imposition of the boson-fermion mass relations dictated by supersymmetry.

The resulting theory is therefore the supersymmetric completion of the large- d bosonic Gaussian saddle. The qualifier “large- d ” refers to the origin of the dynamically generated bosonic masses, whereas the supersymmetric completion is performed at each admissible fixed value of d . This construction is implemented explicitly by the three-step procedure 1, 2, and 3 given below.

The starting point is then BFSS_3 matrix quantum mechanics with mass deformation defined by the action

$$S_{\text{BFSS}_3} = N \int dt \text{Tr} \left[\frac{1}{2} (D_t X_a)^2 - \frac{1}{2} [X_1, X_2]^2 + \frac{i}{2} \bar{\Psi} \gamma_E^0 D_t \Psi + \frac{1}{2} \bar{\Psi} \gamma_E^a [X_a, \Psi] + \frac{\mu^2}{72} X_a^2 - \frac{i\mu}{8} \bar{\Psi} \Psi \right]. \quad (8.41)$$

The simplest approximation is obtained by dropping the interaction terms, leading to the Gaussian theory

$$S_{\text{BFSS}_3}^{\text{Gaussian}} = N \int dt \text{Tr} \left[\frac{1}{2} (D_t X_a)^2 + \frac{i}{2} \bar{\Psi} \gamma_E^0 D_t \Psi + \frac{\mu^2}{72} X_a^2 - \frac{i\mu}{8} \bar{\Psi} \Psi \right], \quad (8.42)$$

which is still supersymmetric. Writing the fermions in terms of a complex spinor ψ , this action takes the form

$$S_{\text{BFSS}_3}^{\text{Gaussian}} = N \int dt \text{Tr} \left[\frac{1}{2} (D_t X_a)^2 + \frac{1}{2} \psi^\dagger D_t \psi + \frac{\mu^2}{72} X_a^2 + \frac{\mu}{8} \psi^\dagger \psi \right]. \quad (8.43)$$

This action should in fact be compared with the BFSS_2 model defined by the action

$$S_{\text{BFSS}_2} = N \int dt \text{Tr} \left(\frac{1}{2} (D_t X)^2 + \frac{1}{2} \psi D_t \psi - \frac{1}{2} \Lambda(t) X^2 - \rho(t) X \right), \quad (8.44)$$

with $\rho(t) = 0$ and $-\Lambda = \mu^2/36$. Identifying $m = m_b^2$, one finds the mass relation

$$m_b = \frac{\mu}{6}, \quad m_f = \frac{\mu}{4} = \frac{3}{2} m_b. \quad (8.45)$$

It follows that the supersymmetric analogue of the large- d single-mass Gaussian model (8.34) is

$$S_{\text{BFSS}_3}^{\text{Gaussian/SUSY}} = N \int dt \text{Tr} \left[\frac{1}{2} (D_t X_a)^2 + \frac{1}{2} s_b^2 X_a^2 + \frac{i}{2} \bar{\Psi} \gamma_E^0 D_t \Psi - \frac{i s_f}{2} \bar{\Psi} \Psi \right], \quad (8.46)$$

with

$$s_b^2 = k_0 + m = k_0 + m_b^2, \quad s_f = \frac{3}{2} s_b. \quad (8.47)$$

An intermediate model between BFSS_3 and BFSS_2 consists of a single bosonic matrix (as in BFSS_2) but a complex fermion (as in BFSS_3):

$$S_{\text{BFSS}_{2/3}}^{\text{Gaussian}} = N \int dt \text{Tr} \left[\frac{1}{2} (D_t X)^2 + \frac{1}{2} \psi^\dagger D_t \psi - \frac{1}{2} \Lambda(t) X^2 - \rho(t) X + \frac{\mu}{8} \psi^\dagger \psi \right]. \quad (8.48)$$

Introducing the fugacities

$$x_b = e^{-\beta m_b}, \quad x_f = e^{-\beta m_f}, \quad (8.49)$$

the quadratic bosonic and fermionic degrees of freedom can be integrated out exactly using the Molien–Weyl formula [60, 61]. For BFSS_3 one obtains

$$Z_N^{(3)}(x_b, x_f) = \frac{1}{N!} \frac{1}{2^{N-1}} \frac{(1+x_f)^{N-1}}{(1-x_b)^{N-1}} \oint \prod_{i=1}^N \frac{dz_i}{2\pi i z_i} \Delta_A(-1, z) \frac{\Delta_f(x_f, z)}{[\Delta_b(-x_b, z)]^2}. \quad (8.50)$$

Here, $(m_b, m_f) = (\frac{\mu}{6}, \frac{\mu}{4})$ or $(m_b, m_f) = (s_b, s_f)$.

BFSS_2 is special in that it contains a massless real fermion, yet its Hamiltonian formulation still involves a complex spinor (like BFSS_3). Thus, its Molien–Weyl partition function reads

$$Z_N^{(2)}(x_b, x_f) = \frac{1}{N!} \frac{1}{2^{N-1}} \frac{(1+x_f)^{N-1}}{(1-x_b)^{N-1}} \oint \prod_{i=1}^N \frac{dz_i}{2\pi i z_i} \Delta_A(-1, z) \frac{\Delta_f(x_f, z)}{\Delta_b(-x_b, z)}, \quad (8.51)$$

with $(m_b, m_f) = (\sqrt{-\Lambda}, 0)$. In fact, this partition functions holds also for the BFSS_{2/3} model with $(m_b, m_f) = (\sqrt{-\Lambda}, \mu/4)$.

For a general BMN _{$d+1$} model, the construction of its supersymmetric large- d Gaussian approximation follows a simple three-step procedure:

1. determine the large- d bosonic gap mass s_1 , or the pair s_1, s_2 in the split-mass case;
2. analyze the large-mass limit of the supersymmetric theory to fix the form of the fermionic sector and the boson-fermion mass relation that preserve supersymmetry at the Gaussian level;
3. replace the bare bosonic masses by their large- d gap values and add the fermionic completion dictated by the large-mass Gaussian theory, while keeping the boson-fermion mass relation fixed.

Thus, for bosonic masses m_1, m_2 with multiplicities n_1, n_2 and n_f complex fermions of mass m_f , the normal-ordered $SU(N)$ partition function takes the Molien-Weyl form [60, 61]

$$Z_{N,d}(x) = \frac{1}{N!} \frac{(1+x_f)^{n_f(N-1)}}{(1-x_1)^{n_1(N-1)}(1-x_2)^{n_2(N-1)}} \oint \prod_{i=1}^N \frac{dz_i}{2\pi i z_i} \Delta_A(1, z) \frac{\Delta_f^{n_f}(x_f, z)}{\Delta_b^{n_2}(-x_2, z) \Delta_b^{n_1}(-x_1, z)}. \quad (8.52)$$

Equivalently, this can be written as a purely holonomy model,

$$Z_{N,d}(x) = \oint \prod_{i=1}^N \frac{d\theta_i}{2\pi} e^{-S_{\text{eff}}(\theta)}, \quad (8.53)$$

which defines the gauged supersymmetric matrix harmonic oscillator (SMHO), generalizing the gauged matrix harmonic oscillator (MHO) defined by equation (8.36). The single-flavor case is recovered by setting $s_1 = s_2$.

Alternatively, one may integrate out only the Gaussian fermionic degrees of freedom using the Molien-Weyl formula, leading to a purely bosonic holonomy model, which we refer to as the *Bosonic Molien-Weyl* model. In this current formulation the bosonic sector is Gaussian, while the full bosonic action—including the commutator interaction terms, both Yang-Mills and possible Chern-Simons contributions—may also be retained explicitly yielding a genuine *Bosonic Molien-Weyl* model [65].

Accordingly, the coordinate and gauge fields are treated as in the standard lattice formulation of BFSS _{$d+1$} matrix models [65]. In particular, we non-perturbatively fix the local $U(N)$ gauge symmetry on the lattice by assigning the holonomy matrix $D = \text{diag}(e^{i\theta_1}, \dots, e^{i\theta_N})$ to the final temporal link. This procedure leads to a bosonized matrix quantum mechanics of the form

$$\begin{aligned}
S_{\text{eff}} &= N \sum_{n=0}^{\Lambda-1} \left[\frac{1}{a} \text{Tr} X_a^2(n) + \frac{as_1^2}{2} \sum_{a_1=1}^{n_1} \text{Tr} X_{a_1}^2(n) + \frac{as_2^2}{2} \sum_{a_2=1}^{n_2} \text{Tr} X_{a_2}^2(n) \right] \\
&- \frac{1}{a} \sum_{n=0}^{\Lambda-2} \text{Tr} X_a(n) X_a(n+1) - \frac{N}{a} \sum_{i,j} e^{-i(\theta_i - \theta_j)} (X_a(\Lambda-1))_{ij} (X_a(0))_{ji} \\
&- \frac{1}{2} \sum_{i \neq j} \ln \sin^2 \frac{\theta_i - \theta_j}{2}, \text{ Vandermonde determinant} \\
&- \frac{n_f}{2} \sum_{i \neq j} \ln \left[(1 + x_f)^2 - 4x_f \sin^2 \frac{\theta_i - \theta_j}{2} \right], \text{ fermion determinant.} \tag{8.54}
\end{aligned}$$

The lattice has spacing a , with the inverse temperature defined by $\beta = \Lambda a$. Again, the single-flavor case is recovered by setting $s_1 = s_2$, which indeed coincide as $s_1 \sim s_2 \sim d^{1/3}$ in the large- d limit. In the absence of mass deformation, corresponding to the true BFSS $_{d+1}$ model, we have exactly $s = d^{1/3}$. The values of n_i , s_i , and n_f for the models considered here are summarized in Table (2). Monte Carlo simulations of these models, either in the purely holonomic form (8.53) or in the bosonized lattice form (8.54), will be reported in detail elsewhere [65].

BFSS _{d+1} QM	BPS configuration	Spectrum	Deformation Parameter
BMN ₁₀	$X_a = 0$, $a = 1, \dots, 6$ $D_t X^p = 0$, $p = 7, 8, 9$ $[X^p, X^q] = i \frac{1}{3} \mu \epsilon^{pqr} X_r$	$6B$: $m_{b1} = \mu/6$, $n_{b1} = 6$ $3B$: $m_{b2} = \mu/3$, $n_{b2} = 3$ $8F$: $m_f = \mu/4$, $n_f = 8$.	μ
BMN ₆ type I	$X_1 = R \cos\left(\frac{1}{6}t\mu\right)$, $X_2 = R \sin\left(\frac{1}{6}t\mu\right)$, $D_t X^p = 0$, $p = 3, 4, 5$ $[X^p, X^q] = i \frac{1}{3} \mu \epsilon^{pqr} X_r$	$2B$: $m_{b1} = \mu/6$, $n_{b1} = 2$ $3B$: $m_{b2} = \mu/3$, $n_{b2} = 3$ $4F$: $m_f = \mu/4$, $n_f = 4$.	μ
BMN ₆ type II	None	$1B$: $m_{b1} = \mu/2$, $n_{b1} = 1$ $4B$: $m_{b2} = \mu/6$, $n_{b2} = 4$ $4F$: $m_f = \mu/4$, $n_f = 4$.	μ
BMN ₄ type I	$D_t X^a = 0$, $a = 1, 2, 3$ $[X^a, X^b] = i \frac{1}{3} \mu_2 \epsilon^{abc} X_c$	$3B$: $m_b = \sqrt{\mu_1^2 + \mu_2^2}/3$, $n_b = 3$ $2F$: $m_f = \sqrt{\mu_1^2 + \mu_2^2}/2$, $n_f = 2$.	$\mu_1 \neq 0, \mu_2$
BMN ₄ type I	$D_t X^a = 0$, $[X^a, X^b] = i \frac{1}{3} \mu_2 \epsilon^{abc} X^c$	$3B$: $m_b = \mu_2/3$, $n_b = 3$ $2F$: $m_f = \mu_3/2$, $n_f = 2$.	μ_2, μ_3 is fictitious
BMN ₄ type II	$X_1 = R \cos\left(\frac{1}{6}t\mu_1\right)$, $X_2 = R \sin\left(\frac{1}{6}t\mu_1\right)$, $X_3 = 0$	$2B$: $m_{b1} = \mu/6$, $n_{b1} = 2$ $1B$: $m_{b2} = \mu/3$, $n_{b2} = 1$ $2F$: $m_f = \mu/4$, $n_f = 2$.	μ
BMN ₃	$X_1 = R \cos\left(\frac{1}{6}t\mu\right)$, $X_2 = R \sin\left(\frac{1}{6}t\mu\right)$.	$2B$: $m_{b1} = \mu/6$, $n_b = 2$ $1F$: $m_f = \mu/4$, $n_f = 1$.	μ
BMN ₂	AdS ²	$1B$: $m_{b1} = \Lambda(t)$, $n_b = 1$ $1/2F$: $m_f = 0$, $n_f = 1$.	$\Lambda(t), \rho(t)$

Table 2: Summary of deformation parameters, spectra and classical BPS solutions of the BFSS/BMN models.

9 Conclusion

We have analyzed the large- d dynamics of the mass-deformed bosonic BFSS $_{d+1}$ matrix model in a correlated double-scaling limit in which

$$d \longrightarrow \infty, \quad m \longrightarrow \infty, \quad \kappa \equiv \frac{m^{3/2}}{\alpha d} \text{ fixed.} \quad (9.1)$$

Equivalently,

$$m = (\kappa \alpha d)^{2/3}, \quad (9.2)$$

so that, after setting $\alpha = 1$, the ratio $m/d^{2/3}$ is held fixed. This is precisely the scaling in which the explicit mass deformation and the dynamically generated Yang–Mills mass shift remain parametrically balanced.

The analysis is controlled by a Gaussian saddle, equivalently a gauged matrix harmonic oscillator, whose self-consistent frequency s is determined by the large- d gap equation. In the double-scaling limit this saddle remains valid over a parametrically wide range of temperatures, covering both the enlarged uniform-holonomy regime and the high-temperature Gaussian branch.

Our first main result is that, in this double-scaling limit, the mass-deformed BFSS $_{d+1}$ model is genuinely dominated by the uniform, center-symmetric holonomy phase. In particular, the effective holonomy action takes the universal Gross–Witten–Wadia form with coefficients $1 - dq^n$, where

$$q = e^{-\beta s}. \quad (9.3)$$

Since the saddle frequency scales as $s \sim d^{1/3}$, the deconfinement instability is pushed to parametrically higher temperature,

$$T_c \sim \frac{d^{1/3}}{\log d}. \quad (9.4)$$

Thus the uniform holonomy region expands parametrically with d . This is the only regime observed in BFSS $_2$ matrix quantum mechanics, where no finite-temperature deconfinement transition exists. As a consequence, the double-scaled BFSS $_{d+1}$ saddle does not merely produce a Gaussian approximation: it opens a parametrically large BFSS $_2$ -like, or AdS $_2$ -like, stringy regime.

In this regime the bulk dynamics reduces to d effectively independent gauged matrix harmonic oscillators. The holonomy-mediated interaction between different matrix directions is weak because the system lies deep in the uniform holonomy region at large d , where

$$dq = d e^{-\beta s} \ll 1. \quad (9.5)$$

Our second main result concerns the structure of the theory in the overlap window

$$T_c^{\text{high}} < T < T_c^{\text{low}}, \quad (9.6)$$

where both the low-temperature and high-temperature saddle-point analyses are simultaneously valid. In this window, explicit evaluation of the extent of space and of the Yang–Mills observable shows that the basic quantities per matrix and per matrix pair are parametrically suppressed at large d . In particular, the commutator contribution between two fixed matrix directions satisfies

$$-\left\langle \text{Tr}[X_1, X_2]^2 \right\rangle \sim d^{-2/3} \longrightarrow 0 \quad (d \rightarrow \infty). \quad (9.7)$$

Thus noncommutativity is weak at the saddle point, and the dynamics is dominated by approximately commuting matrices. The resulting physics is therefore IKKT-like: spacetime emerges from nearly commuting matrices, while interactions are encoded in collective large- N effects rather than in strong matrix noncommutativity.

Importantly, this overlap window widens with increasing d . Together with the parametrically enlarged uniform-holonomy regime described above, this provides a clear signal that the double-scaling limit of the mass-deformed BFSS $_{d+1}$ model interpolates between two complementary descriptions: a BFSS $_2$ -like, AdS $_2$ -like stringy regime governed by weakly coupled single-matrix gauged harmonic-oscillator sectors, and an IKKT-type commuting-matrix regime in which the matrices are localized and almost commuting. This identifies the double-scaled large- d model as a natural arena in which both descriptions coexist and can be treated in a unified and controlled framework.

Finally, we explained how the large- d Gaussian framework extends to mass-deformed supersymmetric BFSS/BMN models and their Molien–Weyl approximations. In the split-mass case, different bosonic sectors acquire different oscillator gaps, but these gaps are tied together by a single self-consistent Hubbard–Stratonovich saddle and become degenerate at leading order in large d . The resulting Gaussian theory admits two useful formulations: a purely holonomic Molien–Weyl model, obtained by integrating out all Gaussian bosonic and fermionic degrees of freedom, and a bosonized lattice model, in which only the fermions are integrated out while the bosonic matrices remain explicit. These formulations provide practical Gaussian benchmarks for Monte Carlo simulations of the full interacting BFSS/BMN theory.

10 Acknowledgments

The author would like to acknowledge helpful discussions with Denjoe O’Connor from the Dublin Institute for Advanced Studies. The author is especially grateful for Denjoe O’Connor’s continued institutional hosting and generous support over the years, including travel, accommodation, and living expenses.

The author also acknowledges the use of ChatGPT-5.5, as well as previous versions, in several auxiliary capacities: (1) as a language editor; (2) as a LaTeX generator; (3) as a Mathematica-like symbolic tool; (4) as an assistant in searching for and reviewing references; and, more importantly, (5) as an “artificial” sounding board for testing, organizing, and refining ideas, effectively replacing in this role the function often played by human collaborators. However, the scientific vision, concept, design, direction, final scientific and mathematical editing, and all intellectual responsibility for this work remain solely with the author.

A Fourier derivation of the holonomic effective action

A.1 Coordinate contribution

The X_a determinant contributes the action

$$S_X[\theta] = \frac{d}{2} \sum_{i,j=1}^N P_{ij,ji} \log \left(\cosh(\beta s) - \cos(\theta_i - \theta_j) \right), \quad (\text{A.1})$$

up to θ -independent constants.

For $x > 0$ one has the identity

$$\log(\cosh x - \cos \phi) = \text{const} - 2 \sum_{n \geq 1} \frac{e^{-nx}}{n} \cos(n\phi), \quad x > 0, \quad (\text{A.2})$$

where again “const” is independent of ϕ . To derive this relation, we start from the elementary factorization

$$\cosh x - \cos \phi = \frac{1}{2} (e^x + e^{-x} - e^{i\phi} - e^{-i\phi}) = \frac{1}{2} e^x (1 - e^{-x} e^{i\phi}) (1 - e^{-x} e^{-i\phi}), \quad x > 0. \quad (\text{A.3})$$

Taking the logarithm yields

$$\log(\cosh x - \cos \phi) = x - \log 2 + \log(1 - qe^{i\phi}) + \log(1 - qe^{-i\phi}), \quad q \equiv e^{-x} \in (0, 1). \quad (\text{A.4})$$

Using the Taylor expansion

$$\log(1 - z) = - \sum_{n \geq 1} \frac{z^n}{n}, \quad |z| < 1, \quad (\text{A.5})$$

we then obtain

$$\log(1 - qe^{i\phi}) = - \sum_{n \geq 1} \frac{q^n}{n} e^{in\phi}, \quad \log(1 - qe^{-i\phi}) = - \sum_{n \geq 1} \frac{q^n}{n} e^{-in\phi}. \quad (\text{A.6})$$

Adding these two series gives immediately our desired identity (A.2).

Apply now this identity (A.2) with $x = \beta s$ and $\phi = \theta_i - \theta_j$ we get the action

$$\begin{aligned} S_X[\theta] &= \frac{d}{2} \sum_{i,j} P_{ij,ji} \left[\text{const} - 2 \sum_{n \geq 1} \frac{e^{-n\beta s}}{n} \cos(n(\theta_i - \theta_j)) \right] \\ &= \text{const} - d \sum_{n \geq 1} \frac{q^n}{n} \sum_{i,j=1}^N P_{ij,ji} \cos(n(\theta_i - \theta_j)), \quad q \equiv e^{-\beta s}. \end{aligned} \quad (\text{A.7})$$

Next, we introduce the Fourier modes of the holonomy defined by the Polyakov moments

$$u_n \equiv \frac{1}{N} \sum_{j=1}^N e^{in\theta_j}, \quad u_{-n} = u_n^*. \quad (\text{A.8})$$

Then

$$\sum_{i,j=1}^N e^{in(\theta_i - \theta_j)} = \left(\sum_i e^{in\theta_i} \right) \left(\sum_j e^{-in\theta_j} \right) = N^2 u_n u_{-n} = N^2 |u_n|^2, \quad (\text{A.9})$$

and since the result is real we also have

$$\sum_{i,j=1}^N \cos(n(\theta_i - \theta_j)) = \sum_{i,j=1}^N \text{Re} \left(e^{in\theta_i} e^{-in\theta_j} \right) = N^2 |u_n|^2. \quad (\text{A.10})$$

We also include the adjoint projector factor:

$$\begin{aligned} \sum_{i,j} P_{ij,ji} \cos(n(\theta_i - \theta_j)) &= \sum_{i,j} \cos(n(\theta_i - \theta_j)) - \frac{1}{N} \sum_i \cos(0), \quad P_{ij,ji} = 1 - \frac{1}{N} \delta_{ij} \\ &= N^2 |u_n|^2 - 1. \end{aligned} \quad (\text{A.11})$$

Dropping the θ -independent constants in (A.7) using (A.11), one obtains

$$S_X[\theta] = -d N^2 \sum_{n \geq 1} \frac{q^n}{n} |u_n|^2, \quad q = e^{-\beta s}. \quad (\text{A.12})$$

This term is attractive as it favors eigenvalue clumping.

A.2 Vandermonde contribution

After fixing to the static diagonal (Polyakov) gauge, the Haar measure of $SU(N)$ induces the Vandermonde determinant, which can be written as the effective action

$$S_{\text{Vdm}}[\theta] = - \sum_{i < j} \log \left[4 \sin^2 \left(\frac{\theta_i - \theta_j}{2} \right) \right]. \quad (\text{A.13})$$

This contribution is purely entropic and universal, as it arises solely from the group measure and is present even in the absence of matter fields.

To analyze this term, we use the Fourier identity

$$-\log\left(2\sin\frac{\phi}{2}\right) = \sum_{n\geq 1} \frac{\cos(n\phi)}{n} + \text{const}, \quad 0 < \phi < 2\pi, \quad (\text{A.14})$$

This identity follows from a standard Fourier expansion. For $|z| < 1$ one has

$$-\log(1-z) = \sum_{n\geq 1} \frac{z^n}{n}. \quad (\text{A.15})$$

Setting $z = re^{i\phi}$ with $0 < r < 1$ and taking the real part yields

$$-\text{Re} \log(1 - re^{i\phi}) = \sum_{n\geq 1} \frac{r^n}{n} \cos(n\phi). \quad (\text{A.16})$$

Since $\text{Re} \log w = \log |w|$, this can be written as

$$-\log |1 - re^{i\phi}| = -\frac{1}{2} \log(1 - 2r \cos \phi + r^2) = \sum_{n\geq 1} \frac{r^n}{n} \cos(n\phi), \quad 0 < r < 1. \quad (\text{A.17})$$

Finally, taking the Abel limit $r \rightarrow 1^-$ gives, for $0 < \phi < 2\pi$, the desired Fourier identity (A.14).

Using now the Fourier identity (A.14), one finds

$$S_{\text{Vdm}}[\theta] = \sum_{i<j} 2 \sum_{n\geq 1} \frac{\cos(n(\theta_i - \theta_j))}{n} + \text{const}. \quad (\text{A.18})$$

Using

$$\sum_{i<j} \cos(n(\theta_i - \theta_j)) = \frac{N^2}{2} |u_n|^2 - \frac{N}{2},$$

we obtain, up to θ -independent constants,

$$S_{\text{Vdm}}[\theta] = N^2 \sum_{n\geq 1} \frac{1}{n} |u_n|^2. \quad (\text{A.19})$$

Adding both contributions (A.12) and (A.19), the leading large- d holonomy effective action is

$$S_{\text{hol}}[\theta] = N^2 \sum_{n\geq 1} \frac{1-dq^n}{n} |u_n|^2 + \text{const}. \quad (\text{A.20})$$

References

- [1] L. Brink, J. H. Schwarz and J. Scherk, “Supersymmetric Yang-Mills Theories,” Nucl. Phys. B **121**, 77-92 (1977)
- [2] M. Baake, M. Reinicke and V. Rittenberg, “Fierz Identities for Real Clifford Algebras and the Number of Supercharges,” J. Math. Phys. **26**, 1070 (1985)
- [3] G. 't Hooft, *A Planar Diagram Theory for Strong Interactions*, Nucl. Phys. B **72**, 461 (1974).
- [4] G. 't Hooft, *Dimensional Reduction in Quantum Gravity*, arXiv:gr-qc/9310026.
- [5] L. Susskind, *The World as a Hologram*, J. Math. Phys. **36**, 6377 (1995).
- [6] T. Banks, W. Fischler, S. H. Shenker and L. Susskind, “M theory as a matrix model: A conjecture,” Phys. Rev. D **55**, 5112-5128 (1997) [arXiv:hep-th/9610043 [hep-th]].
- [7] E. Witten, *Bound States of Strings and p-Branes*, Nucl. Phys. B **460**, 335 (1996).
- [8] N. Itzhaki, J. M. Maldacena, J. Sonnenschein and S. Yankielowicz, *Supergravity and the Large N Limit of Theories with Sixteen Supercharges*, Phys. Rev. D **58**, 046004 (1998).
- [9] J. Polchinski, *Dirichlet Branes and Ramond–Ramond Charges*, Phys. Rev. Lett. **75**, 4724 (1995).
- [10] E. Cremmer, B. Julia, and J. Scherk, *Supergravity Theory in Eleven Dimensions*, Phys. Lett. B **76**, 409 (1978).
- [11] E. Witten, “String theory dynamics in various dimensions,” *Nucl. Phys. B* **443**, 85 (1995).
- [12] Y. Hyakutake, “Quantum M-wave and Black 0-brane,” JHEP **09**, 075 (2014) [arXiv:1407.6023 [hep-th]].
- [13] Y. Hyakutake and S. Ogushi, “Higher derivative corrections to eleven dimensional supergravity via local supersymmetry,” JHEP **02**, 068 (2006) [arXiv:hep-th/0601092 [hep-th]].
- [14] J. Hoppe, *Quantum Theory of a Massless Relativistic Surface and a Two-Dimensional Bound State Problem*, Ph.D. Thesis, MIT (1982).
- [15] J. Hoppe, “Diffeomorphism Groups, Quantization, and $SU(\infty)$,” Int. J. Mod. Phys. A **4**, 5235 (1989).
- [16] B. de Wit, J. Hoppe and H. Nicolai, *On the Quantum Mechanics of Supermembranes*, Nucl. Phys. B **305**, 545 (1988).
- [17] J. Kowalski-Glikman, “Vacuum States in Supersymmetric Kaluza-Klein Theory,” Phys. Lett. B **134**, 194-196 (1984)

- [18] M. Blau, J. M. Figueroa-O’Farrill, C. Hull and G. Papadopoulos, “A New maximally supersymmetric background of IIB superstring theory,” JHEP **01**, 047 (2002) [arXiv:hep-th/0110242 [hep-th]].
- [19] T. Azeyanagi, M. Hanada, T. Hirata and H. Shimada, *On the Shape of a D-Brane Bound State and Its Topology Change*, J. High Energy Phys. **0903**, 121.
- [20] B. Zwiebach, *A First Course in String Theory*, 2nd ed., Cambridge University Press, Cambridge (2009).
- [21] K. Becker, M. Becker and J. H. Schwarz, *String Theory and M-Theory: A Modern Introduction* (Cambridge University Press, Cambridge, 2006).
- [22] J. M. Maldacena, *The Large N Limit of Superconformal Field Theories and Supergravity*, Adv. Theor. Math. Phys. **2**, 231 (1998) [Int. J. Theor. Phys. **38**, 1113 (1999)], arXiv:hep-th/9711200.
- [23] S. S. Gubser, I. R. Klebanov, and A. M. Polyakov, *Gauge Theory Correlators from Non-Critical String Theory*, Phys. Lett. B **428**, 105 (1998), arXiv:hep-th/9802109.
- [24] E. Witten, *Anti-de Sitter Space and Holography*, Adv. Theor. Math. Phys. **2**, 253 (1998), arXiv:hep-th/9802150.
- [25] K. G. Wilson, “Confinement of Quarks,” Phys. Rev. D **10**, 2445-2459 (1974).
- [26] Catterall, S. and Wiseman, T., “Black hole thermodynamics from simulations of lattice Yang–Mills theory,” *Phys. Rev. D* **78**, 041502 (2008).
- [27] Anagnostopoulos, K. N., Hanada, M., Nishimura, J. and Takeuchi, S., “Monte Carlo studies of supersymmetric matrix quantum mechanics with sixteen supercharges at finite temperature,” *Phys. Rev. Lett.* **100**, 021601 (2008).
- [28] Hanada, M., Hyakutake, Y., Ishiki, G. and Nishimura, J., “Holographic description of quantum black hole on a computer,” *Science* **344**, 882 (2014).
- [29] Hanada, M., Hyakutake, Y., Ishiki, G. and Nishimura, J., “Numerical tests of the gauge/gravity duality conjecture for D0-branes at finite temperature and finite N ,” *Phys. Rev. D* **94**, 086010 (2016).
- [30] V. G. Filev and D. O’Connor, “The BFSS model on the lattice,” JHEP **1605**, 167 (2016) [arXiv:1506.01366 [hep-th]].
- [31] Kabat, D. N., Lifschytz, G. and Lowe, D. A., “Black hole thermodynamics from calculations in strongly coupled gauge theory,” *Phys. Rev. Lett.* **86**, 1426 (2001).

- [32] Hanada, M., Hyakutake, Y., Nishimura, J. and Takeuchi, S., “Higher derivative corrections to black hole thermodynamics from supersymmetric matrix quantum mechanics,” *Phys. Rev. Lett.* **102**, 191602 (2009).
- [33] Y. Hyakutake, “Quantum near-horizon geometry of a black 0-brane,” *Progr. Theor. Exp. Phys.* **2014**, 033B04 (2014).
- [34] M. Hanada, *What Lattice Theorists Can Do for Superstring/M-Theory*, *Int. J. Mod. Phys. A* **31**, 1643006 (2016).
- [35] D. E. Berenstein, J. M. Maldacena and H. S. Nastase, “Strings in flat space and pp waves from N=4 superYang-Mills,” *JHEP* **04**, 013 (2002) [arXiv:hep-th/0202021 [hep-th]].
- [36] N. Kim and J. H. Park, “Massive super Yang-Mills quantum mechanics: Classification and the relation to supermembrane,” *Nucl. Phys. B* **759**, 249–282 (2006) [arXiv:hep-th/0607005].
- [37] J. H. Park, “Noncritical $\mathfrak{osp}(1|2, \mathbb{R})$ M-theory matrix model with an arbitrary time-dependent cosmological constant,” *Nucl. Phys. B* **745**, 123–141 (2006) [arXiv:hep-th/0510070].
- [38] R. C. Myers, “Dielectric-branes,” *JHEP* **12**, 022 (1999) [arXiv:hep-th/9910053 [hep-th]].
- [39] H. Lin, O. Lunin and J. M. Maldacena, “Bubbling AdS space and 1/2 BPS geometries,” *JHEP* **10**, 025 (2004) doi:10.1088/1126-6708/2004/10/025 [arXiv:hep-th/0409174 [hep-th]].
- [40] Y. Asano, V. G. Filev, S. Kováčik and D. O’Connor, “The non-perturbative phase diagram of the BMN matrix model,” *JHEP* **07**, 152 (2018) [arXiv:1805.05314 [hep-th]].
- [41] Y. Asano, S. Kováčik and D. O’Connor, “The Confining Transition in the Bosonic BMN Matrix Model,” *JHEP* **06**, 174 (2020) [arXiv:2001.03749 [hep-th]].
- [42] G. Mandal, M. Mahato and T. Morita, “Phases of one dimensional large N gauge theory in a 1/D expansion,” *JHEP* **1002** (2010) 034 doi:10.1007/JHEP02(2010)034 [arXiv:0910.4526 [hep-th]].
- [43] G. Mandal and T. Morita, “Phases of a two dimensional large N gauge theory on a torus,” *Phys. Rev. D* **84**, 085007 (2011) doi:10.1103/PhysRevD.84.085007 [arXiv:1103.1558 [hep-th]].
- [44] D. N. Kabat, G. Lifschytz and D. A. Lowe, “Black hole thermodynamics from calculations in strongly coupled gauge theory,” *Int. J. Mod. Phys. A* **16**, 856 (2001) [*Phys. Rev. Lett.* **86**, 1426 (2001)] [hep-th/0007051].

- [45] D. N. Kabat, G. Lifschytz and D. A. Lowe, “Black hole entropy from nonperturbative gauge theory,” *Phys. Rev. D* **64**, 124015 (2001) [hep-th/0105171].
- [46] K. Furuuchi, E. Schreiber and G. W. Semenoff, “Five-brane thermodynamics from the matrix model,” hep-th/0310286.
- [47] D. J. Gross and E. Witten, “Possible Third Order Phase Transition in the Large N Lattice Gauge Theory,” *Phys. Rev. D* **21**, 446 (1980).
- [48] S. R. Wadia, “N = Infinity Phase Transition in a Class of Exactly Soluble Model Lattice Gauge Theories,” *Phys. Lett.* **93B**, 403 (1980).
- [49] A. M. Polyakov, *Gauge Fields and Strings*, Harwood Academic Publishers, Chur, Switzerland (1987).
- [50] O. Aharony, J. Marsano, S. Minwalla, K. Papadodimas and M. Van Raamsdonk, “The Hagedorn - deconfinement phase transition in weakly coupled large N gauge theories,” *Adv. Theor. Math. Phys.* **8**, 603 (2004) doi:10.4310/ATMP.2004.v8.n4.a1 [hep-th/0310285].
- [51] O. Aharony, J. Marsano, S. Minwalla and T. Wiseman, “Black hole-black string phase transitions in thermal 1+1 dimensional supersymmetric Yang-Mills theory on a circle,” *Class. Quant. Grav.* **21**, 5169 (2004) [hep-th/0406210].
- [52] L. Alvarez-Gaume, C. Gomez, H. Liu and S. Wadia, “Finite temperature effective action, AdS(5) black holes, and 1/N expansion,” *Phys. Rev. D* **71**, 124023 (2005) [hep-th/0502227].
- [53] T. Harmark and N. A. Obers, “New phases of near-extremal branes on a circle,” *JHEP* **0409**, 022 (2004) [hep-th/0407094].
- [54] R. Gregory and R. Laflamme, “Black strings and p-branes are unstable,” *Phys. Rev. Lett.* **70**, 2837-2840 (1993) [arXiv:hep-th/9301052 [hep-th]].
- [55] G. Mandal and T. Morita, “Gregory-Laflamme as the confinement/deconfinement transition in holographic QCD,” *JHEP* **09**, 073 (2011) [arXiv:1107.4048 [hep-th]].
- [56] L. Susskind, “Matrix theory black holes and the Gross-Witten transition,” hep-th/9805115.
- [57] N. Kawahara, J. Nishimura and S. Takeuchi, “Phase structure of matrix quantum mechanics at finite temperature,” *JHEP* **0710**, 097 (2007) doi:10.1088/1126-6708/2007/10/097 [arXiv:0706.3517 [hep-th]].
- [58] J. I. Kapusta and C. Gale, *Finite-Temperature Field Theory: Principles and Applications*, 2nd ed. (Cambridge Univ. Press, 2006).
- [59] M. Le Bellac, *Thermal Field Theory* (Cambridge Univ. Press, 1996).

- [60] D. O'Connor and S. Ramgoolam, “Gauged permutation invariant matrix quantum mechanics: path integrals,” JHEP **04**, 080 (2024) [arXiv:2312.12397 [hep-th]].
- [61] D. O'Connor and S. Ramgoolam, “Permutation invariant matrix quantum thermodynamics and negative specific heat capacities in large N systems,” JHEP **12**, 161 (2024) [arXiv:2405.13150 [hep-th]].
- [62] B. Ydri, R. Khaled and C. Soudani, “Quantized noncommutative geometry from multitrace matrix models,” Int. J. Mod. Phys. A **37**, no.10, 2250052 (2022) [arXiv:2110.06677 [hep-th]].
- [63] B. Ydri, K. Ramda and A. Rouag, “Phase diagrams of the multitrace quartic matrix models of noncommutative Φ^4 theory,” Phys. Rev. D **93**, no.6, 065056 (2016) [arXiv:1509.03726 [hep-th]].
- [64] D. O'Connor, private communication.
- [65] B. Ydri, *unpublished work/work in progress*.
- [66] B. Ydri, *Matrix Models of String Theory*, IOP Expanding Physics, IOP Publishing, Bristol, UK (2018), doi:10.1088/978-0-7503-1726-9.
- [67] E. Sorkin, “A critical dimension in the black-string phase transition,” Phys. Rev. Lett. **93**, 031601 (2004) [arXiv:hep-th/0402216 [hep-th]].
- [68] R. Emparan, R. Luna, M. Martinez, R. Suzuki and K. Tanabe, “Phases and stability of non-uniform black strings,” JHEP **05**, 104 (2018) [arXiv:1802.08191 [hep-th]].
- [69] R. Emparan and C. P. Herzog, “The large D limit of Einstein’s equations,” Rev. Mod. Phys. **92**, no. 4, 045005 (2020) [arXiv:2003.11394 [hep-th]].
- [70] B. Ydri, “A Double-Scaling Large- d Saddle of BFSS/BMN Matrix Quantum Mechanics,” [arXiv:2606.17758 [hep-th]].
- [71] B. Ydri, “Molien–Weyl Singlet Counting and BFSS₂–Factorization in Gaussian Matrix QM,” [arXiv:2605.04621 [hep-th]].
- [72] B. Ydri, *Initiation to Matrix Quantum Gravity and Monte Carlo Simulation of the BFSS₃/BMN₃ System*.
- [73] B. Ydri, “Endpoint formulation and Molien–Weyl structure for the $N = 2$, large- d BFSS/BMN models,” [arXiv:2605.25647 [hep-th]].
- [74] B. Ydri, “Gram–Wishart–Stiefel formulation of the $N = 2$, large- d gauge theory in 1D,” [arXiv:2607.08481 [hep-th]].

- [75] B. Ydri, *Endpoint Entropy and Emergent Grassmannian Geometry in BFSS/BMN Systems*.
- [76] B. Ydri, *Schur–Ingham–Siegel Reduction of Rank–Two Orthogonal HCIZ Integrals in BFSS/BMN Endpoint Geometry*.
- [77] D. O’Connor and S. Ramgoolam, “Negative heat capacities in spherically symmetric sectors of d -matrix quantum mechanics,” [arXiv:2606.09521 [hep-th]].
- [78] Y. Lei and S. Ramgoolam, “Critical dimensions and small cycle dominance from all-orders asymptotics of d -matrix theory,” [arXiv:2603.29610 [hep-th]].